

TU Braunschweig Subject-Specific Qualifications

Application for MSc. Data Science

Category	Course Equivalent	CP (ECTS)
Computer Science (92 ECTS)		
Programming	CSCE 155E - Computer Science I	6
	CSCE 235 - Introduction to Discrete Structures	6
	CSCE 251 - UNIX Programming	2
	CSCE 310 - Data Structures & Algorithms	6
	CSCE 474 - Introduction to Data Mining	6
	CSCE 478 - Introduction to Machine Learning	6
Software Engineering and Database	CSCE 361 - Software Engineering	6
	CSCE 156 - Computer Science II	8
	CSCE 401H - Design Studio I	6
	CSCE 402H - Design Studio II	6
	CSCE 231 - Computer Systems Engineering	8
	CSCE 451 - Operating Systems	6
Theoretical Computer Science	CSCE 322 - Programming Language Concepts	6
	CSCE 423 - Design & Analysis of Algorithms	6
	COMPSCI 136 - Economics & Computation (Algorithmic Game Theory) ¹	8
Mathematics (60 ECTS)		
Analysis	MAT 132 - Calculus & Analytical Geometry I ²	8
	MAT 133 - Calculus & Analytical Geometry II ³	8
	MATH 208 - Calculus III	8
	MATH 325 - Elementary Analysis	6
Linear Algebra Modern Algebra	MATH 314 - Linear Algebra	6
	MATH 310 - Introduction to Modern Algebra	6
	MATH 417 - Group Theory	6
Applied Mathematics	MATH 428 - Introduction to Operations Research	6
	MATH 435 - Math in the City (Mathematical Epidemiology)	6
Statistics and Data Science	ECON 215 - Statistics	6
	STAT 380 - Statistics and Application	6
	ECON 417 - Introduction to Econometrics	6

¹Credits earned at the Harvard Kenneth C. Griffin Graduate School of Arts and Sciences. See [notes on the document](#) for more details.

²Credits transferred from INTI International College Penang, which is deemed equivalent to MATH106 at the University of Nebraska-Lincoln. See [notes on the document](#) for more details.

³Credits transferred from INTI International College Penang, which is deemed equivalent to MATH107 at the University of Nebraska-Lincoln. See [notes on the document](#) for more details.

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Notes on the Document

This is a compilation of the syllabi that are relevant to the requirements of the program for the admissions committee. A document outline is provided for ease of access; the courses are grouped by major and sorted by course number.

At the University of Nebraska-Lincoln, a comprehensive list of all courses offered in my majors, including brief descriptions, can be found in the course catalog at the following links:

1. [Computer Science](#)
2. [Mathematics](#)
3. [Economics](#)

Additionally, I transferred credits from INTI International College Penang, where I obtained credits for MATH 133 and MATH 134. These were recognized as equivalent credits at the University of Nebraska-Lincoln for MATH 106 (Calculus I) and MATH 107 (Calculus II).

However, since INTI International College Penang does not retain its previous syllabi, I have used the syllabi from the University of Nebraska-Lincoln instead.

CS 136 was taken for graduate credit at Harvard University during my time as a Research Associate.

All other credits listed here were obtained at the University of Nebraska-Lincoln.

Computer Science I

CSCE 155E - Syllabus

School of Computing
College of Engineering
University of Nebraska Lincoln

“If you really want to understand something, the best way is to try and explain it to someone else. That forces you to sort it out in your own mind. . . that’s really the essence of programming. By the time you’ve sorted out a complicated idea into little steps that even a stupid machine can deal with, you’ve certainly learned something about it yourself.”

—Douglas Adams, *Dirk Gently’s Holistic Detective Agency*

In my experience, you assert control over a computer—show it who’s the boss—by making it do something unique. That means programming it. . . If you devote a couple of hours to programming a new machine, you’ll feel better about it ever afterwards”

—Michael Crichton, *Electronic Life*

The readings were the most underrated things ever (I’ve never actually read them but I wish I did) thanks for providing them and leaving the whole course available to us.

—Former student a year after taking this course

Enjoyed the videos. Did not really watch them videos until [later], realized that it was a mistake now.

—Former student a year after taking this course

Course Info

Prerequisites: MATH 103 or equivalent.

Description: Introduction to problem solving with computers. Topics include problem solving methods, software development principles, computer programming, and computing in society.

Credit Hours: 3

Postrequisites: The course after this course, CSCE 156 – Computer Science II requires that you receive a grade of C or better in this course to move on. If you are a Computer Science or Computer Engineering major you will need to receive a C or better in this course to continue in the major.

For all other information, see the course website.

University Policies, Resources & Services

Students are responsible for knowing the university policies and be aware of resources found on this page: <https://go.unl.edu/coursepolicies>

Continuation of Instruction

If in-person classes are canceled by the University during a lecture day, lecture may still be livestreamed. If in-person classes are canceled by the University during a lab/hack day, in-person attendance will not be required, help *may* be made available via zoom and you will still be expected to complete the work remotely.

Skills Objectives

This course has several learning objectives and “skills objectives.” These are the skills that, upon successful completion of this course, you should be able to exhibit.

- You should have a mastery of the fundamentals of programming in a high-level language, including data types and rudimentary data structures, control flow, repetition, selection, input/output, and procedures and functions.
- You should be able to approach a reasonably complex problem, design a top-down solution, and code a program in a high-level programming language that automates solutions.
- You should have a familiarity with problem solving methods, including problem analysis, requirements and specifications, design, decomposition and step-wise refinement, and algorithm development (including recursion).
- You should have a familiarity with software development principles and practices, including data and operation abstraction, encapsulation, modularity, code and artifact reuse, prototyping, iterative development, best practices in coding design, style, and documentation, a good understanding of proper testing and debugging techniques and a familiarity with development tools.
- You should have exposure to algorithms for searching, sorting and other problems, graphical user interfaces, event-driven programming, and database access.
- You should have a foundation for further software development and exploration. You should have a deep enough understanding of at least one high-level programming language that you should be able to learn another programming language with relative ease in a relatively short amount of time.

Schedule

See the course website (canvas).

Grading

Assessment (grading) will be based on weekly readings, labs, “hacks”, attendance as well as exams and a comprehensive final with the following point distributions.

Category	Number	Points Each	Total
Starter Points			30
PDC Module/Surveys			70
Readings (zyBooks)	13	15	195
Labs	13	15	195
Hacks	14	25	350
Attendace	14	10	140
Exams	2	75	150
Final	1	150	150
Total			1,280

Starter Points

It is important to start out positively. Put yourself in the mindset that you *will* succeed in this course and commit yourself to putting in a full effort in every aspect of it. To get you started, we’re giving you **free starter points**. You have a perfect score in this course already! Keep it up!

PDC Modules/Surveys

This course is experimenting with introducing **Parallel and Distributed Computing** (PDC) concepts early in the computing curriculum. These topics are usually covered in depth in advanced elective courses. However, we have developed a series of “codeless” modules that do not require any coding knowledge and cover PDC concepts in an accessible manner.

As part of this study, you will complete pre-module survey(s) which will be graded based on completion. You will then go through several modules on PDC which includes running and observing some simulations, visualizations, etc. You may complete these modules at your own pace. Once completed, you will then take post-module survey(s) again graded based on completion only. Full details and due dates are posted in canvas.

Readings

This course is organized into modules (roughly 1 module per week) and each module has reading associated with it. Some reading is required and other

readings are suggested/recommended and provide a deeper understanding of the topics. Some of the required reading will be assessed and will count toward your grade.

The assessed reading for this course is delivered through an online textbook provided by zyBooks, a web-based interactive text book that has you do some reading and then assesses your understanding through short quiz questions. You may retry these quizzes until you get them correct. These readings are assessed based on completion and are generally due *before* the lectures on the modules in order to better prepare you for the presented material. If you do not complete the readings for each module, you will not receive any credit.

Not every set of required reading represents an equal amount of work. Some modules have more reading than others. However, all assessment is equally weighted since it is based on completion.

Labs

There will be weekly labs that give you hands-on exercises for topics recently covered in lecture. The purpose of lab is not only to give you further working experience with lecture topics, but also to provide you with additional information and details not necessarily covered in lecture. Each lab will have some programming requirements and a supplemental worksheet.

Depending on logistics, those in the on-campus section may be randomly paired with a partner. One of you will be the *driver* and the other will be the *navigator*. The navigator will be responsible for reading the instructions and guiding the driver. The driver will be in charge of the keyboard and will type the code. Both driver and navigator are responsible for developing and working through solutions together. Neither the navigator nor the driver is “in charge,” it is an equal partnership. Beyond your immediate pairing, you are encouraged to help and interact and with other pairs in the lab.

Each lab is assessed based on completion. In general you will need to submit code and (possibly) an electronic writeup of your worksheet through an online grading system. You will have until midnight on the day of the lab (Tuesdays) to submit your solutions. Points will be awarded based on the results of the online grader.

Hacks

There will be weekly Thursday *hack sessions* that will provide an opportunity to start working on exercises in an open, collaborative environment. Each hack session is a simple program or exercise. You may not necessarily complete the entire exercise during the hack session.

Further details are provided in the handouts, but you are highly encouraged to collaborate with any individual and to receive as much help as you desire on the exercises.

You may complete hacks on your own, but you are encouraged to pair up and collaborate with (**at most**) one other student. If you choose to pair up with another student, you must form a pair using canvas (People -> Hack Pairs).

Attendance

To ensure participation your attendance **and** engagement in your weekly lab/hack sessions will be graded. Each session will be worth 10 points. To earn the full 10 points you must:

- Attend **both** the lab and hack session for the entire time. Do not be more than 5 minutes late.
- Put forth a full effort. You must work on the lab/hack for that week or for other material in the course; you may not work on other material or remain idle.
- You may only leave the lab/hack session early if you have already completed both the lab and hack for that module.

Alternative

It is understandable that you may not be able to attend every lab/hack session or that your background or approach to this material means you don't need the structured environment that the lab/hack session provides. As an alternative, *if you can prove yourself capable* of completing labs and hacks on your own you may choose not to attend the lab/hack session.

If you attend as outlined above, you get the full 10 points. If you choose not to attend, you may still earn the 10 points **provided** that you earn at least 75% of the remaining points for each module (Reading/Lab/Hack). If you do not attend and you earn less than 75% of the points for the module, you will get a zero for the attendance portion of the module.

Exams

There will be several exams as well as a comprehensive final exam. These will be open-book, open-note, *required computer* exams. The exams consist of live coding exercises for which you will need your own machine as you will be coding and submitting programs online for grading. More details will be announced closer to the exam dates. No collaboration is allowed on exams.

15th Week Policy Notification

A per UNL's 15th Week Policy (also known as "dead week") available here:

<https://registrar.unl.edu/academic-standards/policies/fifteenth-week-policy/>

we are required to serve written notice that the final assignment as well as the final lab, hack, and assignment will be due during the 15th week.

Scale

Final letter grades will be awarded based on the following standard scale. This scale may be adjusted upwards if the instructor deems it necessary based on the final grades only. No scale may be made for individual assignments or exams.

Letter Grade	Percent
A+	$\geq 97\%$
A	$\geq 93\%$
A-	$\geq 90\%$
B+	$\geq 87\%$
B	$\geq 83\%$
B-	$\geq 80\%$
C+	$\geq 77\%$
C	$\geq 73\%$
C-	$\geq 70\%$
D+	$\geq 67\%$
D	$\geq 63\%$
D-	$\geq 60\%$
F	$< 60\%$

Grading Policy

If you have questions about grading or believe that points were deducted unfairly, you must first address the issue with the individual who graded it to see if it can be resolved. Such questions should be made within a reasonable amount of time after the graded assignment has been returned. No further consideration will be given to any assignment a week after it grades have been posted. It is important to emphasize that the goal of grading is consistency. A grade on any given assignment, even if it is low for the entire class, should not matter that much. Rather, students who do comparable work should receive comparable grades (see the subsection on the scale used for this course).

Late Work Policy

In general, there will be no make-up exams or late work accepted. Exceptions may be made in certain circumstances such as health or emergency, but you must make every effort to get prior permission. Documentation may also be required.

Assignments have a strict due date/time as defined in canvas and the online handin system. All program files must be handed in as specified in individual assignment instructions. Programs that are even a few seconds past the due date/time will be considered late and you will be locked out of handing anything in after that time.

Grader Policy

Failure to adhere to the requirements of an assignment in such a manner that makes it impossible to grade your program means that a disproportionate amount of time would be spent evaluating your assignment. For this reason, we will not grade any assignment that does not compile and run through the online grading system.

Academic Integrity

All homework assignments, programs, and exams must represent your own work unless otherwise stated. No collaboration with fellow students, past or current, is allowed unless otherwise permitted on specific assignments or problems. The Department of Computer Science & Engineering has an Academic Integrity Policy. All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself. The most recent version of the Academic Integrity Policy can be found at <https://computing.unl.edu/academic-integrity-policy>

Artificial Intelligence Tool Usage

Various Artificial Intelligence (AI) applications have been developed in recent years (technically called Large Language Models) that can generate code based on user interactions. The use of these tools (including but not limited to ChatGPT, GitHub's co-pilot, etc.) in *any* capacity for this course is ***strictly prohibited*** and any such use will be considered a violation of academic integrity. Violations may include zeros on assignments or failure of the course.

These tools have legitimate uses, but not for students in this course. You are here to learn the basics of problem solving, programming and computing. These tools simply give you answers without you having to think, learn or understand the solution(s) that they give you. Often these tools will give you answers that violate academic integrity even if you do not explicitly ask them to do so. It would *clearly* be a violation of academic integrity if you asked another *human* to write a program for you and they did so. Asking an AI to do so is no different. You will not build muscle by having someone else lift weights for you. You will not learn anything by having someone else, human or AI, do the learning for you.

That said, these tools *can* provide a meaningful learning experience if they are properly filtered. For that reason, we will be allowing the use of the built-in "Rubber Duck" debugger chat bot in the CS50 IDE. It has been configured so that it won't provide code directly but you can ask it questions to understand your code and/or resolve problems.

Communication & Getting Help

The primary means of communication for this course is an online message forum system designed for college courses. You should have received an invitation to join. If you have not, contact the instructor immediately. On this forum, you can ask questions anonymously, remain anonymous to your classmates, or choose to be identified. Using this open forum system the entire class benefits from the instructor and TA responses. In addition, you and other students can also answer each other's questions (again you may choose to remain anonymous or identify yourself to the instructors or everyone). You may still email the instructor or TAs, but more than likely you will be redirected to this message board for help.

Learning Assistant Program (LAP)

This course is supported by the CSE Learning Assistant Program (LAP). The mission of the LAP is to improve student comprehension and retention in computing fields by focusing on the learner's experience. This course will be supplemented by Learning Assistants (LAs) and Course Leaders (CLs) to help improve your learning. LAs and CLs are other undergraduate students who have taken the same or similar courses and have been trained to help you succeed in this course. Your LAs and CLs will hold regular office hours, help with grading, and assist you with labs/assignments.

I strongly encourage you to utilize the LAs and CLs when you are completing coursework. More information can be found in the Learning Assistant Program Module on Canvas.

Getting Help

Your success in this course is ultimately your responsibility. Your success in this course depends on how well you utilize the opportunities and resources that we provide. There are numerous outlets for learning the material and getting help in this course:

- **Lectures:** attend lectures regularly and when you do use the time appropriately. Do not distract yourself with social media or other time wasters. Actively take notes (electronic or hand written). It is well-documented that good note taking directly leads to understanding and retention of concepts.
- **Lecture Videos:** Lecture videos are intended as a supplement that mirrors lecture material but that may not cover everything. Watch them at your own pace on a regular basis for reiteration or in case you missed something in lecture.
- **Required Reading:** do the required reading on a regular basis. The readings provide additional details and depth that you may not necessarily get directly in lecture.

- Labs & Hack Sessions: use your time during lab and hack sessions wisely. Engage with your lab instructors, teaching assistants, your partner(s) and other students. Be sure to adequately prepare for labs by reading the handouts before coming to lab. Get started and don't get distracted.
- Online message board: if you have questions post to our online message board. It is the best and likely fastest way to get help with your questions. Also, be sure to read other student's posts and questions and feel free to answer yourself!
- Office Hours & Student Resource Center: the instructor and teaching assistants hold regular office hours throughout the week as posted on the course website. Attend office hours if you have questions or want to review material. The Student Resource Center (SRC, <https://computing.unl.edu/current-undergraduate#SRC>) Monday through Friday. Even if your TAs are not scheduled during that time, there are plenty of other TAs and students present that may be able to help. And, you may be able to help others!
- Don't procrastinate. The biggest reason students fail this course is because they do not give themselves enough opportunities to learn the material. Don't wait to the last minute to start your assignments. Many people wait to the last minute and flood the TAs and SRC, making it difficult to get help as the due date approaches. Don't underestimate how much time your assignment(s) will take and don't wait to the week before hand to get started. Ideally, you should be working on the problems as we are covering them.
- Get help in the *right way*: when you go to the instructor or TA for help, you must demonstrate that you have put forth a good faith effort toward understanding the material. Asking questions that clearly indicate you have failed to read the required material, have not been attending lecture, etc. is *not acceptable*. Don't ask generic questions like "I'm lost, I don't know what I'm doing". Instead, explain what you have tried so far. Explain why you think what you have tried doesn't seem to be working. Then the TA will have an easier time to help you identify misconceptions or problems. This is known as "Rubber Duck Debugging" where in if you try to explain a problem to someone (or, lacking a live person, a rubber duck), then you can usually identify the problem yourself. Or, at the very least, get some insight as to what might be wrong.

Computer Science II

CSCE 156 - Syllabus - Spring 2025

School of Computing
College of Engineering
University of Nebraska-Lincoln
University of Nebraska-Omaha

“Smart data structures and dumb code works a lot better than the other way around.”

—Eric S. Raymond, *The Cathedral and the Bazaar*

Course Info

Prerequisites: MATH 106 or equivalent and a grade of C or better in CSCE 155

Description: Data structures, including linked lists, stacks, queues, and trees; algorithms, including searching, sorting, and recursion; programming language topics, including object-oriented programming; pointers, references, and memory management; design and implementation of a multilayer application with SQL database.

Credit Hours: 4

Postrequisites: This course is a prerequisite for CSCE 310 - Data Structures & Algorithms. A grade of C or better is required to move on to CSCE 310.

University Policies, Resources & Services

Students are responsible for knowing the university policies and resources found on this page: <https://go.unl.edu/coursepolicies>

Learning & Skills Objectives

Throughout the duration of this course, you will be presented with a lot of information, some of which will have immediate application and some whose importance is not immediately apparent. As a different measure of your success in this course, by the end, you should have the following skills.

- Be able to more easily pick up and learn a new programming language on your own
- Be able to recognize, use, design and appreciate programs with an object-oriented flavor

- Be able to design a database-backed application from scratch given a general problem statement
- Be able to understand and implement basic data structures as well as recognize how and when to leverage appropriate simple data structures
- Be able to identify and analyze basic algorithms

Overall, CS2 is a continuation of CS1. At the end of a typical CS1 course, you should be able to approach a reasonably complex problem, design a top-down solution, and code a program in a high-level programming language. By the end of this course your approach to problem solving should be modified to ask and answer design questions first:

- What entities/objects would be appropriate to model this problem?
- How would these objects interact to solve the problem?
- What data structures would be the most appropriate or efficient to use?

Tentative Topics

- Introduction to Java / Python (Honors section)
- Object-Oriented Programming, OOP in Java
- Database Design, SQL, Database Connectivity, multi-tier design
- Lists (array-based, linked; circular doubly-linked), Stacks & Queues
- Algorithm Analysis, Recursion, Searching & Sorting
- Trees; Binary Search Trees

Continuation of Instruction

If in-person classes are canceled by the University during a lecture day, lecture may still be livestreamed.

Assessment

Assessment (grading) will be based on labs and assignments. To start you out on the right foot, you have been awarded 20 free starter points. You are currently earning an A in this course; keep it up!

Category	Number	Points
Starter Pts		20
Surveys		30
Labs	14	350
Assignments	8	950
Total		1350

The honor section point distribution is similar but there are more total points due to additional required points on the assignments.

Scale

Final letter grades will be awarded based on the following standard scale. This scale may be adjusted upwards if the instructor deems it necessary based on the final grades only. No scale will be made for individual assignments.

Letter Grade	Percent
A+	$\geq 97\%$
A	$\geq 93\%$
A-	$\geq 90\%$
B+	$\geq 87\%$
B	$\geq 83\%$
B-	$\geq 80\%$
C+	$\geq 77\%$
C	$\geq 73\%$
C-	$\geq 70\%$
D+	$\geq 67\%$
D	$\geq 63\%$
D-	$\geq 60\%$
F	$< 60\%$

Labs & Attendance

There are 14 labs. They are graded based on completion which means *all* of the test cases must compile, run, **and pass**. If any test(s) fail no points will be awarded. Each lab is worth 25 points: 20 for the actual lab and 5 points for attendance at lab. However, if you cannot attend or choose not to attend you can still earn the 5 points back by earning full credit on the lab itself.

Labs give you hands-on exercises for topics recently covered in lecture. The purpose of lab is not only to give you further working experience with lecture topics, but also to provide you with additional information and details not necessarily covered in lecture. Each lab will come with some starter code and exercises that you have to complete. You will submit your code as specified through the handin and we will grade your lab using the webgrader.

Labs are setup as a *peer programming* experience. If you are attending lab *in person*, you may be randomly paired with a partner. One of you will be the *driver* and the other will be the *navigator*. The navigator will be responsible for reading the instructions and guiding the driver. The driver will be in charge of the keyboard and will type the code. Both driver and navigator are responsible for developing and working through solutions *together*. Neither the navigator nor the driver is “in charge.” This is an equal collaboration and all efforts should

be equally shared. Beyond your immediate pairing, you are encouraged to help and interact and with other pairs in the lab. If you are attending lab remotely or online, you may be paired up with another person virtually if you choose or you may complete the lab on your own.

Assignments

The majority of your assessment will be based on assignments whose point values may vary. There are 8 assignments as well as a design document. Assignments will be due 1 second before midnight Friday evening the week that they are due. Code and other relevant *artifacts* must be submitted using handin unless otherwise stated.

In this course, most of the assignments build on prior assignments. Therefore, it is **extremely important** for you to stay on task and not fall behind with assignments. You should start on assignments early and resolve any and all issues (bugs, design flaws) as soon as possible so they do not carry over to the next assignment. To facilitate good time management, we have provided a **Gantt Chart** with a recommended schedule of work.

15th Week Policy Notification

A per UNL's 15th Week Policy (available here: <https://registrar.unl.edu/academic-standards/policies/fifteenth-week-policy/>) we are required to serve written notice that the final assignment as well as the final lab will be due during the 15th week or "dead week."

Grading Policy

If you have questions about grading or believe that points were deducted unfairly, you must first address the issue with the individual who graded it to see if it can be resolved. Such questions should be made within a reasonable amount of time after the graded assignment has been returned. No further consideration will be given to any assignment a week after it grades have been posted. It is important to emphasize that the goal of grading is consistency. A grade on any given assignment, even if it is low for the entire class, should not matter that much. Rather, students who do comparable work should receive comparable grades (see the subsection on the scale used for this course).

Late Work Policy

In general, there will be no make-up or late work accepted. Exceptions may be made in certain circumstances such as health or emergency, but you must make every effort to get prior permission. Documentation may also be required.

Homework assignments have a strict due date/time as defined by the server's system clock. All program files must be handed in using handin as specified in individual assignment handouts. Programs that are even a few seconds past

the due date/time will be considered late and you will be locked out of handing anything in after that time.

Webgrader Policy

Failure to adhere to the requirements of an assignment in such a manner that makes it impossible to grade your program via the webgrader means that a disproportionate amount of time would be spent evaluating your assignment. For this reason, we will not grade any assignment that does not compile and run through the webgrader.

Academic Integrity

All homework assignments, programs, and exams must represent your own work unless otherwise stated. No collaboration with fellow students, past or current, is allowed unless otherwise permitted on specific assignments or problems. The School of Computing has an Academic Integrity Policy. All students enrolled in any School of Computing course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself.

Use of AI The use of any AI tools for the design or implementation of programs is strictly prohibited for this course.

Honors Section

For those in the Honors section of this course topics will be covered at a faster pace so that additional “advanced” topics can also be covered.

In addition, each assignment may contain additional requirements or problems that must be completed. There may also be additional assignments. Bottom line, there will be higher expectations with regard to the quality and quantity of work in this course compared to the regular section. As a student in this course, you will be expected to meet or exceed those expectations which may require more of your time and a more proactive approach to studying and greater diligence in completing assignments.

Communication & Getting Help

The primary means of communication for this course is Piazza, an online forum system designed for college courses. We have established a Piazza group for this course and you should have received an invitation to join. If you have not, contact the instructor immediately. With Piazza you can ask questions anonymously, remain anonymous to your classmates, or choose to be identified. Using this open forum system the entire class benefits from the instructor and TA responses. In addition, you and other students can also answer each other's

questions (again you may choose to remain anonymous or identify yourself to the instructors or everyone). You may still email the instructor or TAs, but more than likely you will be redirected to Piazza for help.

In addition, there is an anonymous suggestion box that you may use to voice your concerns about any problems in the course if you do not wish to be identified.

Learning Assistant Program (LAP)

This course is supported by the CSE Learning Assistant Program (LAP).

The mission of the LAP is to improve student comprehension and retention in computing fields by focusing on the learner's experience. This course will be supplemented by Learning Assistants (LAs) and Course Leaders (CLs) to help improve your learning. LAs and CLs are other undergraduate students who have taken the same or similar courses and have been trained to help you succeed in this course. Your LAs and CLs will hold regular office hours, help with grading, and assist you with labs/assignments.

I strongly encourage you to utilize the LAs and CLs when you are completing coursework. More information can be found in the Learning Assistant Program Module on Canvas.

UNL Writing Center

The Writing Center can provide you with meaningful support as you write for this class as well as other courses. Trained undergraduate and graduate peer consultants are available to talk with you about all forms of communication. You are welcome to bring in everything from lab reports, presentations, and research papers to cover letters, application essays, and graduate theses and dissertations. Writing Center Consultants can work with you at any stage of the writing process, from brainstorming and organizing your ideas through polishing a final draft.

There are three ways you can connect with a Consultant:

- Online (a real-time, video conversation)
- eTutoring (email feedback)
- In person in Andrews 102 To learn more about online options and view video tutorials, visit <https://www.unl.edu/writing/online-writing-center-services>.

Sign up for an appointment any time by visiting <https://unl.mywconline.com>. For other information about the Writing Center, please visit <https://unl.edu/writing>.

Getting Help

Your success in this course is ultimately your responsibility. Your success in this course depends on how well you utilize the opportunities and resources that we

provide. There are numerous outlets for learning the material and getting help in this course:

- Lectures: attend lectures regularly and when you do use the time appropriately. Do not distract yourself with social media or other time wasters. Actively take notes (electronic or hand written). It is well-documented that good note taking directly leads to understanding and retention of concepts.
- Lecture Videos: Lecture videos are intended as a supplement that mirrors lecture material but that may not cover everything. Watch them at your own pace on a regular basis for reiteration or in case you missed something in lecture.
- Required Reading: do the required reading on a regular basis. The readings provide additional details and depth that you may not necessarily get directly in lecture.
- Labs: use your time during labs wisely. Engage with your lab instructors, teaching assistants, your partner(s) and other students. Be sure to adequately prepare for labs by reading the handouts before coming to lab. Get started and don't get distracted.
- Piazza: if you have questions ask them on Piazza. It is the best and likely fastest way to get help with your questions. Also, be sure to read other student's posts and questions and feel free to answer yourself!
- Office Hours & Student Resource Center: the instructor and teaching assistants hold regular office hours throughout the week as posted on the course website. Attend office hours if you have questions or want to review material. The Student Resource Center is open Monday through Friday. Even if your TAs are not scheduled during that time, there are plenty of other TAs and students present that may be able to help. And, you may be able to help others!
- Don't procrastinate. The biggest reason students fail this course is because they do not give themselves enough opportunities to learn the material. Don't wait to the last minute to start your assignments. Many people wait to the last minute and flood the TAs and SRC, making it difficult to get help as the due date approaches. Don't underestimate how much time your assignment(s) will take and don't wait to the week before hand to get started. Ideally, you should be working on the problems as we are covering them.
- Get help in the *right way*: when you go to the instructor or TA for help, you must demonstrate that you have put forth a good faith effort toward understanding the material. Asking questions that clearly indicate you have failed to read the required material, have not been attending lecture, etc. is *not acceptable*. Don't ask generic questions like "I'm lost, I don't know what I'm doing". Instead, explain what you have tried so far. Explain

why you think what you have tried doesn't seem to be working. Then the TA will have an easier time to help you identify misconceptions or problems. This is known as "Rubber Duck Debugging" where in if you try to explain a problem to someone (or, lacking a live person, a rubber duck), then you can usually identify the problem yourself. Or, at the very least, get some insight as to what might be wrong.

CSCE 231: Computer Systems Engineering

Fall 2020

Schedule

All times are U.S. Central Time (UTC-5 until November 1; UTC-6 after November 1)

Lecture

- Section 150: Monday, Wednesday, Friday via prerecorded videos

Recitation

- Section 153: Monday 3:30pm-4:20pm in Zoom
- Section 151: Tuesday 3:30pm-4:20pm in Zoom
- Section 152: Wednesday 8:30am-9:20am in Zoom

Laboratory

- Section 165: Monday 4:30pm-6:20pm in Avery 021 & Zoom
- Section 161: Tuesday 4:30pm-6:20pm in Avery 021 & Zoom
- Section 162: Tuesday 6:30pm-8:20pm in Avery 021 & Zoom
- Section 163: Wednesday 9:30am-11:20am in Avery 021 & Zoom
- Section 164: Wednesday 1:30pm-3:20pm in Avery 021 & Zoom

The Zoom link for recitation & laboratory will be available on Canvas.

Instructor

Dr. Christopher Bohn bohn@unl.edu

Office Hours: TR 9:30-11:00am Avery 262

Zoom office hours link: https://go.unl.edu/bohn_office_hours

TAs

Name	Office Hours	email
Aaron Johnson	M,W 11:30am-12:30pm	aaronjohnson14325@gmail.com
Brennan Rhoadarmer	M,F 9:30-11:00am	rhoadarmerb@gmail.com
Katie Gerot	M,T 9:00-10:00am	katie.gerot@huskers.unl.edu
Christine Discenza	R 12:30-2:30pm	cdiscenza3@gmail.com
Thomas Hillebrandt	M,F 1:30-2:30pm T,R 10:00-11:00am	thomas.hillebrandt@huskers.unl.edu
Adrian Pilkington	M 7:00-9:00pm	apilkington@huskers.unl.edu

TA office hours are in the CSE Virtual Student Resource Center (vSRC).

Catalog Description

Computer Systems Engineering (4 cr) Lec 3, rct 1, lab 2. Introduction to organization, structure, and applications of computer systems. Boolean Logic, Digital Arithmetic, Processor Organization, C Programming, Machine Language Programming, Input/Output, Memory Organization and Management, Building Embedded System Application.

Prerequisites

Grade of “P” or “C” or better in CSCE 235, CSCE 235H, or RAIK 184H.

Prerequisites by Topic

- *Mastery of* mathematical problem solving skills, mathematical maturity and competence to the level of introductory calculus
- *Familiarity with* programming, logarithms, integer and floating point numbers
- *Exposure to* recursion, operating systems services

Course Objectives

- *Mastery of* Boolean algebra, logic equations, binary numbers (including negatives), fixed-point binary arithmetic, hexadecimal notation, exponential numbers
- *Mastery of* C programming, accessing I/O devices, processor organization, stacks, subroutines, instruction set architecture (with hands- on experience in embedded system design and implementation)
- *Familiarity with* logic gates and diagrams, floating point arithmetic, memory devices and hierarchies, micro-operations & micro- architectures
- *Familiarity with* assembler language principles, flow of control, assemblers, linkers, loaders, the syntax & semantics of short assembler programs
- *Familiarity with* memory errors such as buffer overflows and segmentation faults, security vulnerabilities

Course Goals

How does a modern digital computer interpret programs written in a high-level programming language into the streams of electronic signals that actually do the thinking, the arithmetic, the keeping track of data, the sequencing of actions, and the input/output? What is the hardware-software interface and how can we use it to build modern computer systems and develop software? What system design considerations go into defining and designing it? In this course you will gain an understanding of the interface by first writing programs in a low-level assembly language that is close to the machine language,

developing the logic design of basic processor components, and then designing and implementing an embedded system. In the process you will gain practical experience with computer-aided design tools and C programming environments. The course comprises of lectures, recitation, and labs. The recitation sessions are used to prepare students to tackle the implementation and experimentation aspects of concepts discussed during lectures. The labs provide time for students to implement and experiment with these topics.

Topics Covered

1. *System Level Organization*: CPU, memory systems (main memory, cache, virtual memory), storage technologies, I/O devices & processes
2. *Arithmetic and Logic Level*
 - (a) Boolean algebra, truth tables, logic equations, logic gates, combinatorial logic, sequential logic & memory devices, clocks, waveform diagrams
 - (b) Signed & unsigned binary numbers, floating-point numbers, BCD, hexadecimal, ASCII
 - (c) Signed & unsigned fixed-point arithmetic, floating-point arithmetic, ALU structures
3. *Micro-Architecture Level*: data paths & components, micro- operations, memory interfacing, the fetch/execute cycle
4. *Instruction Set Architecture Level*: instruction types & formats, opcodes, operands, immediate values, addressing modes, flow of control, branching & procedure calls
5. *Assembler Language level*: syntax, directives vs. instructions, assemblers, linkers, loaders, semantics of simple programs, stack management, procedure calls
6. *High-Level Programming Level*: C programming syntax, pointer arithmetic, I/O access, polling & interrupt handling, dynamic memory management, memory errors, security vulnerabilities

Lecture, Recitation, and Laboratory

While the lecture will occasionally address practical matters, the principal purpose of the lectures is to provide you with a theoretical understanding of computer systems engineering. Through the individual labs, the group lab, and the group project, you will learn to apply this theory to practice. The recitation will help you transition from the theory of the lectures to the practice of the labs.

It is a common misunderstanding that the purpose of this course is to program in C. That is not the case. A course such as this requires working with computer systems much closer to the hardware level than most software developers typically do. Few high-level languages afford access to the underlying computer system, and of those that do, C is by far the most commonly- used.

Grading

Your grade will be calculated based on your performance in the following areas:

- *Homework* (20%) Homework will help cement the concepts from the lecture and the textbook. Doing so will pay off in the other components of the course grade. Because group study is an effective learning technique, I encourage you to work with your peers to discuss the homework, but **be sure to write your own solutions and indicate whom you worked with.**
- *Labs* (30%) The labs will provide hands-on practice to provide better understanding of what you've learned in the lectures. Some labs will be programmed in C; others entail assembly programming.
- *Midterm Exams* (30%) We will have two midterm exams, that you will take during lab time. They are scheduled for **September 21-23** and **October 26-28**; if there are changes, I'll announce them at least a week in advance. *Unless I explicitly specify otherwise, the exams will be closed book/closed note/closed neighbor/closed internet; and programmable calculators, laptops, tablets, and smartphones will be disallowed.* You may have one $8\frac{1}{2}'' \times 11''$ sheet of notes, double-sided, printed or hand-written. Alternate exams will be available for students who cannot attend lab in-person.
- *Final Exam* (20%) This will be a comprehensive exam, scheduled for **Sunday, November 25 at 3:30-5:30pm**. As with the midterm exams, unless I explicitly specify otherwise, the exams will be closed book/closed note/ closed neighbor/ closed internet.
- Final grades will be assigned based on the traditional cutoff percentages.

Grade	A+	A	A-	B+	B	B-	C+	C	C-	D	F
Score	97	93	90	87	83	80	77	73	67	60	<60

- *Rounding of final grades* I am not in the habit of *giving* grades, but I am sympathetic to students who put in effort. In lieu of rounding, after calculating each student's score and adding extra credit points, I will make one final adjustment to scores near a letter-grade cutoff:

$$adjustment = \begin{cases} 0.50 & \text{if all assignments completed} \\ 0.015 \times \text{number of assignments completed} & \text{otherwise} \end{cases}$$

Textbooks

- *Programming at the Hardware-Software Interface*, 1st ed., Christopher A. Bohn
- Other useful references available in the "Pages" section of the course's Canvas site, including *The C Programming Language*, 2nd ed., Brian W. Kernigan & Dennis M. Ritchie ("K&R")

Notional Sequence

Topics		Chapter
Week 1: August 17-21		
lecture	Intro, Prerequisite review, Digital data	1, 2
lab	Lab 0 (Simple C refresher, environment setup)	
Week 2: August 24-28		
lecture	Digital data, Integer Arithmetic	2, 3
lab	Lab 1 (Advanced C refresher)	
Week 3: August 31 - September 4		
lecture	Integer, Floating Point Arithmetic	3, 4
lab	Lab 2 (Integer Lab)	
Week 4: September 7-11		
lecture	Floating Point Arithmetic	4
lab	Lab 3 (Float Lab)	
Week 5: September 14-18		
lecture	Simple Assembly Language	5
lab	Lab 3, continued	
Week 6: September 21-15		
lecture	Simple & Structured Assembly Language	5, 6
lab	Exam 1 (weeks 1-5)	
Week 7: September 28 - October 2		
lecture	Structured Assembly Language	6
lab	Lab 4 (“Bomb” Lab)	
Week 8: October 5-9		
lecture	Assembly Language, Buffer Overflow	6, first half of ch11
lab	Lab 4, continued	
Week 9: October 12-16		
lecture	Concurrency, Processor Design	7, first half of ch8
lab	Lab 5 (“Attack” Lab)	
Week 10: October 19-23		
lecture	Simple Processor Design	first half of ch8
lab	Lab 5, continued	
Week 11: October 26-30		
lecture	Exceptional Control Flow, System I/O	9
lab	Exam 2 (weeks 6-10)	
Week 12: November 2-6		
lecture	Advanced Processor Design	second half of ch8
lab	Lab 6 (TBD)	
Week 13: November 9-13		
lecture	Advanced Processor Design, Memory	10, second half of ch8
lab	Lab 6, continued	
Weeks 14-15: November 16-20		
lecture	Memory, Tying it all together	10, 12, second half of ch11
lab	no lab currently scheduled	
Final Exam: Sunday, November 25, 3:30-5:30pm		

Policies and Resources

- *Attendance:* Lecture this semester will be prerecorded and will be available through VidGrid; you may view them at any time. I plan to release a week's worth of lectures prior to the start of that week. We will cover some material in class that is not covered in the text; your assignments and test questions may be based on that material. In particular, many of the details necessary to complete assignments will be presented in class. In the interest of maximizing knowledge retention, do not binge-watch the recordings. It is to your benefit to watch the recordings during their nominal week.
- *Class Preparation and Participation:* I expect you to be prepared for class, participate in discussions, and ask and answer questions.
- *Assignments:* All work must be completed when due, and all quizzes and exams must be taken when given. **No make-up quizzes or exams will be given, and with rare exception no late homework or labs will be accepted.** If you have a valid excuse (e.g., illness or injury of yourself or your dependent, an academic conference, a job interview, military obligation, jury duty, bereavement, religious observances, participation in university-sponsored intercollegiate athletic events, etc.), the score for a missed quiz or exam score will be replaced by the average of the remaining quizzes/exams. Similarly, if you have a valid excuse then at my option you will either be given an extension on a homework or lab assignment or will have the missed score replaced by the average of the remaining assignments in the same category.
 - If you wish to challenge the grade you received on an assignment, you must demonstrate that the answer you provided is correct; *i.e.*, that your grade is erroneous. We will not adjust your grade if your answer is too illegible for us to read. You will have two class-days in which to do so. We will not adjust grades for an assignment after two days past the day on which it was returned to you.
 - I will, at a minimum, honor the guidance for excused absences in the UNL Faculty Senate's Class Attendance Policy. See <https://www.unl.edu/facultysenate/classattendancepolicy%20%20April%203%2C%202018.pdf>
 - * Notify me of planned absences in advance, and preferably early. I will not excuse missed assignments due to planned absences if you notify me after-the-fact.
 - * Notify me of unplanned absences when you are able. In general, I will not require proof of the excuse; however, I will require proof of the excuse for missed exams, and I may require proof of the excuse if absences are extended or common.
 - * If missed assignments due to excused absences become frequent enough that I cannot gauge how well a student has met the course objectives, then at my discretion I may award an "Incomplete" for the course and require the student to complete the assignments before receiving a letter grade.
 - We will provide information regarding the exams at a future date.
- *Services for Students with Disabilities:* The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on

your disability (including mental health, chronic or temporary medical conditions), please let me know immediately so that we can discuss options privately. To establish reasonable accommodations, I may request that you register with Services for Students with Disabilities (SSD). If you are eligible for services and register with their office, make arrangements with me as soon as possible to discuss your accommodations so they can be implemented in a timely manner. SSD contact information: 232 Canfield Admin. Bldg.; 402- 472-3787.

- *Counseling and Psychological Services* UNL offers a variety of options to students to aid them in dealing with stress and adversity. Counseling and Psychological Services (CAPS, <https://caps.unl.edu/>) is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological and emotional well-being when issues arise. CAPS can be reached by calling 402-472-7450. Big Red Resilience & Well- Being (<https://resilience.unl.edu/big-red-resilience-well-being>) provides fun events, innovative education, and dynamic services to help students understand emotions, manage stress, build strength, connect with others, develop grit and navigate transitions.
- *Diversity and Inclusion* The University of Nebraska-Lincoln does not discriminate on the basis of race, ethnicity, color, national origin, sex (including pregnancy), religion, age, disability, sexual orientation, gender identity, genetic information, veteran status, marital status, and/ or political affiliation.
- *Student Resource Center:* The CSE Department has established a Virtual Student Resource Center which you can access through Canvas. See <https://canvas.unl.edu/courses/95368>.
- *Check Your Email* The CSE Department's policy is that all students in CSE courses should regularly check their email so they don't miss important announcements.
 - We will make announcements through Piazza; make sure that Piazza is configured to forward these announcements to your preferred the email address.
 - When email is appropriate, we will contact you through the email in Canvas; make sure that Canvas is has your preferred the email address.
 - If you aren't already doing so, I strongly encourage you to check your huskers.unl.edu email daily. The university has indicated that any announcements it makes will be to your huskers.unl.edu email.
- *Departmental Contact Form:* The CSE Department has an Anonymous Online Contact Form that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified. See <http://cse.unl.edu/contact-form>.
- *Academic Integrity Policy:* Academic honesty is essential to the existence and integrity of an academic institution. The responsibility for maintaining that integrity is shared by all members of the academic community. The University's Student Code of Conduct addresses academic dishonesty. Students who commit acts of academic dishonesty are subject to disciplinary action and are granted due process and the

right to appeal any decision. See <https://stuaafs.unl.edu/DeanofStudents/Student%20Code%20of%20Conduct%20May%20Rev%202014%20a.pdf>.

All homework assignments, exams, etc. must be your own work. The Computer Science and Engineering department has an Academic Integrity Policy. All students enrolled in any CSE course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself. See http://cse.unl.edu/ugrad/resources/academic_integrity.php.

The Department requires me to report every offense to the Chair for further consideration. The key to avoiding cheating is to be totally open and transparent about any and all collaborations, noting that appropriate teamwork and collaboration will be highly encouraged. Here is some elaboration on the examples listed on the department's academic integrity webpage:

- I encourage discussions of *what* and *why*, but discussions of specific solutions or implementations are prohibited.
- Being in possession of a worked or partially-worked solution to an assignment (whether from a fellow student, from the internet, or from another source) before you have completed the assignment is unauthorized collaboration on the assignment.
- If another student has a copy of your worked solution to an assignment before s/he has completed the assignment, I will assume that you facilitated their cheating unless it can be demonstrated that they obtained the copy despite your reasonable precautions to prevent them from doing so. Students who share a computer should protect their files either by using separate accounts or by placing their coursework in password-protected folders.
- If we detect academic misconduct on group assignments, I will hold the full group responsible unless there is compelling evidence that only a proper subset of the group committed the misconduct and that the remainder of the group was unaware of the misconduct.

We will use software, including but not limited to MOSS, to help us detect academic integrity violations; however, we will also apply human judgement. We will retain your assignment submissions to be compared with future students' submissions.

- *Student Concerns and Feedback:* Your experience with remote learning in this course is important to me. If you have questions, concerns, or positive feedback, please contact me at bohn@unl.edu or 472-1803. If I am unable to respond, or you feel I've not adequately addressed your concerns, you can contact Dr. Marilyn Wolfe at mwolf@unl.edu or 472-2401. If your concern is still not resolved, please contact Dean Lance Pérez at lcperetz@unl.edu or 472-5259.
- *Required Use of Face Coverings for On-Campus Shared Learning Environments:* As of July 17, 2020 and until further notice, all University of Nebraska–Lincoln (UNL) faculty, staff, students, and visitors (including contractors, service providers, and others) are required to use a facial covering at all times when indoors except under specific conditions outlined in the COVID 19 face covering policy found at:

<https://covid19.unl.edu/face-covering-policy>. This statement is meant to clarify classroom policies for face coverings:

To protect the health and well-being of the University and wider community, UNL has implemented a policy requiring all people, including students, faculty, and staff, to wear a face covering that covers the mouth and nose while on campus. The classroom is a community, and as a community, we seek to maintain the health and safety of all members by wearing face coverings when in the classroom. Failure to comply with this policy is interpreted as a disruption of the classroom and may be a violation of UNL's Student Code of Conduct.

Individuals who have health or medical reasons for not wearing face coverings should work with the Office of Services for Students with Disabilities (for students) or the Office of Faculty/Staff Disability Services (for faculty and staff) to establish accommodations to address the health concern. Students who prefer not to wear a face covering should work with their advisor to arrange a fully online course schedule that does not require their presence on campus.

Students in the classroom:

1. If a student is not properly wearing a face covering, the instructor will remind the student of the policy and ask them to comply with it.
2. If the student will not comply with the face covering policy, the instructor will ask the student to leave the classroom, and the student may only return when they are properly wearing a face covering.
3. If the student refuses to properly wear a face covering or leave the classroom, the instructor will dismiss the class and will report the student to Student Conduct & Community Standards for misconduct, where the student will be subject to disciplinary action.

Instructors in the classroom:

1. If an instructor is not properly wearing a face covering, students will remind the instructor of the policy and ask them to comply with it.
2. If an instructor will not properly wear a face covering, students may leave the classroom and should report the misconduct to the department chair or via the TIPS system for disciplinary action through faculty governance processes.

Last updated: 09:14-2020/08/27

Introduction to Discrete Structures

“Computer Science is no more about computers than astronomy is about telescopes.”

– Edsger Dijkstra

Course Information

Prerequisites	A grade of 'P' or 'C' or better in CSCE 155 (A, E, H, N, T) / SOFT 160 (H) / RAIK 183H and MATH 106.
Lectures	Tuesday and Thursday (12:30:PM to 1:45:PM) Brace Lab-206
Instructor	Dr. Shruti Bolman <i>née</i> Daggumati sdaggumati@unl.edu Avery Hall 369
Office Hours	Please check “Syllabus” on Canvas for updates and details
Textbook (suggested)	<i>Discrete Mathematics and Its Applications</i> Kenneth H. Rosen, McGraw Hill (7 th Edition)
TAs	Please check “Syllabus” on Canvas for updates and details
Homepage	We will use Canvas (http://canvas.unl.edu) to: • send out announcements • post lecture notes • assign homework

Course Details

Computer Science is *not programming*. Rather, Computer Science is the mathematical modeling and study of what computation is – that is, what problems have a computational solution and how efficient that solution can be. Thus, a strong foundation in mathematics is essential to your success as a computer scientist. At the heart of computer science are fundamental discrete structures that we will study in this course. Specifically, you will learn many of the mathematical definitions, techniques, and ways of thinking that will be useful in Computer Science.

Catalog Description

Survey of elementary discrete mathematics. Elementary graph and tree theories, set theory, relations and functions, propositional and predicate logic, methods of proof, induction, recurrence relations, principles of counting, elementary combinatorics, and asymptotic notations.

Tentative Schedule of Topics

Below is a list of topics I intend to cover along with the relevant sections of the text. This schedule is tentative and may be changed or topics added/removed as time dictates. Furthermore, though we will follow the book, additional material may be introduced (with sufficient resources) while the depth of each topic may vary. A complete schedule of topics as well as the associated required reading is available on the course webpage. The topic outline is roughly as follows.

- Propositional Logic, Predicate Logic
- Proofs
- Sets
- Functions
- Relations
- Algorithms & Algorithm Analysis
- Induction
- Recurrence Relations
- Combinatorics
- Graphs & Trees

Teaching Style

I teach using power point slides, white board and class discussions. Handouts of the slides are available, but you are ultimately responsible for the material, thus regular attendance is strongly encouraged. Furthermore, you will be expected to read the relevant sections of the textbook before coming to class. Several pop quizzes (though not necessarily announced) will also be given in the lecture and recitation class. Since make-up quizzes will not be given, attendance is highly recommended.

The text offers good examples of the materials and numerous opportunities to work out examples. The answers to all odd numbered questions are in the back of the book and provide a good opportunity to work problems on your own. Homework and exams are used to assess your progress in this course and should be considered the *minimal* amount of work required to learn the material. You should work additional examples and exercises from the book in order to master the material.

Discussion Through Piazza

We will be using Piazza for class discussion. The system is highly catered to provide you help fast and efficiently from classmates, TAs, and myself. Rather than emailing questions to the teaching staff, I encourage you to post your questions on Piazza. Please note that posted questions or discussion will be moderated if appeared to be inappropriate or irrelevant. Also, if you have a **question for a specific instructor (me or a TA)**, you must send an email to that person, otherwise don't expect to receive a quick response. Sign up for the class page at: <https://piazza.com/unl/spring2020/csce235>

Grading

Grading will be based on homework, quizzes, and exams with the following contributions.

Category	Weight
Homework	45%
Quizzes	20%
Midterm Exam	15%
Final Exam	20%

Scale

Letter grades will be awarded based on the following scale. I will use an **absolute grading policy**. For example, scores within the range between 89.00 and 89.49 will be rounded to 89.00 while scores within the range between 89.50 and 89.99 will be rounded to 90.00. Therefore, a score of 89.49 will earn you a B+ grade but 89.50 will earn an A-(minus) grade.

A+	>= 97	B+	>= 87	C+	>= 77	D+	>= 67	F	<60
A	>= 93	B	>= 83	C	>= 73	D	>= 63		
A-	>= 90	B-	>= 80	C-	>= 70	D-	>= 60		

Homework

There will be 6 homework assignments, about one every other week. Homework will consist of selected exercises from the text as well as original problems. You will be expected to follow all instructions on the homework assignments. Clarity and legibility are of great importance. If homework is sloppy or unclear, points may be deducted. Each assignment will have a fixed deadline (beginning of class). Homework can be completed in pairs, if students choose to do so. The responsibility to learn and understand the material falls on the student.

Quizzes

There will be several pop quizzes (i.e., they may or may not be announced in advance) given during the lecture/recitation. They will generally be short and will cover recent topics.

Exams

There will be one midterm exam (in class) and one final exam. These will be **closed notes/book** exams and comprehensive.

Grading Policy

If you have questions about grading or believe that points were deducted unfairly, you must first address the one who graded it to see if it can be resolved. Such questions should be made within a reasonable amount of time after the graded assignment has been returned. **No further consideration will be given to any assignment a week after it has been graded and returned (regardless of whether you fail to pick it up when handed back).** It is your responsibility to **check Canvas regularly** and report missing/incorrect entry **within one week of submission** of your assignment or exam. You need to report questions related to grading **before I post your final grade** in Canvas.

Late Work Policy

Homework assignments have a strict *in-class* (at the beginning) due date.

It is understandable that unforeseen events may interfere with your ability to hand in all homework assignments on time. As such, I've adopted a policy of allowing **one late homework**. You may turn in a single assignment **up to** a week late. Any submissions after a week will not be considered. Any late submissions after using your one "free pass" will not be considered.

Exceptions: You are NOT ALLOWED to use a late pass on either the first or the last homework assignments.

In general, there will be no make-up exams or quizzes. Exceptions may be made in certain circumstances such as health or emergency, but you must make every effort to get prior permission. Documentation may also be required.

University and Departmental Policies

Dead Week Policy

In conformance with UNL's 15th Week Policy (see Registration and Records main webpage, <http://www.unl.edu/regrec/>), be aware that the final homework may be due during the final week of classes. Finally, all assignments, homework, labs or otherwise, will have a strict final due date during the final week of classes. This supersedes any unused late or screw-up passes that you may have (that is, such passes cannot be used to extend the due date of any assignment past the last week of classes). Also, please be aware that there may be quizzes during the final week of semester.

Academic Integrity

All homework assignments, quizzes, exams, etc. must be your own work. No direct collaboration with fellow students, past or current, is allowed unless otherwise stated. The Computer Science & Engineering department has an [Academic Integrity Policy](#). All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself. **If any student performs academic dishonesty (complete or partial)**, as defined by the UNL academic integrity policy, following actions will be taken.

- 1) For a first offense where the student actions were isolated and/or they are remorseful, they will receive no credit for the assignment in question.
- 2) For a first offense that is considered egregious, e.g., their actions impacted many others, they persist in making untruthful statements in light of overwhelming evidence, etc., they will receive a failing grade for the class and will not be permitted to drop the class or change to pass/no-pass grading.
- 3) For a second offense, they will receive a failing grade for the class and will not be permitted to drop the class.

- 4) For a third offense, they will be expelled from their CSCE major or minor and will not be permitted to subsequently enroll in any CSCE course.

It is acceptable to discuss the meaning of assignments with fellow students. Discussing general approaches and strategies for solutions may be permissible, but unless specifically allowed, such communications should not include written material or code and should not transmit substantive or specific elements of a solution. Any cooperation beyond discussing general approaches and general strategies is not allowed unless specifically permitted by the instructor. You may not develop joint solutions, share work, or copy anything. You are also responsible for safeguarding your own work. All external contributions must be acknowledged, including help from others or from non-course materials such as websites. If in doubt, ask.

Accommodations for Students with Disabilities

Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodations to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 232 Canfield Administration, 472-3787 voice or TTY.

Sources for Help and Assistance

UNL offers a variety of options to students to aid them in dealing with stress and adversity. Counseling and Psychological Services (CAPS) (<https://caps.unl.edu>) is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological, and emotional well-being when issues arise. CAPS can be reached by calling 402-472-7450.

Big Red Resilience & Well-Being (<https://resilience.unl.edu/big-red-resilience-well-being>) provides fun events, innovative education, and dynamic services to help students understand emotions, manage stress, build strength, connect with others, develop grit, and navigate transitions

CSE Department

The CSE Department has an anonymous contact form (<http://cse.unl.edu/contact-form>) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.

It is CSE Department policy that all students in CSE courses are expected to regularly check their email so they do not miss important announcements. You are ultimately responsible for your success in this course.

If you have questions on material covered or assigned in class, it is up to you to seek out assistance from the course instructors or TAs. Staff in the CSE Student Resource Center may also be able to assist you with general questions. The CSE Student Resource Center (<http://cse.unl.edu/src>) is located at Avery 12. The CSE Department also maintains a Frequently Asked Questions page (<https://cse.unl.edu/faq>).

Finally, I will hold regular office hours and will make myself available by appointment; please email me to set one up. **However, note that you may not receive any response during weekends/holidays and after office hours.**

Help

Your success in this course is ultimately your responsibility. That said, there are several outlets for you to seek help and assistance.

1. Your Instructor – Attend lecture regularly and engage in class discussions, ask questions in class, visit me during my office hours or setup a meeting time to see me
2. Your TAs – TAs hold regular weekly office hours, visit with them and ask questions, ask for examples, etc.
3. Student Resource Center – Though they may not be your direct TA, all Graduate Teaching Assistants (and some advanced undergraduates) hold regular office hours in the Student Resource Center (<http://cse.unl.edu/src/index.shtml>) open Monday thru Friday 9AM to 7PM and should be staffed most hours. Ask for help from anyone in the SRC.
4. Course materials – start on assignments early, attend lectures and labs, work extra problems from the book, read all required (and optional!) materials.
5. Your colleagues – Chances are, if you are having problems, your classmates are having them too. Discussion and dialog among students is encouraged (within the parameters set by CSE's academic integrity policy, this course's policy, and policies set for individual assignments).

CSCE 251, Sec 700: UNIX Programming

University of Nebraska – Lincoln

Spring 2020

Instructor: Garrett Wirka

E-mail: Through Canvas

Course Location: Online

Website: <https://canvas.unl.edu/>

Course Description

This online course introduces the basics of the Unix operating system, including commands and script programming. The goal of this course is for you to become familiar with Unix, to learn the art of scripting, and to support other courses in programming languages, data structures, and operating systems.

The topics covered in this course include: Unix history, accessing Unix systems, different programs/utilities found in Unix, the Unix file system, file processing, ownership and permissions, editing files with vi and emacs, filters and pipes, wildcards and regular expressions, command shells, AWK programming, Unix processes, and shell scripting.

Required Materials

There are no required materials for this course.

Online Course

This course is an online-only course. There will not be any regular meeting times. All materials will be posted to Canvas. You are responsible for checking Canvas regularly to keep up with any information and content that is posted. If you have any problems or questions, please feel free to contact the instructor via Canvas.

Course Objectives

After you complete this course, you will have:

1. Developed a good understanding of the Unix operating system

2. Gained an understanding of and experience with basic Unix commands and the bash programming environment
3. Used regular expressions and gained an understanding of AWK programming
4. Obtained a basic understanding of Unix processes
5. Perhaps most importantly, taken your first steps towards learning effective shell scripting

Course Structure

Homework Assignments

There will be three required assignments to complete during this course. These will be used to help you gain hands-on experience with Unix, Unix programming, and shell scripting. In total, assignments are worth 55% of your final grade, though they may not necessarily be weighted equally.

Assignments may be composed of written questions, scripting exercises, or some combination of the two. The solutions to written work must be submitted as a text file, and scripting exercise solutions as shell scripts. More detailed instructions will be given with the individual assignments. Both text files and scripts must be submitted via handin (cse.unl.edu/handin). Scripts will be graded using a webgrader (cse.unl.edu/~cse251/grade), which you will have access to in order to verify your solutions. It is strongly recommended that you take advantage of this.

Lastly, constructive collaboration on homework assignments is encouraged. All students are expected to do and submit their own work on homework assignments, though you are allowed to ask each other questions and bounce ideas off of one another, especially for the purpose of debugging. **If you collaborate with another student, prominently indicate their name(s) in your solution (commented out, in the case of scripts).** Failure to do so will leave you vulnerable to accusations of violating the department's academic integrity policy, discussed below.

Quizzes

There will be 3 quizzes for this course. Each quiz is worth 10% of your final grade. These quizzes will all be multiple choice and taken via Canvas. Each quiz will have questions randomly selected (and in random order) from a large question bank, so that no two quizzes will be the same. The quizzes will be open everything (books, notes, etc.). During the quizzes, any communication regarding the quiz contents with other students is strictly prohibited. You must complete the quizzes by yourself. The three quizzes cover material from lectures 1–9, 10–14, and 15–22 respectively.

Final Exam

The final exam for this course will be open all semester as a quiz on Canvas, but must be completed by 11:59:00 pm on **Wednesday, May 6th**. The final exam will essentially be a big quiz (worth 15% of your final grade) with the questions randomly drawn from the question banks of

the previous 3 quizzes. Some of the questions may be ones you've had before, while others may be new. There will be 60 questions total, and you will have 2 hours to complete it.

Grading Policy

Your final grade for this course will be determined according to your grades on quizzes, homework assignments, and the final. Each quiz will be worth 10% of your grade, and the final will be worth 15%. The homework assignments will make up the remaining 55% of the points, but they may be weighted differently.

Computing letter grades starts with this base scale, where s is the final score:

$$\begin{aligned}s &\geq 90\% \Rightarrow A \\ 80\% &\leq s < 90\% \Rightarrow B \\ 70\% &\leq s < 80\% \Rightarrow C \\ 60\% &\leq s < 70\% \Rightarrow D \\ s &< 60\% \Rightarrow F\end{aligned}$$

You will receive a "+" with your grade if the last digit of your score is ≥ 7 , and a "-" if the last digit is < 3 . I will scale up from this base scale if warranted. So if you get an 87% in this course you are guaranteed at least a B+ (similarly, an 82% guarantees at least a B-).

Course Policies

Late Assignments / Quizzes

All due dates in this course are at exactly 11:59:00 pm on the specified date. CSE's system clock, which timestamps your submissions, is the official clock for this course. Late submissions for quizzes and the final will not be accepted. Late submissions for assignments are allowed within **one week** of the deadline, for at most half credit.

Late Pass

Life happens, and sometimes deadlines can't be met. To accommodate life, each student may use a late pass for one **homework** during this course, which will allow for one assignment to be submitted up to one week late with no penalty. This pass will be automatically applied if you submit an assignment within a week after the deadline has passed, provided you have not yet used your late pass. This late pass **may not** be used on any of the quizzes or the final.

Academic Integrity

All homework assignments, scripts, and quizzes must be your own work. No collaboration with fellow students, past or current, is allowed unless otherwise permitted. The Computer Science & Engineering department has an Academic Integrity Policy. All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself. The most recent version of the Academic Integrity Policy can be found at <http://cse.unl.edu/academic-integrity-policy>.

It is worth noting that all scripts submitted in this course will be compared with those of other students.

Accommodations for Disabilities

Students with disabilities are encouraged to contact us for a confidential discussion of their individual needs for academic accommodation. This includes students with mental health disabilities like depression and anxiety. It is the policy of the University of Nebraska-Lincoln to provide individualized accommodations to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 117 Louise Pound Hall, (402) 472-3787.

CSE Anonymous Contact Form

The CSE Department has an anonymous contact form that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified. The form can be found at <http://cse.unl.edu/contact-form>.

Additional Information

It is CSE Department policy that all students in CSE courses are expected to regularly check their email so they do not miss important announcements.

Computer Science III

Data Structures & Algorithms Syllabus

Department of Computer Science & Engineering
University of Nebraska–Lincoln

CSCE 310 – Summer 2021

“Computer Science is no more about computers than astronomy is about telescopes.”
—Edsger Dijkstra

“If you want to be a good programmer you just program every day for two years. If you want to be a world class programmer you can program every day for ten years, or you could program every day for two years and take an algorithms class.”
—Charles E. Leiserson

1 Course Info

Prerequisites: CSCE 156 (Computer Science II) and CSCE 235 (Discrete Math)

Description: A review of algorithm analysis, asymptotic notation, and solving recurrence relations. Advanced data structures and their associated algorithms, heaps, priority queues, hash tables, trees, binary search trees, and graphs. Algorithmic techniques, divide and conquer, transform and conquer space-time trade-offs, greedy algorithms, dynamic programming, randomization, and distributed algorithms. Introduction to computability and NP-completeness.

Credit Hours: 3

Textbook: The *recommended* text book for this course is *Introduction to the Design and Analysis of Algorithms* (any edition) by Anany Levitin. However, no text book is required as there are plenty of free online Data Structures and Algorithms resources:

- My lecture notes: cse.unl.edu/~cbourke/ComputerScienceThree.pdf
- Open DSA: <https://opendatastructures.org/>
- *Algorithms* by Jeff Erickson <http://jeffe.cs.illinois.edu/teaching/algorithms/>

Postrequisites: If you are a Computer Science or Computer Engineering major you will need to receive a C or better in this course to continue in the major.

2 Course Overview

Computer Science is not programming. Rather, Computer Science is the mathematical modeling and study of what computation is—what problems have a computational solution and how efficient that solution can be. Thus, a strong foundation in mathematics is essential to your success as a computer scientist. At the heart of computer science are fundamental, discrete structures which we will study in this course. Specifically, you will learn many of the mathematical definitions, techniques, and ways of thinking that will be useful in Computer Science.

2.1 Topics

- A review of algorithms, algorithm analysis and asymptotics
- Brute Force algorithms, backtracking, generating combinatorial objects
- Divide & conquer techniques, repeated squaring, Karatsuba multiplication, Strassen's matrix multiplication, etc.
- Algorithms for linear systems
- Greedy Algorithms: Huffman coding
- Balanced Trees: Heaps, AVL Trees, 2-3 Trees
- Hash-based data structures
- Graph algorithms: DFS, BFS, MSTs, path finding, shortest path
- Dynamic Programming
- Computation and computability

3 Schedule

See Canvas

4 Course Delivery

For summer sessions this course is delivered online only in a (more-or-less) asynchronously manner.

- Daily lectures will be live streamed via YouTube (<https://www.youtube.com/c/ChrisBourkeUNL/live>), however the time is yet to be determined.
 - Recordings of the lectures will be available immediately following so you can watch/rewatch at your convenience
 - During the live broadcast, Piazza will be used for questions/answers
- For assignments that allow collaboration, you may use any medium you choose.
 - You can establish your own Zoom rooms to talk back and forth and share a screen
 - You may use discord or slack instead
 - You can (in fact should) be using git to share code (but only use private repos)
 - There are (free) online IDEs that allow you both to type in the same editor at the same time: <https://repl.it>, <https://ide.cs50.io/>; a more extensive list: <https://gist.github.com/rouzbeh84/4bafc9fe4fe02edf506d11997c4674b0>
- Written solutions must be submitted through canvas as PDFs
- Live office hours will be held online via zoom
- Exams will still be run, but asynchronously as “take home” exams that will be released the day-of. You will have a limited but *flexible* time period to complete them and submit it electronically. No collaboration will be allowed on the exams.

5 Accommodations for Students with Disabilities

The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let me know immediately so that we can discuss options privately. To establish reasonable accommodations, I may request that you register with Services for Students with Disabilities (SSD). If you are eligible for services and register with their office, make arrangements with me as soon as possible to discuss your accommodations so they can be implemented in a timely manner. SSD contact information: 117 Louise Pound Hall.; 402-472-3787

6 Grading

Grading will be based on assignments (both written and programming portions) as well as two exams.

Category	Number	Points Each	Total
Assignments	4	200	800
Midterm	100	1	100
Final	100	1	100
Total			1,000

6.1 Scale

Final letter grades will be awarded based on the following standard scale. This scale may be adjusted upwards if the instructor deems it necessary based on the final grades only. No scale will be made for individual assignments or exams.

Letter Grade	Percent
A+	≥ 97
A	≥ 93
A-	≥ 90
B+	≥ 87
B	≥ 83
B-	≥ 80
C+	≥ 77
C	≥ 73
C-	≥ 70
D+	≥ 67
D	≥ 63
D-	≥ 60
F	< 60

6.2 Assignments

There will be 4 assignments that will consist of both written exercises as well as *substantial* programming problems. You will be expected to follow all instructions on the assignments. Clarity and legibility are of great importance. If homework is sloppy or unclear, points may be deducted. You are not required to typeset your written solutions; however, it is strongly recommended that you do so using L^AT_EX, markdown or similar typesetting system. Resources for L^AT_EX are available on the course web page. Source code and all relevant files for programming portions must be handed in using the CSE web handin program. Each assignment will have a fixed deadline based on CSE's server time. No late assignments will be accepted.

Further, programming solutions will be graded using our online webgrader system. Failure to submit compilable or runnable code may result in a zero. You are expected to do your

own substantial testing (and to submit valid, working test cases as well), but it is essential that your submissions work on the webgrader.

6.3 Exams

There will be two exams, both of which will be open-book, open-note, open-computer but you may *not* collaborate with anyone in or outside the class on the solutions.

6.4 Grading Policy

If you have questions about grading or believe that points were deducted unfairly, you must first address the issue with the individual who graded it to see if it can be resolved. Such questions should be made within a reasonable amount of time after the graded assignment has been returned. No further consideration will be given to any assignment a week after it grades have been posted. It is important to emphasize that the goal of grading is consistency. A grade on any given assignment, even if it is low for the entire class, should not matter that much. Rather, students who do comparable work should receive comparable grades (see the subsection on the scale used for this course).

6.5 Late Work Policy

In general, there will be no make-up exams or late work accepted. Exceptions may be made in certain circumstances such as health or emergency, but you must make every effort to get prior permission. Documentation may also be required.

Homework assignments have a strict due date/time as defined by the CSE server's system clock. All program files must be handed in using CSE's webhandin as specified in individual assignment handouts. Programs that are even a few seconds past the due date/time will be considered late and you will be locked out of handing anything in after that time.

6.6 Webgrader Policy

Failure to adhere to the requirements of an assignment in such a manner that makes it impossible to grade your program via the webgrader means that a disproportionate amount of time would be spent evaluating your assignment. For this reason, we will not grade any assignment that does not compile and run through the webgrader.

6.7 Academic Integrity

All homework assignments, programs, and exams must represent your own work unless otherwise stated. No collaboration with fellow students, past or current, is allowed unless otherwise permitted on specific assignments or problems. The Department of Computer Science & Engineering has an Academic Integrity Policy. All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself. The most recent version of the Academic Integrity Policy can be found at <http://cse.unl.edu/academic-integrity>

7 Summer Session Policy

As a summer session, the course pace and presentation is accelerated. What would normally be covered over 15 weeks is compressed into less than five. Your success in this course depends on your acceptance of this fact and a commitment to putting in the extra work necessary to understand this material in the time that we do have. This means extensive daily review of materials outside of lecture and a diligent attitude toward completing assignments. As such, no late work will be accepted and no makeup quizzes or exams will be given. The compressed time period and logistics of offering summer courses make it extraordinarily difficult to make such considerations.

In addition, the summer version of this course lacks the same resources that would be available during the regular academic year. In particular, the Student Resource Center is closed and there is no recitation section. This may make getting additional help more difficult and you should make the appropriate adjustments or reconsider taking this course during the regular academic year.

8 Communication & Getting Help

The primary means of communication for this course is Piazza, an online forum system designed for college courses. We have established a Piazza group for this course and you should have received an invitation to join. If you have not, contact the instructor immediately. With Piazza you can ask questions anonymously, remain anonymous to your classmates, or choose to be identified. Using this open forum system the entire class benefits from the instructor and TA responses. In addition, you and other students can also answer each other's questions (again you may choose to remain anonymous or identify yourself to the instructors or everyone). You may still email the instructor or TAs, but more than likely you will be redirected to Piazza for help.

In addition, there are two anonymous suggestion boxes that you may use to voice your concerns about any problems in the course if you do not wish to be identified. My personal

box is available at <https://cse.unl.edu/~cbourke/email/>. The department also maintains an anonymous suggestion box available at <https://cse.unl.edu/contact-form>.

8.1 Getting Help

Your success in this course is ultimately your responsibility. Your success in this course depends on how well you utilize the opportunities and resources that we provide. There are numerous outlets for learning the material and getting help in this course:

- Lectures: attend lectures regularly and when you do use the time appropriately. Do not distract yourself with social media or other time wasters. Actively take notes (electronic or hand written). It is well-documented that good note taking directly leads to understanding and retention of concepts.
- Required Reading: do the required reading on a regular basis. The readings provide additional details and depth that you may not necessarily get directly in lecture.
- Piazza: if you have questions ask them on Piazza. It is the best and likely fastest way to get help with your questions. Also, be sure to read other student's posts and questions and feel free to answer yourself!
- Office Hours: the instructor and GTA(s) hold regular office hours throughout the week as posted on the course website. Attend office hours if you have questions or want to review material.
- Don't procrastinate. The biggest reason students fail this course is because they do not give themselves enough opportunities to learn the material. Don't wait to the last minute to start your assignments. Many people wait to the last minute and flood the TAs and SRC, making it difficult to get help as the due date approaches. Don't underestimate how much time your assignment(s) will take and don't wait to the week before hand to get started. Ideally, you should be working on the problems as we are covering them.
- Get help in the *right way*: when you go to the instructor or TA for help, you must demonstrate that you have put forth a good faith effort toward understanding the material. Asking questions that clearly indicate you have failed to read the required material, have not been attending lecture, etc. is *not acceptable*. Don't ask generic questions like "I'm lost, I don't know what I'm doing". Instead, explain what you have tried so far. Explain why you think what you have tried doesn't seem to be working. Then the TA will have an easier time to help you identify misconceptions or problems. This is known as "Rubber Duck Debugging" where in if you try to explain a problem to someone (or, lacking a live person, a rubber duck), then you can usually identify the problem yourself. Or, at the very least, get some insight as to what might be wrong.



CSCE 322: Programming Language Concepts (3 credits)

Fall 2022 – Section 001

Lecture: 10:30am – 11:20am MWF in Avery 106

Contact Information:

Instructor

Marcus Gubanyi
"Office:" Avery 352
Email: marcus.gubanyi@cune.edu

Teaching Assistants

Jared Soundy and Mrinal Rawool
Office Hours in the Student Resource Center
Email: jared.soundy@huskers.unl.edu and mrawool2@huskers.unl.edu

Office hours for the instructor and TA's are up-to-date on Canvas.

Prerequisite: CSCE 156/H or RAIK 184H or CSCE 311 or SOFT 161/H

Course Materials: Students are expected to obtain a copy of *Concepts of Programming Languages* by Robert W. Sebesta, 11th edition (2016). ISBN-13: 978-0133943023. Other materials will be available on Canvas.

Course Description: List-processing, string-processing, and other types of high-level programming languages. Fundamental concepts of data types, control structures, operations, and programming environments of various programming languages. Analysis, formal specification, and comparison of language features.

Objectives: Students successfully completing this course will be able to:

1. Understand different programming language paradigms,
2. Understand formal language theory as it relates to programming languages,
3. Evaluate programming language design choices,
4. Appreciate the complexity of the implementation of programming language features,
5. Implement solutions with functional programming.

Grading: Grading will be based on a weighted system outlined below. Late submissions will be accepted up to one week after the deadline, penalized 30%.

Item	Weight
3 Midterm Exams	3 × 10%
Final Exam (Cumulative)	20%
Homework	20%
Programming Assignments	30%

Letter grades will be awarded based on the following percentages:

A+ ≥ 97	B+ ≥ 87	C+ ≥ 77	D+ ≥ 67	F < 60
A ≥ 93	B ≥ 83	C ≥ 73	D ≥ 63	
A- ≥ 90	B- ≥ 80	C- ≥ 70	D- ≥ 60	

Course Schedule: The course is split into 4 modules, each spanning three to four weeks. Each module has a homework assignment, programming assignment, and an exam. Homework will be due a week prior to the exam and is directly related to exam content. Programming assignments will be due after the exam, except for the final. ***All dates are tentative and subject to change.***

1. Programming Language Concepts

- Lecture Topics: Programming Paradigms, Language Evaluation, Design and Implementation, Evolution of Programming Languages, Functional Programming
- Homework 1: Due September 9
- Midterm 1: September 16
- Program 1: Due September 23

2. Formal Language Theory and Syntax

- Lecture Topics: Formal Language Theory, Chomsky's Hierarchy, Grammars, Automata, Parsing
- Homework 2: Due September 5
- Midterm 2: October 12
- Program 2: October 19

3. Variables

- Lecture Topics: Variables, Binding, Scope, Data Types, Memory Management
- Homework 3: Due November 2
- Midterm 3: November 9
- Program 3: Due November 16

4. "Control Structures"

- Lecture Topics: Expressions, Statements, Selection, Iteration, Subprograms, Abstract Data Types, Object-Oriented Programming
- Homework 4: Due December 2
- Program 4: Due December 2
- Final Exam: Finals Week

UNL Course Policies and Resources: Students are responsible for knowing the university policies and resources found on this page: <https://go.unl.edu/coursepolicies>.

- University-wide Attendance Policy
- Academic Honesty Policy
- Services for Students with Disabilities
- Mental Health and Well-Being Resources
- Final Exam Schedule
- Fifteenth Week Policy
- Emergency Procedures
- Diversity & Inclusiveness
- Title IX Policy
- Other Relevant University-Wide Policies

SoC Academic Integrity: The Computer Science & Engineering department has an [Academic Integrity Policy](#). Please note that by staying on the course you are abiding by the rules and regulations as described on the page. This is non-negotiable.

Miscellaneous

- The SoC has an [anonymous suggestion box](#) that you may use to voice your concerns about any problems in the course or school if you do not wish to be identified.
- The SoC has a [Student Resource Center](#) in Avery 12. This is where you can get additional help from the TA during his/her office hours and additional help with coursework and assignments. This is also the place where you can get technical support with SoC Computer Systems.
- All students enrolled in CSCE/SOFT courses will be enrolled in the [Virtual SRC](#). You can access the virtual SRC via canvas. All SRC tutors will have experience up through CSCE 310 and/or SOFT 262. The SRC tutors will serve as hosts to move students into break-out rooms while the specific course TAs are holding office hours and be available to answer general questions. For our TA hours, please check our office hours page on canvas.

CSCE 361: Software Engineering

Course Description

Catalog Description

Computer Systems Engineering (3 cr). Techniques used in the disciplined development of large software projects. Software requirements analysis and specifications, program design, coding and integration testing, and software maintenance. Software estimation techniques, design tools, and complexity metrics. *Requires participation in a group design and implementation of a software project.*

Prerequisites

Grade of “P” or “C” or better in CSCE 310, CSCE 310H, CSCE 311, SOFT 260, SOFT 260H, or RAIK 283H.

Prerequisites by Topic

- *Mastery of* A high-level programming language
- *Mastery of* Basic data structures and algorithm design
- *Familiarity with* Two or more programming languages

Course Objectives

- *Mastery of* Concepts and techniques for large-scale software development.
- *Familiarity with* Design techniques, such as structured and object-oriented analysis
- *Familiarity with* The software life cycle and the software development process
- *Familiarity with* Pragmatic aspects of developing software systems
- *Exposure to* Software testing techniques
- *Exposure to* Developing and maintaining large-scale software systems
- *Exposure to* Using existing software systems or legacy code when developing a software system
- *Exposure to* Working in a team of software developers
- *Exposure to* Communication of technical results (done in context of projects)
- *Exposure to* Ethical issues in computing

Topics Covered

- Programming vs. large-scale software development
- Software process and life cycle models
- Requirements engineering
- Specification techniques
- Structured analysis and design
- Object-oriented design techniques

- Software testing (black box and white box; unit and system)
- System maintenance issues and techniques
- Software architecture

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Design Studio

Design Studio is a two semester (yearlong) sequence of courses:

First (fall) semester (Design Studio I & III) consists of the following course numbers:

RAIK 401H, RAIK 403H, CSCE 401H, CSCE 403H

Second (spring) semester (Design Studio II & IV) consists of the following course numbers:

RAIK 402H, RAIK 404H, CSCE 402H, CSCE 404H

Staff and Faculty:

Cheryl Nelson	Kauffman #108B	Interim Director	cheryl.nelson@unl.edu
Professor Justin Firestone	Kauffman #140	Tribe Lead	justin.firestone@unl.edu
Professor David Keck	Kauffman #145	Tribe Lead	dkeck@unl.edu
Professor Robert Mackalski	Kauffman #139	Tribe Lead	robert.mackalski@unl.edu
Professor Stephanie Valentine	Kauffman #144	Tribe Lead	valentine@unl.edu
Rachel Bradley	Kauffman #108C	Grad TA Tech Resource	rbradley4@huskers.unl.edu

Faculty and staff are available to meet by request through an appointment.

Prerequisites to start sequence: Good standing in the University Honors Program and admission to the Jeffrey S. Raikes School of Computer Science and Management or by invitation; RAIK 383H or equivalent. See below for class sequencing.

Meeting: TR 2:00pm – 3:15pm, KAUF Great Hall (122)

Finals Time:

- First semester = Tuesday, December 14th: 1:00pm – 3:00pm, KAUF Great Hall (122)
- Second semester = Wednesday, May 11th: 1:00pm – 3:00pm, KAUF Great Hall (122)

Showcase: Friday, April 29th: 2:00pm – 4:30pm, Nebraska Innovation Campus

Course Objectives

At the end of this course, students will have gathered practical real-world product development experience with interdisciplinary team-based roles, environments, best practices, and processes by working with industry, government, and non-profit partners.

There are five main learning objectives where the students shall be able to:

- A. Apply, demonstrate, and integrate learned professional skills including:
 - Communications
 - Leadership
 - Ethics
 - Teamwork
 - Critical Thinking
 - Problem Solving
- B. Discover and manage diverse expectations including:
 - Client Relationships
 - Accountability
 - Conflict Resolution
 - Self-management
 - Self-motivation
 - Time Management

- C. Apply, demonstrate, and integrate learned business skills including:
 - Product Management (requirements analysis; marketing; story mapping, intellectual property)
 - Project Management and Planning
 - Risk Management
 - Visioning
 - Design Thinking
- D. Apply, demonstrate, and integrate learned software engineering skills including:
 - Testing/Quality Assurance
 - DevOps (working build processes; static code analysis; continuous integration)
 - Tech Proficiencies/Dependability Engineering
 - Architecture and Design
 - Release-driven Iterative Methodologies
- E. Learn beyond project/practice with:
 - Continual Learning
 - Shared Learning
 - Self-improvement
 - Intellectual Curiosity (creativity; innovation)

Course Sequencing

Students in the Raikes School traditionally take Design Studio for two years (four consecutive semesters). We offer the opportunity for students outside of the Raikes School, called Associates, to also take Design Studio by invitation for either one or two years. Here are the course progressions for the different type of students/options:

- Raikes School Student: RAIK 401H → RAIK 402H → RAIK 403H → RAIK 404H
- Raikes School Student with Research Studio Options:
 - RAIK 401H → RAIK 402H → RAIK 405H → RAIK 406H
 - RAIK 405H → RAIK 406H → RAIK 401H → RAIK 402H
- Associate: CSCE 401H → CSCE 402H [→ CSCE 403H → CSCE 404H] (*Optional for second year)
- Software Engineering Associate Options:
 - CSCE 401H → CSCE 402H → CSCE 403H → CSCE 404H
 - CSCE 486 → CSCE 488 → CSCE 403H → CSCE 404H
 - CSCE 401H → CSCE 402H → SOFT 403 → SOFT 404

The reason for this sequencing relates to ACE requirements. The RAIK 401H-402H (and CSCE 486-488) sequence offers both ACE 8 and ACE 10. The course progressions above preserve the student's ability to earn both of those graduation requirements with their initial year of Design Studio (or Senior Design).

Class Time

Design Studio is made up of both hands-on product development and review sessions with faculty and staff. Class time is allocated each week for lectures and workshops designed to support the work that you are doing in teams. We will continue to have guest lectures from the community on various topics. In addition to lectures and workshops, class-time occasionally will be used for teams to present work for review and feedback.

Class topics and content will be organized by weeks and made available in Canvas modules throughout both semesters. The fall and spring schedules will be provided in a separate Design Studio Course Schedule document.

Snow Days

If in-person classes are canceled, you will be notified through a Canvas announcement on alternative measures taken by DS to hold the class session via Zoom or if the class session will be canceled or rescheduled. Students will be expected to check these notifications for continuity of instruction.

Team Time

The focus of each student team is to achieve a successful project by delivering a working solution that delights the customer. Each team is responsible for organizing itself effectively to complete the project. Team time ideally will be scheduled in 3-4 hour blocks but may not be less than 2 hours without the permission of the program lead and tribe lead. For team time, the longer the block the more productive the team. Depending on the skill and efficiency of the student team, it may take more or less time. At least one block of team time must be scheduled on a weekday between 9am and 5pm (beginning at 4:30pm does not fulfill this requirement). Due to the nature of software development, some weeks may require more work than others. Students are encouraged to use all available resources to work “smarter” and not necessarily just “harder.” Team time cannot be scheduled during class time when a class session is scheduled to be held.

Students are expected to spend at least 12 hours in team time each week and are encouraged to spend around 15 hours doing project work during any given week. Most of these hours should have the entire team present and no hours with only first-year or only second-year students. Schedules must be approved by the project’s leadership team, sponsor and tribe lead. The team needs to send a finalized schedule (including weekly meetings) to all stakeholders and post and pin the schedule in their Slack project channel.

Your team will be supervised by a tribe lead (faculty instructor) who will meet with teams throughout the semester to review work and provide feedback. The program lead role which advises teams on DS policies and procedures will be handled by Cheryl Nelson on an as-needed basis.

Grading

Grades will be based on both exhibited practices and delivered work product. Points can be earned in the course as assigned by the tribe lead in the following areas and scaled based on the percentages below.

- Product release evaluations from sponsors (graded 3 times per semester) – 45%
- Best Practices and Process – 30%
- Special Topics Presentation – 10%
- Executive Summary and Project Retrospective – 5%
- Course Participation and Engagement – 5%
- Individual Student Assessment – 5%

Students are expected to participate and be engaged with the class content and activities. This includes with in-class discussions, review of posted videos, completion of assigned activities, etc. While attendance may not be taken on every occasion, instructors will continually make note of who is and is not attending class. It is not acceptable to miss class without a University approved excused absence. Exceeding two unexcused absences will result in reduction of course participation and engagement grade.

Project Releases

A major component of Design Studio is delivery of a software product that delights the customer. Students will be regularly evaluated by their sponsor and points will be based on performance in their development teams. At the end of each release, sponsors will evaluate teams based on four criteria: 1) vision, 2) planning, 3) execution, and 4) communication.

The following points will be awarded for each criterion as follows:

<i>Release</i>	Communication	Execution	Planning	Vision
1 - October 8	3	2	5	5
2 - November 5	3	6	3	3
3 – December 10	3	8	2	2

4 - February 11	3	6	3	3
5 - March 11	3	8	2	2
6 - April 15	3	8	3	1

Sponsors will evaluate the entire team and assign ratings based on their interactions with the team during the release development cycle. Tribe leads will assign the team points based on sponsor feedback and their own view of the project. Points from releases will be distributed in conjunction with a peer review process with each member of the team assigning some number of points to all team members. The total points earned by a student for each release will be assigned by the tribe lead in consideration of the average of all points distributed to that student by the members of the team. Points will be deducted from a student for non-completion of the peer review process within 5 days of notification by DS system.

Team Best Practices

New in Semester Two is the process of Best Practices. One of the goals of Design Studio is to prepare students to be successful members of development project teams by exposing them to best practices that are associated with managing, building, and releasing high-quality products. Students will set goals in the following categories which will be reviewed with Tribe Leads in their weekly interactions.

SEMESTER 1 - Completed

1. Product Management – Customer Discovery, Understanding Requirements, Ideation
2. Story Mapping – Vision, Release Planning
3. Project Management – Initiation, Planning, Estimating, Logistics
4. Teamwork – Leadership, Inclusion, Accountability
5. DevOps – such as, but not limited to:
 - Working Build
 - Static Code Analysis
 - Continuous Integration
 - Continuous Deployment
6. Releases – Process, Commitments
7. Risk Management

SEMESTER 2

8. Communications – Presentations, Technical Writing
9. Testing – Quality Assurance, Test Strategy, User Testing
10. Security
11. Technical Proficiencies – such as, but not limited to:

○ Accessibility	○ Scalability
○ Efficiency/Performance	○ Telemetry
○ Interoperability/Compatibility	○ Testability
○ Privacy	○ Usability
○ Reliability/Dependability	○ Others not listed
12. Expectation Management – Improving Estimation, Velocity, Re-setting Expectations
13. Transition

For each team, the points associated with the goals in the above areas will be determined by the team. The degree to which each of these items are performed will be evaluated by the tribe lead assigned to your team. Be advised that you should ensure that you **always** conform to these practices. If you are unsure of how to put practices in place on your team, **communicate with your tribe lead** who is here to help you.

Policies

You **will never** be downgraded or negatively impacted for asking for help in Design Studio. Late work will receive an automatic 50% deduction and will not be accepted more than one week past the due date. Grades will be assigned based on the following scale:

A+	*	A	>= 93.0000	A-	>= 90.0000
B+	>= 87.0000	B	>= 83.0000	B-	>= 80.0000
C+	>= 77.0000	C	>= 73.0000	C-	>= 70.0000
D+	>= 67.0000	D	>= 63.0000	D-	>= 60.0000
F	< 60.0000				

* Grades of A+, at the discretion of the faculty and staff, will be reserved for students who demonstrate an exemplary performance on all practice and academic tasks.

Student Expectations

Students are expected to follow all sponsor, coach, and faculty and staff instructions during the completion of their assigned projects. When expectations are unclear or unknown, it is the responsibility of the student to seek clarification by consulting the appropriate parties. As students will be collaborating closely with industry professionals, all students are expected to be respectful and interact with the highest levels of professionalism in all aspects of this experience. Students are required to attend (either virtually or in-person) all scheduled class events and participate in activities.

First year students are generally expected to be team members, contributing fully to the team while gaining experience in the Design Studio expectations and process. First year students have the opportunity to provide leadership in a specific subject area or skill within a team once those areas are identified. Second year students are generally expected to be team leaders, fulfilling one of the two manager roles on the team and using their experience to lead and direct the team through Design Studio.

Team Expectations

Projects are completed over the course of the academic year. Projects will be completed using an iterative development methodology based on Scrum as outlined in course materials. Each project team, in conjunction with the sponsor, coach, and faculty and staff, is expected to operate using the processes prescribed by Design Studio. Deviation from the processes should be discussed with the Design Studio director and tribe lead and agreed to by all parties; deviations from the Design Studio process rarely will be approved. Teams also are responsible for reporting their status to all stakeholders on at least a weekly basis. Expectations of the team will come primarily from the sponsor and be supported by class content.

Conflicts

Design Studio includes a significant amount of software development work for sponsors who pay a fee for development. The primary purpose of the course is to provide an educational opportunity for the students and an opportunity for sponsors to recruit interns and new post-graduate hires. At the same time, teams are developing intellectual property that will be owned by the sponsor at the end of the Design Studio year. As a result, Design Studio requires some conflict management policies similar to those that you would find in an internship or after you accept a job.

Your Time

What you do in your free time is up to you; however, when you are working on your project during scheduled team time, that is the sponsor's time. When in team time you should not be working on tasks that are unrelated to your Design Studio project. There is always plenty to do and team time is not homework time. If you have your own business, you should **never under any circumstances use team time or Design Studio equipment or resources for any work related to your business.**

Design Studio Resources

We will provide you with all the tools and resources necessary to successfully complete your project. This includes a space to work, a machine to develop on, and devices needed to adequately test your product. You should not use any of these resources for tasks that are not related to your Design Studio project. While team rooms are primarily for Design Studio work, other quiet uses of the space that don't impact the other Design Studio teams are acceptable (such as using it as a quiet place to do homework outside of team time). You should not use personal computers for Design Studio related development and vice versa. If you choose to do Design Studio work on a personally owned computer, be aware that your computer and all its content may be subject to review by the sponsor. Also, be aware that anything you put on a University-owned computer is not private and may be reviewed by UNL faculty and staff, or a member of the sponsor organization.

Employment

To balance time spent on Design Studio with all the other activities in which students are involved, and to prevent undue influence that may disrupt that balance, working for a Design Studio sponsor during the academic year is prohibited. Working for means being paid in any way for work that is being done during either the fall or spring semester. In a limited number of cases, exceptions to this policy can be made with the understanding that **no student may work for the sponsor of the project to which they are assigned under any circumstances.** This policy does not prevent you from seeking post-graduate employment or internships or accepting any offer of employment outside of the Design Studio year; you are encouraged to take advantage of the opportunities that are created by working with sponsors in Design Studio. Requests for an exemption to this policy must be made to the Director of Design Studio in writing (email is fine).

Sponsor Meetings

Weekly team meetings with sponsors are required. They may occur during regularly scheduled team times but may not be scheduled during class time when a class session is scheduled to be held. All team members must be present either in-person or virtually.

Release Meetings

Release meetings allow teams to hand off, demonstrate, and defend the current results while providing direction and planning for project completion. The goal of each release meeting is to provide the sponsor with releasable work product. Releasable work product is some iteration of the project that the sponsor can interact with outside of the meeting, share with others, and use to provide feedback independent of interaction with the team. Teams are encouraged to plan features around making each release as functional as possible and push aggressively towards production-ready releases. All Design Studio teams will release work product in a coordinated effort across all teams and tribes. See the Canvas course calendar or Slack for release dates.

Each release will involve the Design Studio project team, the sponsor team (including the product owner), and the assigned Design Studio tribe lead working together to demonstrate and discuss their progress on the project to the project executive sponsors. Release meetings must be scheduled within one week (before or after) of the scheduled release date and must take place between the hours of 9am and 5pm; without permission from Design Studio, a release meeting may not be scheduled to run later than 5pm. If a sponsor requires a release meeting that will run late, it should be scheduled with your team's tribe lead. Unless otherwise requested by and prearranged with the sponsor, in-person release meetings need to take place in Kauffman.

Teams are **strongly encouraged** to schedule release meetings as early as possible after the backlog for the release has been set. Tribe leads attend multiple meetings, and it may be difficult to coordinate schedules with them. There are a limited number of conference rooms available for in-person meetings or video conference rooms for online meetings with the team members taking place in Kauffman. A calendar meeting invitation should be sent to all required meeting participants.

Online Tools

Online management and collaboration tools will be detailed during the first several weeks of class. Teams are expected to utilize Design Studio supplied project resources for collaboration, issue tracking, and source code management. We use Slack, Canvas, SharePoint, GitHub, and ZenHub for these purposes. In cases where sponsors wish to provide their own tools, teams are expected to use those and ensure that the tribe lead and Design Studio staff have access during the project.

Meeting and Team Rooms

Each team will be assigned a team room and will have access to space for the various meetings that take place in Design Studio. Please be sure to choose meeting spaces most appropriate for the size and technology needs (i.e. video conferencing) of your planned meeting. The following spaces are available to Design Studio teams this year for meetings:

- Kauffman 127A – Small Video Conference Room
- Kauffman 127B – Small Video Conference Room
- Kauffman 127C – Small Video Conference Room
- Kauffman 127D – Large Video Conference Room
- Kauffman 127E – Large Video Conference Room
- Kauffman 102 – Large Conference Room

The following spaces are available to Design Studio teams this year for working team times:

- Kauffman 103 – 3 spaces
- Kauffman 104 – 3 spaces
- Kauffman 114 – 2 spaces
- Kauffman 115 – 2 spaces
- Kauffman 116 – 2 spaces
- Kauffman 125 – 3 spaces
- Kauffman 137/A – 5 spaces

Conference rooms in Kauffman can be scheduled for sponsor meetings throughout the semester using the UNL scheduling tool – instructions will be provided.

SoC Student Resource Center (SRC)

The Computer Science Resource Center (SRC) is committed to continuing to provide support for students enrolled in Computer Science and Software engineering courses. The Student Resource Center is staffed by student tutors and teaching assistants who are available to help you with course work. Teaching assistants for all courses hold office hours in the Student Resource Center.

A virtual Student Resource Center has been set up via Canvas (<https://canvas.unl.edu/courses/120620>) to accommodate students during the COVID-19 pandemic.

HOURS: Monday – Friday, 10 a.m. – 5 p.m.

LOCATION: Avery Hall Room #12

The SoC Student Resource Center is staffed by student tutors and teaching assistants who are available to help you with course work. The SRC is a community hub for SoC students. Even if you don't need help, the Student Resource Center is a great place to study with your peers.

Writing Center

At the Writing Center, our undergraduate and graduate Writing Consultants work with writers at all levels, from all disciplines, at all stages of the writing process. All members of the UNL community (students, faculty, and staff) are welcome. Whether you are brainstorming or organizing ideas or polishing a final draft, we look forward to discussing your writing with you.

All forms of communication are welcome, from essays, lab reports, research papers, and journal articles to presentations, cover letters, personal statements, and theses/dissertations. Writing Center Consultants can work with you at any stage of the writing process, from brainstorming and organizing your ideas through polishing a final draft.

The Writing Center's main location is 102 Andrews Hall. During the 2021-22 academic year, offering online and in-person appointments. Please schedule your appointments as usual through WOnline. Visit Online Writing Center Services page to learn more about our online formats. Sign up for an appointment any time by visiting unl.mywonline.com. For other information about the Writing Center, please visit unl.edu/writing.

Course Policies

Huskers Email Required

@huskers.unl.edu accounts became the primary email for all university communications to students beginning May 20, 2019. This email will ensure that students never miss important university communications and it will be connected to MyRED and Canvas accounts. Additionally, students will use it to access university library services and software — like Adobe Creative Cloud, Matlab, VidGrid, Zoom, Office 365 and more.

A @huskers.unl.edu address has been assigned to you. This will be the primary email for all Design Studio communications emailed to students this year. To access your @huskers.unl.edu email, simply log in to mymail.unl.edu to an external site. with your My.UNL username and password. If you aren't sure what your @huskers.unl.edu email address is, you'll find it listed under the Profile tab in MyRED or MyNCTA.

It's important to check your @huskers.unl.edu email frequently — we recommend daily — so you don't miss any time-critical messages from the university, especially from the offices of Scholarships and Financial Aid, University Registrar, Admissions, and Student Accounts. Instructors will communicate with you primarily through Canvas and Slack, but may use this email, as well.

Recording of Class-Related Activity

We invite all of you to join us in actively creating and contributing to a positive, productive, and respectful classroom culture. Each student contributes to an environment that shapes the learning process. Any work and/or communication that you are privy to as a member of this course should be treated as the intellectual property of the speaker/creator and is not to be shared outside the context of this course.

Students may not make or distribute screen captures, audio/video recordings of, or livestream, any class-related activity, including lectures and presentations, without express prior written consent from the DS Director or an approved accommodation from Services for Students with Disabilities. If you have (or think you may have) a disability such that you need to record or tape class-related activities, you should contact Services for Students with Disabilities. If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Failure to follow this policy on recording or distributing class-related activities may subject you to discipline under the Student Code of Conduct.

ACE Requirements

The CSCE 401H course will satisfy ACE Learning Outcome #8: Explain ethical principles, civics, and stewardship, and their importance in society. Learning opportunities for ACE8 will include:

- reading, viewing, and discussing the required materials,
- an investigative case study,
- hands-on design of your capstone project.

In this course, you will be given opportunities to learn, be assessed on what you have learned, and have specific critical skills reinforced. Specifically, through the required materials, you will learn about events due to poor ethics, lack of civic judgments, and inadequate stewardship exercised by the environment, and how these events impacted the society. The readings and other materials have been specifically chosen for that purpose. After going through these materials, quizzes will be given. First, you will be required to learn about general ethical principles expanded into applying those principles via several real-world case studies. Second, you will be quizzed regarding the application of the ethical principles. Quizzes related to these activities will count towards 10% of your semester grade. As part of the overall course grade, you will be assessed via the quizzes, your project reports, and how you practice ethical principles in the way you document your work and acknowledge others'.

The RAIK 402H and CSCE 402H courses will satisfy ACE Learning Outcome #10: Generate a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.

Design Studio is the capstone learning experience of the Jeffrey S. Raikes School in Computer Science and Management. In Design Studio, student teams partner with sponsoring businesses and government agencies to develop real-world, software-based solutions meeting their sponsor's needs. Design Studio gives students and sponsors the ability to interact, while benefiting from the support of Raikes School faculty and facilities. Students have opportunities to demonstrate their abilities, knowledge, and capacities by completing consulting based technology projects. Throughout these consulting based projects, student teams will be required to determine and meet the expectations set forth by the project sponsors. Each project goes through all phases of the development process including initiation, execution, and closing.

Students are evaluated based on 1) individual and team improvement, 2) achievement of learning objectives, and 3) project success. Graded assignments used to assess achievement of this outcome include:

- Presentation to the sponsor at completion of several checkpoints throughout the project to assess appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.

- Iteration commitment/satisfaction forms to assess appropriate technical proficiency, synthesis, interpretation, and reflection.
- Sponsor documents to assess appropriate technical proficiency, information collection, synthesis, and interpretation.
- Project sponsor surveys and sign-off documents to assess broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.
- Project retrospective to assess information collection, synthesis, interpretation, presentation, and reflection.

SoC Policies

Design Studio follows all of the School of Computing course policies:

- The SoC has an anonymous contact form (<http://cse.unl.edu/contact-form>) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.
- All homework assignments, quizzes, exams, etc. must be your own work. No direct collaboration with fellow students, past or current, is allowed unless otherwise stated. The SoC has an Academic Integrity Policy (<http://cse.unl.edu/academic-integrity-policy>). All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case-by-case basis and may result in a failing assignment or a failing grade for the course itself.

Additional UNL Course Policies and Resources

Students are responsible for knowing the university policies and resources found on this page.

(<https://go.unl.edu/coursepolicies>)

Satisfying Senior Project (Thesis) Requirements

UNL Honors Senior Project

Participation in Design Studio will satisfy the requirement for an undergraduate thesis in the UNL Honors program. The use of the Design Studio experience in satisfying these requirements is subject to all policies, deadlines, and requirements of the respective program. Teams will complete two documents in the spring (Project Retrospective and Executive Summary) that will be approved by Design Studio faculty and staff and can then be submitted to the Honors program to complete the thesis requirement. Additional instructions will be provided in the spring semester with the two documents on exactly how to submit your work for Honors thesis credit.

UNL Honors Senior Project:

<https://honors.unl.edu/academics/senior-project>

College of Arts & Sciences Undergraduate Thesis for Degrees with Distinction

Participation in Design Studio may be used to partially satisfy requirements for an undergraduate thesis for graduation with some level of distinction in the College of Arts & Sciences. The use of the Design Studio experience in partially satisfying these requirements is subject to all policies, deadlines, and requirements of the college. Students intending to complete a thesis are advised to familiarize themselves with the program requirements.

UNL College of Arts and Sciences:

<http://cas.unl.edu/degrees-distinction>

The work of writing a thesis must be supervised by a faculty member from the college (faculty advisor). If you do not have a preferred faculty advisor for your thesis, Dr. Steve Cooper, Executive Director of the Raikes School, will serve as your faculty advisor.

Requirements for an Undergraduate Thesis for Degrees with Distinction

The thesis must be the result of independent, sustained thought and intellectual curiosity. A survey of the literature about a particular topic is not sufficient however novel it may be. Ideally, there should be a clear formulation of a problem or question, scholarly study which illuminates or addresses it supported by the work done in Design Studio, and a conclusion supported by evidence. A bibliography and reference to existing literature in the field should be included where appropriate. The thesis should be a minimum of 20 pages. Students, working closely with their two thesis advisors, may develop a project that takes a form other than a traditional thesis (e.g. they may develop video, audio, or web-based project), but the research, knowledge, effort, and quality of work should be comparable to that of a thesis in the discipline.

Students should work closely with their faculty advisor as they complete their thesis. A final copy of the thesis must be submitted to their faculty advisor no later than one month prior to the program deadline for final thesis submission.

Design Studio

Design Studio is a two semester (yearlong) sequence of courses:

First (fall) semester (Design Studio I & III) consists of the following course numbers:

RAIK 401H, RAIK 403H, CSCE 401H, CSCE 403H

Second (spring) semester (Design Studio II & IV) consists of the following course numbers:

RAIK 402H, RAIK 404H, CSCE 402H, CSCE 404H

Faculty and Staff:

Mark Antonson	Kauffman #108A	402-472-7465	mark.antonson@unl.edu
Professor Justin Firestone	Kauffman #140	402-472-6000	justin.firestone@unl.edu
Professor David Keck	Kauffman #145	402-472-6000	dkeck@unl.edu
Professor Robert Mackalski	Kauffman #139	402-472-6000	robert.mackalski@unl.edu
Cheryl Nelson	Kauffman #108B	402-472-6003	cheryl.nelson@unl.edu
Jeremy Suing	Kauffman #108D	402-472-1658	jsuing@unl.edu
Professor Stephanie Valentine	Kauffman #144	402-472-6000	valentine@unl.edu
Architecture and Engineering Lead – TBD	Kauffman #108C	402-472-5052	

Faculty and staff are available to meet by request through an appointment.

Prerequisites to start sequence: Good standing in the University Honors Program and admission to the Jeffrey S. Raikes School of Computer Science and Management or by invitation; RAIK 383H or equivalent. See below for class sequencing.

Meeting: TR 2:00pm – 3:15pm, KAUF Great Hall (122)

Finals Time:

- First semester = Tuesday, December 14th: 1:00pm – 3:00pm, KAUF Great Hall (122)
- Second semester = TBD

Showcase: Friday, April 29th: 1:30pm – 4:30pm, Nebraska Innovation Campus

Course Objectives

At the end of this course, students will have gathered practical real-world product development experience with interdisciplinary team-based roles, environments, best practices, and processes by working with industry, government, and non-profit partners.

There are five main learning objectives where the students shall be able to:

- A. Apply, demonstrate, and integrate learned professional skills including:
 - Communications
 - Leadership
 - Ethics
 - Teamwork
 - Critical Thinking
 - Problem Solving
- B. Discover and manage diverse expectations including:
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 - Conflict Resolution
 - Self-management
 - Self-motivation
 - Time Management

- C. Apply, demonstrate, and integrate learned business skills including:
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 - Risk Management
 - Visioning
 - Design Thinking
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 - Intellectual Curiosity (creativity; innovation)

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- Raikes School Student with Research Studio Options:
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 - RAIK 405H → RAIK 406H → RAIK 401H → RAIK 402H
- Associate: CSCE 401H → CSCE 402H [→ CSCE 403H → CSCE 404H] (*Optional for second year)
- Software Engineering Associate Options:
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 - CSCE 486 → CSCE 488 → CSCE 403H → CSCE 404H
 - CSCE 401H → CSCE 402H → SOFT 403 → SOFT 404

The reason for this sequencing relates to ACE requirements. The RAIK 401H-402H (and CSCE 486-488) sequence offers both ACE 8 and ACE 10. The course progressions above preserve the student's ability to earn both of those graduation requirements with their initial year of Design Studio (or Senior Design).

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Design Studio is made up of both hands-on product development and review sessions with faculty and staff. Class time is allocated each week for lectures and workshops designed to support the work that you are doing in teams. We will continue to have guest lectures from the community on various topics. In addition to lectures and workshops, class-time occasionally will be used for teams to present work for review and feedback. Students in this course must work in close physical proximity to one another for extended periods of time in order to achieve the academic goals of the course. For this reason, the Raikes School has determined that face coverings will be required in this course. If you are unwilling to comply with this requirement, please visit with your advisor about possible alternative courses that you might take in lieu of this one.

Class topics and content will be organized by weeks and made available in Canvas modules throughout both semesters. The fall and spring schedules will be provided in a separate Design Studio Course Schedule document.

Team Time

The focus of each student team is to achieve a successful project by delivering a working solution that delights the customer. Each team is responsible for organizing itself effectively to complete the project. Team time ideally will be scheduled in 3-4 hour blocks but may not be less than 2 hours without the permission of the program lead and tribe lead. For team time, the longer the block the more productive the team. Depending on the skill and efficiency of the student team, it may take more or less time. At least one block of team time must be scheduled on a weekday between 9am and 5pm (beginning at 4:30pm does not fulfill this requirement). Due to the nature of software development, some weeks may require more work than others. Students are encouraged to use all available resources to work “smarter” and not necessarily just “harder.” Team time cannot be scheduled during class time when a class session is scheduled to be held.

Students are expected to spend at least 12 hours in team time each week and are encouraged to spend around 15 hours doing project work during any given week. Most of these hours should have the entire team present and no hours with only first-year or only second-year students. Schedules must be approved by the project’s leadership team, sponsor, program lead, and tribe lead. The team needs to send a finalized schedule (including weekly meetings) to all stakeholders and post and pin the schedule in their Slack project channel.

Each team will be assigned a program lead. The program lead is the primary connection with the Design Studio organization. Program leads will spend time each week in team time with you, so you must schedule at least some team time that can accommodate the program lead’s schedule. In addition to the program lead, your team will be supervised by a tribe lead (faculty instructor) who will meet with teams throughout the semester to review work and provide feedback.

Grading

Grades will be based on both exhibited practices and delivered work product. Points can be earned in the course as assigned by the tribe lead in the following areas and scaled based on the percentages below.

- Product release evaluations from sponsors (graded 3 times per semester)
 - 45% for RAIK403H/CSCE403H and RAIK404H/CSCE404H
 - 30% for RAIK401H/CSCE401H and RAIK402H/CSCE402H
- Best practices and process (graded 3 times per semester)
 - 30% for RAIK403H/CSCE403H and RAIK404H/CSCE404H
 - 45% for RAIK401H/CSCE401H and RAIK402H/CSCE402H
- 10% – ethics (CSCE401H)/core values (RAIK401H/RAIK403H) assignments (fall semester)
- 10% – special topics presentations (spring semester)
- 8% – course participation and engagement
- 7% – individual student assessment (17% for CSCE403H in fall semester)

Students are expected to participate and be engaged with the class content and activities. This includes with in-class discussions, review of posted videos, completion of assigned activities, etc.

Project Releases

A major component of Design Studio is delivery of a software product that delights the customer. Students will be regularly evaluated by their sponsor and points will be based on performance in their development teams. At the end of each release, sponsors will evaluate teams based on four criteria: 1) vision, 2) planning, 3) execution, and 4) communication. The following points will be awarded for each criterion as follows:

<i>Release</i>	Communication	Execution	Planning	Vision
1 - October 8	3	2	5	5
2 - November 5	3	6	3	3
3 – December 10	3	8	2	2

4 - February 11	3	6	3	3
5 - March 11	3	8	2	2
6 - April 15	3	8	3	1

Sponsors will evaluate the entire team and assign ratings based on their interactions with the team during the release development cycle. Tribe leads will assign the team points based on sponsor feedback and their own view of the project. Points from releases will be distributed in conjunction with a peer review process with each member of the team assigning some number of points to all team members. The total points earned by a student for each release will be assigned by the tribe lead in consideration of the average of all points distributed to that student by the members of the team. Points will be deducted from a student for non-completion of the peer review process within a given timeframe.

Team Best Practices

One of the goals of Design Studio is to prepare students to be successful members of development project teams by exposing them to best practices that are associated with managing, building, and releasing high-quality products. Students will be evaluated throughout the semester on how they exhibit these practices and processes. Tools and techniques for implementing these will be discussed in presentations and workshops by faculty and staff and industry partners within class sessions and the Canvas modules. Assessment will be based on several activities related to the practices and processes. The following areas will be evaluated for each team over the course of the academic year:

1. Product Management – Customer Discovery, Understanding Requirements, Ideation
2. Story Mapping – Vision, Release Planning
3. Project Management – Initiation, Planning, Estimating, Logistics
4. Teamwork – Leadership, Inclusion, Accountability
5. DevOps – such as, but not limited to:
 - Working Build
 - Static Code Analysis
 - Continuous Integration
 - Continuous Deployment
6. Releases – Process, Commitments
7. Risk Management
8. Communications – Presentations, Technical Writing
9. Testing – Quality Assurance, Test Strategy, User Testing
10. Security
11. Technical Proficiencies – such as, but not limited to:
 - Accessibility
 - Efficiency/Performance
 - Interoperability/Compatibility
 - Privacy
 - Reliability/Dependability
 - Scalability
 - Telemetry
 - Testability
 - Usability
 - Others not listed
12. Expectation Management – Improving Estimation, Velocity, Re-setting Expectations
13. Transition

For each team, the following points will be assessed at the given releases for each of the evaluated best practices during the course of the two semesters:

#	Best Practice	R1	R2	R3	R4	R5	R6
1	Product Management	3	7				
2	Story Mapping	3	7				
3	Project Management	3	7				
4	Teamwork		3	7			

5	DevOps		3	7		
6	Releases		3	7		
7	Risk Management		3	7		
8	Communications			3	7	
9	Testing			3	7	
10	Security				3	7
11-1	Technical Proficiencies				3	7
11-2	Technical Proficiencies				3	7
12	Expectation Management				3	7
13	Transition					3 7

The degree to which each of these items is performed will be evaluated by the tribe lead assigned to your team twice during the year. Be advised that you should ensure that you **always** conform to these practices. If you are unsure of how to put practices in place on your team, **communicate with your tribe lead** who is here to help you.

Policies

You **will never** be downgraded or negatively impacted for asking for help in Design Studio. Late work will receive an automatic 50% deduction and will not be accepted more than one week past the due date. Grades will be assigned based on the following scale:

A+	*	A	>= 93.0000	A-	>= 90.0000
B+	>= 87.0000	B	>= 83.0000	B-	>= 80.0000
C+	>= 77.0000	C	>= 73.0000	C-	>= 70.0000
D+	>= 67.0000	D	>= 63.0000	D-	>= 60.0000
F	< 60.0000				

* Grades of A+, at the discretion of the faculty and staff, will be reserved for students who demonstrate an exemplary performance on all practice and academic tasks.

Student Expectations

Students are expected to follow all sponsor, coach, and faculty and staff instructions during the completion of their assigned projects. When expectations are unclear or unknown, it is the responsibility of the student to seek clarification by consulting the appropriate parties. As students will be collaborating closely with industry professionals, all students are expected to be respectful and interact with the highest levels of professionalism in all aspects of this experience. Students are required to attend (either virtually or in-person) all scheduled class events and participate in activities.

First year students are generally expected to be team members, contributing fully to the team while gaining experience in the Design Studio expectations and process. First year students have the opportunity to provide leadership in a specific subject area or skill within a team once those areas are identified. Second year students are generally expected to be team leaders, fulfilling one of the two manager roles on the team and using their experience to lead and direct the team through Design Studio.

Team Expectations

Projects are completed over the course of the academic year. Projects will be completed using an iterative development methodology based on Scrum as outlined in course materials. Each project team, in conjunction with the sponsor, coach, and faculty and staff, is expected to operate using the processes prescribed by Design Studio. Deviation from the processes should be discussed with the program lead and tribe lead and agreed to by all parties; deviations from the Design Studio process rarely will be approved. Teams also are responsible for reporting their status to all stakeholders on at least a weekly basis. Expectations of the team will come primarily from the sponsor and be supported by class content.

Conflicts

Design Studio includes a significant amount of software development work for sponsors who pay a fee for development. The primary purpose of the course is to provide an educational opportunity for the students and an opportunity for sponsors to recruit interns and new post-graduate hires. At the same time, teams are developing intellectual property that will be owned by the sponsor at the end of the Design Studio year. As a result, Design Studio requires some conflict management policies similar to those that you would find in an internship or after you accept a job.

Your Time

What you do in your free time is up to you; however, when you are working on your project during scheduled team time, that is the sponsor's time. When in team time you should not be working on tasks that are unrelated to your Design Studio project. There is always plenty to do and team time is not homework time. If you have your own business, you should **never under any circumstances use team time or Design Studio equipment or resources for any work related to your business.**

Design Studio Resources

We will provide you with all the tools and resources necessary to successfully complete your project. This includes a space to work, a machine to develop on, and devices needed to adequately test your product. You should not use any of these resources for tasks that are not related to your Design Studio project. While team rooms are primarily for Design Studio work, other quiet uses of the space that don't impact the other Design Studio teams are acceptable (such as using it as a quiet place to do homework outside of team time). You should not use personal computers for Design Studio related development and vice versa. If you choose to do Design Studio work on a personally owned computer, be aware that your computer and all its content may be subject to review by the sponsor. Also, be aware that anything you put on a University owned computer is not private and may be reviewed by UNL faculty and staff, or a member of the sponsor organization.

Employment

To balance time spent on Design Studio with all the other activities in which students are involved, and to prevent undue influence that may disrupt that balance, working for a Design Studio sponsor during the academic year is prohibited. Working for means being paid in any way for work that is being done during either the fall or spring semester. In a limited number of cases, exceptions to this policy can be made with the understanding that **no student may work for the sponsor of the project to which they are assigned under any circumstances.** This policy does not prevent you from seeking post-graduate employment or internships or accepting any offer of employment outside of the Design Studio year; you are encouraged to take advantage of the opportunities that are created by working with sponsors in Design Studio. Requests for an exemption to this policy must be made to the Director of Design Studio in writing (email is fine).

Sponsor Meetings

Weekly team meetings with sponsors are required. They may occur during regularly scheduled team times but may not be scheduled during class time when a class session is scheduled to be held. All team members must be present either in-person or virtually.

Release Meetings

Release meetings allow teams to hand off, demonstrate, and defend the current results while providing direction and planning for project completion. The goal of each release meeting is to provide the sponsor with releasable work product. Releasable work product is some iteration of the project that the sponsor can interact with outside of the meeting, share with others, and use to provide feedback independent of interaction with the team. Teams are encouraged to plan features around making each release as functional as possible and push aggressively towards production-ready releases. All Design Studio teams will release work product in a coordinated effort across all teams and tribes. See the Canvas course calendar or Slack for release dates.

Each release will involve the Design Studio project team, the sponsor team (including the product owner), and the assigned Design Studio program lead and/or tribe lead working together to demonstrate and discuss their progress on the project to the project executive sponsors. Release meetings must be scheduled within one week (before or after) of the scheduled release date and must take place between the hours of 9am and 5pm; without permission from Design Studio, a release meeting may not be scheduled to run later than 5pm. If a sponsor requires a release meeting that will run late, it should be scheduled with your team's program lead and tribe lead. Unless otherwise requested by and prearranged with the sponsor, in-person release meetings need to take place in Kauffman.

Teams are **strongly encouraged** to schedule release meetings as early as possible after the backlog for the release has been set. Program leads and tribe leads attend multiple meetings, and it may be difficult to coordinate schedules with them. There are a limited number of conference rooms available for in-person meetings or video conference rooms for online meetings with the team members taking place in Kauffman. A calendar meeting invitation should be sent to all required meeting participants.

Online Tools

Online management and collaboration tools will be detailed during the first several weeks of class. Teams are expected to utilize Design Studio supplied project resources for collaboration, issue tracking, and source code management. We use Slack, Canvas, SharePoint, GitHub, and ZenHub for these purposes. In cases where sponsors wish to provide their own tools, teams are expected to use those and ensure that the tribe lead and program lead have access during the project.

Meeting and Team Rooms

Each team will be assigned a team room and will have access to space for the various meetings that take place in Design Studio. Please be sure to choose meeting spaces most appropriate for the size and technology needs (i.e. video conferencing) of your planned meeting. The following spaces are available to Design Studio teams this year for meetings:

- Kauffman 127A – Small Video Conference Room
- Kauffman 127B – Small Video Conference Room
- Kauffman 127C – Small Video Conference Room
- Kauffman 127D – Large Video Conference Room
- Kauffman 127E – Large Video Conference Room
- Kauffman 102 – Large Conference Room (not equipped for remote participants, use only if no 127 rooms are available and all participants are in-person)

The following spaces are available to Design Studio teams this year for working team times:

- Kauffman 103 – 3 spaces
- Kauffman 104 – 3 spaces
- Kauffman 114 – 2 spaces
- Kauffman 115 – 2 spaces
- Kauffman 116 – 2 spaces
- Kauffman 125 – 3 spaces
- Kauffman 137/A – 5 spaces

Conference rooms in Kauffman can be scheduled for sponsor meetings throughout the semester using the UNL scheduling tool – instructions will be provided.

CSE Student Resource Center (SRC)

The Computer Science Resource Center (SRC) is committed to continuing to provide support for students enrolled in Computer Science and Software engineering courses.

HOURS: Monday – Friday, 10 a.m. – 5 p.m.

LOCATION: Avery Hall Room #12 with a Canvas (<https://canvas.unl.edu/courses/95368>) page for access

The CSE Student Resource Center is staffed by student tutors and teaching assistants who are available to help you with course work. The SRC is a community hub for CSE students. Even if you don't need help, the Student Resource Center is a great place to study with your peers.

Writing Center

The Writing Center can provide you with meaningful support as you write for this class as well as every course in which you enroll. Trained undergraduate and graduate peer consultants are available to talk with you about all forms of communication. You are welcome to bring in everything from lab reports, presentations, and research papers to cover letters, application essays, and graduate theses and dissertations. Writing Center Consultants can work with you at any stage of the writing process, from brainstorming and organizing your ideas through polishing a final draft.

In 2021-22, there are **three** ways you can connect with a Consultant: **Online** (a real-time, video conversation), **eTutoring** (email feedback), and **in person**. To learn more about online options and view video tutorials, visit <https://www.unl.edu/writing/online-writing-center-services>. For information about the Writing Center's COVID-19 policies, visit <https://www.unl.edu/writing/covid-19-policies>.

Sign up for an appointment any time by visiting unl.mywconline.com. For other information about the Writing Center, please visit unl.edu/writing.

Course Policies

Required Use of Face Coverings

Students in this course must work in close physical proximity to one another for extended periods of time in order to achieve the academic goals of the course. For this reason, the Raikes School has determined that face coverings will be required in this course. If you are unwilling to comply with this requirement, please visit with your advisor about possible alternative courses that you might take in lieu of this one.

Huskers Email Required

@huskers.unl.edu accounts became the primary email for all university communications to students beginning May 20, 2019. This email will ensure that students never miss important university communications and it will be connected to MyRED and Canvas accounts. Additionally, students will use it to access university library services and software — like Adobe Creative Cloud, Matlab, VidGrid, Zoom, Office 365 and more.

A @huskers.unl.edu address has been assigned to you. This will be the primary email for all Design Studio communications emailed to students this year. To access your @huskers.unl.edu email, simply log in to mymail.unl.edu to an external site. with your My.UNL username and password. If you aren't sure what your @huskers.unl.edu email address is, you'll find it listed under the Profile tab in MyRED or MyNCTA.

It's important to check your @huskers.unl.edu email frequently — we recommend daily — so you don't miss any time-critical messages from the university, especially from the offices of Scholarships and Financial Aid, University Registrar, Admissions, and Student Accounts. Instructors will communicate with you primarily through Canvas and Slack, but may use this email, as well.

Recording of Class-Related Activity

We invite all of you to join us in actively creating and contributing to a positive, productive, and respectful classroom culture. Each student contributes to an environment that shapes the learning process. Any work and/or

communication that you are privy to as a member of this course should be treated as the intellectual property of the speaker/creator and is not to be shared outside the context of this course.

Students may not make or distribute screen captures, audio/video recordings of, or livestream, any class-related activity, including lectures and presentations, without express prior written consent from DS Director Mark Antonson or an approved accommodation from Services for Students with Disabilities. If you have (or think you may have) a disability such that you need to record or tape class-related activities, you should contact Services for Students with Disabilities. If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Failure to follow this policy on recording or distributing class-related activities may subject you to discipline under the Student Code of Conduct.

ACE Requirements

The CSCE 401H course will satisfy ACE Learning Outcome #8: Explain ethical principles, civics, and stewardship, and their importance in society. Learning opportunities for ACE8 will include:

- reading, viewing, and discussing about the required materials,
- an investigative case study,
- hands-on design of your capstone project.

In this course, you will be given opportunities to learn, be assessed on what you have learned, and have specific critical skills reinforced. Specifically, through the required materials, you will learn about events due to poor ethics, lack of civic judgments, and inadequate stewardship exercised by the environment, and how these events impacted the society. The readings and other materials have been specifically chosen for that purpose. After going through these materials, quizzes will be given. First, you will be required to learn about general ethical principles expanded into applying those principles via several real-world case studies. Second, you will be quizzed regarding the application of the ethical principles. Quizzes related to these activities will count towards 10% of your semester grade. As part of the overall course grade, you will be assessed via the quizzes, your project reports, and how you practice ethical principles in the way you document your work and acknowledge others'.

The RAIK 402H and CSCE 402H courses will satisfy ACE Learning Outcome #10: Generate a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.

Design Studio is the capstone learning experience of the Jeffrey S. Raikes School in Computer Science and Management. In Design Studio, student teams partner with sponsoring businesses and government agencies to develop real-world, software-based solutions meeting their sponsor's needs. Design Studio gives students and sponsors the ability to interact, while benefiting from the support of Raikes School faculty and facilities. Students have opportunities to demonstrate their abilities, knowledge, and capacities by completing consulting based technology projects. Throughout these consulting based projects, student teams will be required to determine and meet the expectations set forth by the project sponsors. Each project goes through all phases of the development process including initiation, execution, and closing.

Students are evaluated based on 1) individual and team improvement, 2) achievement of learning objectives, and 3) project success. Graded assignments used to assess achievement of this outcome include:

- Presentation to the sponsor at completion of several checkpoints throughout the project to assess appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.
- Iteration commitment/satisfaction forms to assess appropriate technical proficiency, synthesis, interpretation, and reflection.

- Sponsor documents to assess appropriate technical proficiency, information collection, synthesis, and interpretation.
- Project sponsor surveys and sign-off documents to assess broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.
- Project retrospective to assess information collection, synthesis, interpretation, presentation, and reflection.

CSCE Policies

Design Studio follows all Department of Computer Science and Engineering course policies:

- The CSE Department has an anonymous contact form (<http://cse.unl.edu/contact-form>) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.
- All homework assignments, quizzes, exams, etc. must be your own work. No direct collaboration with fellow students, past or current, is allowed unless otherwise stated. The Computer Science & Engineering department has an Academic Integrity Policy (<http://cse.unl.edu/academic-integrity-policy>). All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case-by-case basis and may result in a failing assignment or a failing grade for the course itself.

Accommodations of Students with Disabilities

The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let one of the instructors know immediately so that we can discuss options privately. To establish reasonable accommodations, the instructor may request that you register with Services for Students with Disabilities (SSD). If you are eligible for services and register with their office, make arrangements with one of the instructors as soon as possible to discuss your accommodations so they can be implemented in a timely manner. SSD contact information: 117 Louise Pound Hall; 402-472-3787.

Counseling and Psychological Services

UNL offers a variety of options to students to aid them in dealing with stress and adversity. Counseling and Psychological & Services (CAPS) (<https://caps.unl.edu/>); is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological and emotional well-being when issues arise. CAPS can be reached by calling 402-472-7450. Big Red Resilience & Well-Being (BRRWB) (<https://resilience.unl.edu/big-red-resilience-well-being>) provides one-on-one well-being coaching to any student who wants to enhance their well-being. Trained well-being coaches help students create and be grateful for positive experiences, practice resilience and self-compassion, and find support as they need it. BRRWB can be reached by calling 402-472-8770.

Diversity and Inclusion

The University of Nebraska-Lincoln does not discriminate on the basis of race, ethnicity, color, national origin, sex (including pregnancy), religion, age, disability, sexual orientation, gender identity, genetic information, veteran status, marital status, and/or political affiliation.

If you have questions or concerns, please contact your instructor (tribe lead). If your instructor cannot answer your question or resolve your concern, please contact the Director of Design Studio. If further escalation is needed, please contact the Executive Director of the Raikes School.

For further information, you can also contact the Institutional Equity and Compliance office, <https://www.unl.edu/equity/>.

Satisfying Thesis Requirements

UNL Honors Program Thesis

Participation in Design Studio will satisfy the requirement for an undergraduate thesis in the UNL Honors program. The use of the Design Studio experience in satisfying these requirements is subject to all policies, deadlines, and requirements of the respective program. Teams will complete two documents in the spring (Project Retrospective and Executive Summary) that will be approved by Design Studio faculty and staff and can then be submitted to the Honors program to complete the thesis requirement. Additional instructions will be provided in the spring semester with the two documents on exactly how to submit your work for Honors thesis credit.

UNL Honors Thesis:

<https://honors.unl.edu/academics/thesis>

College of Arts & Sciences Undergraduate Thesis for Degrees with Distinction

Participation in Design Studio may be used to partially satisfy requirements for an undergraduate thesis for graduation with some level of distinction in the College of Arts & Sciences. The use of the Design Studio experience in partially satisfying these requirements is subject to all policies, deadlines, and requirements of the college. Students intending to complete a thesis are advised to familiarize themselves with the program requirements.

UNL College of Arts and Sciences:

<http://cas.unl.edu/degrees-distinction>

The work of writing a thesis must be supervised by a faculty member from the college (faculty advisor). If you do not have a preferred faculty advisor for your thesis, Dr. Steve Cooper, Executive Director of the Raikes School, will serve as your faculty advisor.

Requirements for an Undergraduate Thesis for Degrees with Distinction

The thesis must be the result of independent, sustained thought and intellectual curiosity. A survey of the literature about a particular topic is not sufficient however novel it may be. Ideally, there should be a clear formulation of a problem or question, scholarly study which illuminates or addresses it supported by the work done in Design Studio, and a conclusion supported by evidence. A bibliography and reference to existing literature in the field should be included where appropriate. The thesis should be a minimum of 20 pages. Students, working closely with their two thesis advisors, may develop a project that takes a form other than a traditional thesis (e.g. they may develop video, audio, or web-based project), but the research, knowledge, effort, and quality of work should be comparable to that of a thesis in the discipline.

Students should work closely with their faculty advisor as they complete their thesis. A final copy of the thesis must be submitted to their faculty advisor no later than **one month** prior to the program deadline for final thesis submission.

Exploring Graph Data Science Methods in Neo4j

Farm Credit Services of America

Design Studio Release 6

1 Introduction

In this document, we wish to explore possible methods to analyze graph-based data with a focus on interpretation. The nature of loan-provision services require some level of understanding of the underlying metadata and relevant domain knowledge to truly understand and make informed decisions. This would allow Farm Credit Services of America to better utilize NEO4J.

Hence, this proof of concept would describe the

1. Dataset used and Code
2. Description of Components and its Interpretation
3. Centrality Measures and what it means
4. Node Similarity Measures for Product Recommendation to be Explored

2 Dataset and Code

The dataset that we used for the following proof of concept is by querying the NEO4J database with the following code:

```
MATCH (c1:Customer)-[:HAS_FINANCIAL]->(f:Financial)<-[:HAS_FINANCIAL]-(c2:Customer)
MERGE (c1)-[:SHARES_FINANCIAL {id: toInteger(c1.P_CustKey)+toInteger(c2.P_CustKey)}]-(c2)
WITH
MATCH (c1:Customer)-[:SHARES_FINANCIAL]-(c2:Customer)
RETURN c1.P_CustKey as customer_one, c2.P_CustKey as customer_two
```

This connects customers that shares financials, which allows us to create a graph that relates different customers based on **shared financial ties**. This is helpful especially for understanding how risks are shared across different customers in a business unit (as we would define later).

The reason behind exploring risks and financials (as apposed to products) is because the measures introduced here are **relatively simple and interpretable measures of node importance**. There are other methods described in Section 6 where we discuss algorithms that can be used for product recommendation.

The relevant code as a Jupyter Notebook and the data file `financial_graph.csv` accompanied.

3 Components of Related Business Units

3.1 Weakly-Connected Components

To define our definition of a business unit, we first have to define a **weakly-connected component**.

Formal Definition - A weakly-connected component is a maximal subgraph of a directed graph such that for every pair of vertices u, v in the subgraph, there is an undirected path from u to v and a directed path from u to v .

Human Definition - If customer A can reach customer B somehow, A and B are in a weakly-connected component.



Figure 1: Weakly-Connected Components of Financial Graph

Figure 1 shows all the weakly-connected components of the financial graph. There are a 156 number of components, with the largest component having 15 number of nodes. This means that the percentage of nodes that are within the largest connected component of this graph is 3.59%.

This is a **relatively disconnected graph** since most social-network graphs has $\approx 99\%$ of nodes within the largest connected component. The observation should make sense since most customers who share financials are usually connected via family ties or are just a part of a bigger business operation.

3.2 Business Units as a Unit of Observation

Based on the observations stated above, we define a **business unit** as all the customers in the same weakly-connected component. Hence, there are 156 business units in this graph.

Note that business unit in this context refer to a grouping of customers based on the rule stated. To see how each customer in a business unit relate one another, we require metadata and additional domain knowledge to make any inference since customers can be connected operations or family members or other relationships.

We also use the business unit as a unit of observation since we are assuming that those who do not share financials do not pose financial risks to each other.

4 Centrality Measures & their Interpretations

Centrality measures are algorithmic methods to provide a metric to **determine the position, role and importance of a node**. There are different centrality measures that provide different understanding since the **notion of importance is dependent on what we consider important**.

4.1 Degree Centrality

Formal Definition - The number of links incident upon a node over the theoretical maximum links.

Human Definition - How many links over all possible links.

This is the simplest measure of the relative importance of a node with respect to other nodes. **A node with a high degree centrality means that it is connected to a lot of other nodes**, which gives a good measure of how important the node is.

From a risk perspective, a node with high degree centrality directly exposes risks (and distribute risks which makes the whole less risky) to the connected nodes.

4.2 Eigenvector Centrality

Formal Definition - A measure of the transitive influence of nodes where relationships originating from high-scoring nodes contribute more to the score of a node than connections from low-scoring nodes.

Human Definition - Am I connected to other highly connected nodes?

The most famous form of an algorithm that uses this measure is Google's PageRank algorithm. Nodes who are connected to other nodes that are more influential would be assign a higher score.

From a risk perspective, **a high eigenvector centrality implies the strong influence of the node since it can spread and distribute risks to a wide part of the network**. This shows the importance of a node (like the main business line/operation) but it also implies that a contagion can occur. Contagion in this context means that if this line of business fails, the whole business unit could fail.

4.3 Betweenness Centrality

Formal Definition - Betweenness centrality measures the extent to which a vertex lies on paths between other vertices. Vertices with high betweenness may have considerable influence within a network by virtue of their control over information passing between others. They are also the ones whose removal from the network will most disrupt communications between other vertices because they lie on the largest number of paths taken by messages.

Human Definition - Am I a bottleneck (or star)?

Betweenness centrality measures a different kind of importance. While it does not imply its influence over the whole network, **a high betweenness centrality means that it exerts a strong influence on its direct neighbors**.

For an example, while the Farm Credit System is not the biggest financial service provider nationwide (low degree/eigenvector centrality) but can exert significant influence on the agriculture market (high betweenness centrality), assuming that the agriculture sector rarely borrows outside of the Farm Credit System.

4.4 Putting Things Together

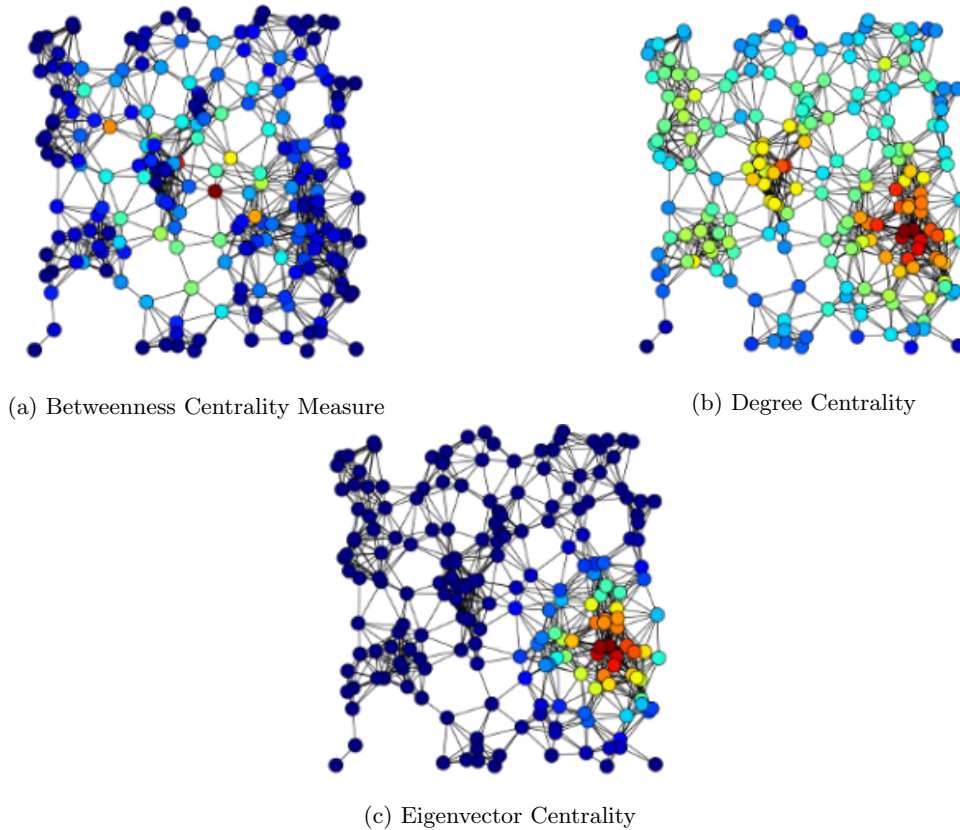


Figure 2: Different Centrality Measures on the Same Graph Reflecting Different Importance

Figure 2 shows the different centrality measures.

In Figure 2a, we see that the node with the highest betweenness centrality measure (the middle dark red node) is not considered important by other measures. However, it serves as a bottleneck (or star) that connects the two important components.

The eigenvector centrality in Figure 2c also differs from degree centrality in Figure 2b since it also accounts how connected the neighbors are. This is reflected in the cluster in the bottom-right corner where the eigenvector centrality has the largest measured value.

5 Case Study of a Business Unit

Looking at one of the business units obtained from the financial graph data. We observe some interesting patterns. Figure 3 provides a succinct visual interpretation of what the measure reflects and how we can understand this from a risk perspective as well as when considering providing a customer a loan.

5.1 Results from the Different Measures

Customer Key	Degree Centrality	Eigenvector Centrality	Betweenness Centrality
448256	0.1	0.047	0.0
5122	0.8	0.42	0.18
66023	0.5	0.309	0
66024	0.5	0.309	0
299688	0.4	0.243	0
111533	0.5	0.309	0
363503	0.3	0.170	0
61719	0.9	0.449	0.281
229913	0.6	0.275	0.215
924	0.5	0.309	0
229917	0.5	0.268	0.015

Table 1: Results from Applying the Measures (**Bolded Red Values** are the Ones In Future Discussion)

Table 1 shows the results from different centrality measure of this business unit. While Customer 61719 (in blue) has the highest measure for three centrality measures, we want to look at Customers 5122 and 229913 where they have a significant difference between degree and betweenness centrality. This would inform us on their different type of importance of the customer in the business unit.

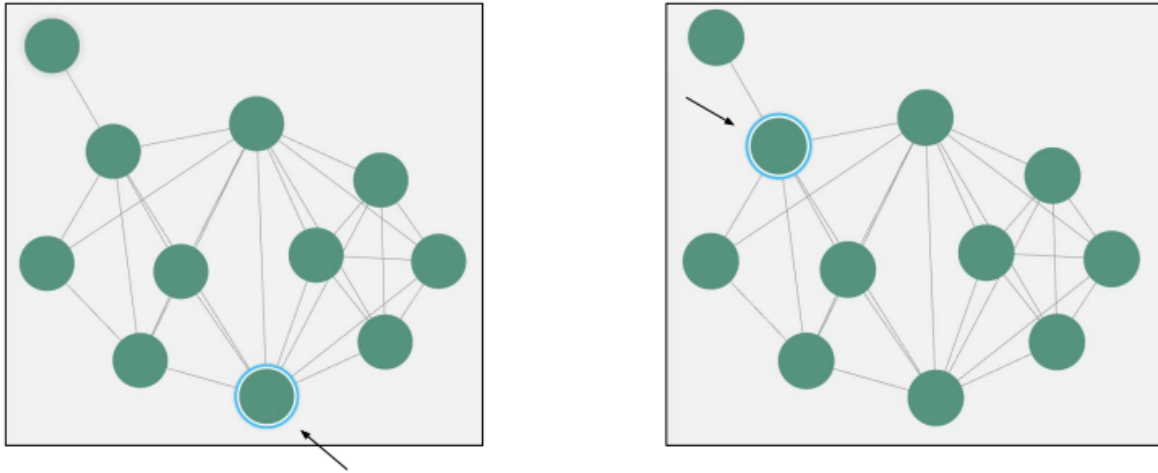
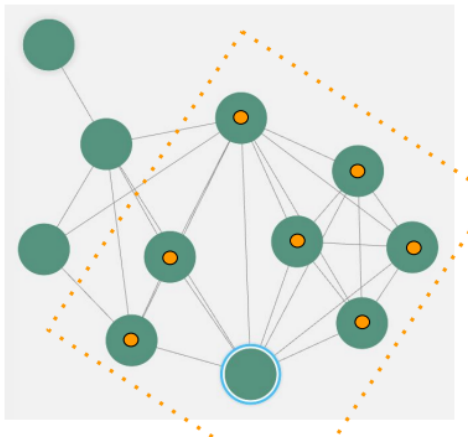


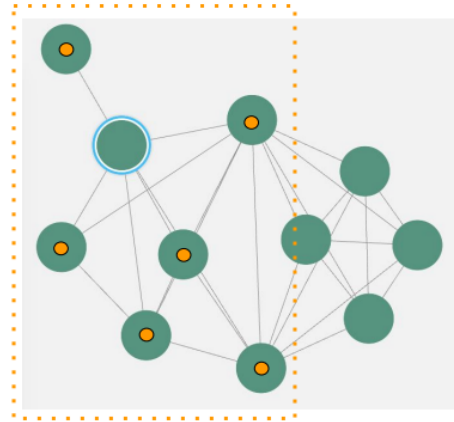
Figure 3: (Left) Customer 5122 has a higher eigenvector and degree centrality whereas (Right) Customer 229913 has a higher betweenness centrality

Note that from Figure 3, we can see that the highlighted node in the *right* frame is **connected to nodes that are less connected customers and one customer is entirely dependent on it**. On the other hand, on the *left*, the node is **connected to a lot of nodes that are strongly connected with each other**.

5.2 Interpreting and Inferring from the Measures



(a) Customer 5122



(b) Customer 229913

To make the example concrete, let us assume that this business unit raises cows and pigs, with cows being the central business. For Customer 5122 (who raises cows, or involved in raising cows like providing feed etc.) as shown in Figure 4a, it is **very well connected to other customers** who are involved in the cow raising business, and so **there is no dependence on a single customer within that group**.

On the other hand, Customer 229913 in Figure 4b might be the only supplier of feed for pigs, and the connected customers to 229913 have pigs as livestock. This means that they are **dependent on Customer 229913** though as a whole, they play a **smaller role in the business unit**.

5.2.1 Risks

With the connection of nodes being the relationship of *sharing financials with one another*, it is natural to first look at the problem from a risk perspective. In economic literature studying financial networks, sharing financial presents a risk (pun intended) and a reward:

1. **Risk** - Contagion, a very well connected and influential customer can cause other customers to fail if it fails, similar to a *domino effect*
2. **Reward** - Diversification, fluctuations in performance of a customer can be 'absorbed' by other customers

Referencing Customer 5122 in Figure 4a, with a lot of connections, **small changes** to one of the nodes **can not truly affect the whole network** since the risk is 'dissipated' across the network. In this scenario, *unsystematic risks* for Customer 5122 in Figure 4a is lowered as a result of the connection. However, any **severe shocks** on one of the nodes **can lead to a contagion**, where all other customers might face financial difficulties as a result of *systemic risks*.

On the other hand, the **likelihood of a contagion** for Customer 229913 in Figure 4b is **lower** since it has a relatively weak effect on the whole network. However, Customer 229913 **exposes a lot of unsystematic risks to its direct neighbors**, where it presents a bottleneck in the network.

5.2.2 Loan Provisions

For loan provisions, the idea behind this discussion stems from the **idea of contribution towards a financial statement** and we would be focusing on the discussion for situations like Customer 229913 in Figure 4b.

Customer 229913 has a 'star' shape within the network. Our intuition suggests that such patterns could arise if **Customer 229913 has a very good financial standing**, whereas the other customers that are

connected are simply **using its reputation to obtain more favorable loans**. This is analogous to Enron's financial statements and its 'special purpose entities' (*to be clear we are not accusing anyone of committing fraud*).

In the event that 'star' shaped financial relationships exists, it suggests that a **closer inspection in the individual contributions to the financial statement** has to be made to ensure that Farm Credit Services of America is not exposed to unexpected risks.

6 Node Similarity Measures

While we mostly described centrality measures and their interpretations, Machine Learning with graphs has other huge developments particularly in the unsupervised and semi-supervised methods. This is important because these developments would be able to help Farm Credit Services of America **recommend products to potential customers** based on certain characteristics that are important.

6.1 State of the Art Algorithms

A huge stride in Graph-based ML has been the ability to **measure the 'similarity' of nodes using unsupervised techniques**. The most famous of which are **DeepWalk, Node2Vec and Graph Neural Networks (and their spin-off architectures)**.

The three methods mentioned all **define the notion of node similarity based on shared neighbors** (nodes that are connected to one another, and on and on). They then **map this similarity to a vector space** where similarity measure such as the dot product or sin function can be applied

6.2 Collaborative Filtering

Node similarity is often used in the context of collaborative filtering, where we **recommend products to customers with some shared characteristics and connections**. In the context of Farm Credit Services of America, this means product recommendation and looking for potential leads.

6.3 Weaknesses of Such Methods

Although these methods have been successfully applied, they face the **weakness in interpretability**. This however, should not be too much of a problem for product recommendations since some recommendation is better than none at all. Furthermore, the mapping of vectors are based on similarity measures, which might not be applicable to measures of risks associations as discussed in the previous section.

CSE 423/823 Design and Analysis of Algorithms

Spring 2021

General Information

Synchronized Conferencing:

Zoom meeting ID: 928 3248 6295; passcode: 423823

Office hours of instructor and TA will use the same ID.

Instructor:

Juan Cui

Avery Hall, 122E

Phone: 402-472-5023

Email: jcui@unl.edu

Office Hours: Tue 11:00-12:00am & Thu 3:30-4:30pm via Zoom

Teaching Assistant:

Maria Salome Perez Rosero <mperezrosero2@huskers.unl.edu>

Office Hours: Fri 9:30 – 11:30 am (or by appointment) via Zoom

Lectures:

Tue/Thu 9:30 am – 10:45 am, meet via Zoom. Lecture slides with audio recording will be posted on Canvas.

Required Text:

Data Mining: Introduction to Algorithms, Third Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, MIT Press, 2009.

Recommended Text:

Introduction to the Theory of Computation, Third Edition by Michael Sipser, Cengage Learning, 2012.

Introduction to the Design and Analysis of Algorithms, Third Edition by Anany Levitin, Pearson, 2011.

Prerequisites:

A grade of "P" or "C" or better in CSCE 310, CSCE 310H, CSCE 311, SOFT 260, SOFT 260H or RAIK 283H

Communication:

A new means of communication for this course is Piazza, an online forum system designed for college courses, in addition to Canvas.

Sign up link: <http://piazza.com/unl/spring2021/cse423823>

With Piazza you can ask questions anonymously, remain anonymous to your classmates, or choose to be identified. Using this open forum system, the entire class benefits from the instructor and TA responses. In addition, you and other students can also answer each other's questions (again you

may choose to remain anonymous or identify yourself to the instructors or everyone). You may still email the instructor or TAs, but more than likely you will be redirected to Piazza for help. You are encouraged to check in and respond to the questions posed by other students. Extra credit will be given to students who actively participate in this activity, particularly to students who answer the questions posed by a student.

Course Overview

Algorithms for problems such as sorting, graph reachability, matching, etc. are at the heart of almost all real-life computer applications. Efficiency and correctness of these algorithms directly influence the overall efficiency and correctness of the applications that use them. Hence it is important to come up with very fast and correct solutions to these basic computational problems. As the name suggests, this course has mainly two aspects: designing algorithms for basic computational problems and analyzing their correctness and efficiency. These two aspects go hand in hand with each other.

Course Bulletin Description

Mathematical preliminaries. Strategies for algorithm design, including divide-and-conquer, greedy, dynamic programming and backtracking. Mathematical analysis of algorithms. Introduction to NP-Completeness theory, including the classes P and NP, polynomial transformations and NP-complete problems.

Course Learning Objectives

At the end of this course, students will be able to:

- Understand main concepts of algorithms
- Implement and apply algorithms
- Design and integrate algorithms
- Analyze correctness and complexity of algorithms

Tentative Course Schedule (All Information Subject to Change):

Week	Date (week of)	Topic	Textbook Coverage
1	1/25	Introduction	n/a
2	2/01	Median and order statistics	Chapter 9
3	2/08	Dynamic programming and Greedy algorithms	Chapter 15 & 16
4	2/15	Elementary graph algorithms	Chapter 22
5	2/22	Minimum span tree	Chapter 23
6	3/01	Single source shortest paths	Chapter 24
7	3/08	Single source shortest paths	Chapter 24
8	3/15	All-pairs shortest paths	Chapter 25

9	3/22	All-pairs shortest paths	Chapter 25
10	3/29	Maximum flow	Chapter 26
11	4/05	Maximum flow	Chapter 26
12	4/12	NP-completeness	Chapter 34
13	4/19	Approximation Algorithms	Chapter 35
14	4/26	Course summary	n/a
Final	Exam 10:00 a.m. to 12:00 p.m. Tuesday, May 4		

Course Elements

Homework

General Guidelines

- *Homework will be assigned and announced. Due date will be clearly stated. Hard copies will not be collected during this semester.*
- *A majority of points will be awarded to the PROCESS of getting to the solution and not on the final answer.*

Format Guidelines

- *Your name should appear on each page.*
- *The pages of your homework (per assignment) must submitted as a single PDF format.*
- *You may include handwriting or pictures as long as they can be assembled into the single PDF file. If you copy and paste some online images or published resources, please refer to the website or resource name to avoid copy right violation.*
- *Poor quality of the document will lead to poor grades if critical steps or answers cannot be recognized. No further argument is allowed.*
- *Final answers must include correct units for full credit when applicable.*

Group Homework Policy

- *You are encouraged to work on your homework assignments in groups, and each student must turn in their homework by themselves or by their teammate (one copy for one team is enough).*
 - *If you do choose to work with a group, the names of the group members should be provided with the homework assignment.*
 - *If you choose to work on your own, please better notify "NO teammates".*
- *Teammates names cannot be added or changed after the homework is submitted and past the deadline. If your name is missed or duplicated and thus your homework becomes invalid, you can still submit a late homework with 50% total points deduction next day.*
- *The team size can be 1-4 students. Each team should consist of only students from CSE 423 or from CSE 823.*
- *The team can be changed dynamically for each homework. You can propose names for future homework, otherwise we can randomly assign.*

Late Homework Policy

- *If a late due date is necessary, permission from the instructor is required.*
- *Late homework will have 50% deducted from the grade per school day.*

- *Under no circumstances will homework assignments be accepted after graded homework has been returned and solutions have been posted on canvas.*
- *Homework turned in late will not always be returned at the same time as the homework turned in on time.*

Quizzes:

Multiple choices or simple questions are provided on Canvas directly. Every student should be responsible to finish it before deadline. The quiz has a time limit.

Exam:

- *See “general guidelines” and “format guidelines” for “homework”.*
- *Due date is stricter. No late exam is allowed.*
- *Make-up exams are not always available unless there is a compelling and documented reason that you miss the exam. In absence of sufficient justification for missing an exam, zero will be assigned for that exam. Whenever possible, notify the instructor before you must miss an exam.*

Communication Expectations:

Expectations for Behavior

- *Students are expected to display tolerance and respect in all communication.*
- *For this class to be effective, you must be an active learner.*
- *Students should learn from each other with respect and dignity, in order to benefit from group studies.*

Grading

Grades for the above course elements will be weighted as follows:

Classroom Participation	0%
Homework	30%
Quiz	25%
Mid-term	20%
Final Exam	25%

Grading Scale:

Numerical score or class curved % (whichever is higher)	Letter grade
97-100	A+

93-97	A
90-93	A-
87-90	B+
83-87	B
80-83	B-
77-80	C+
73-77	C
70-73	C-
67-70	D+
63-67	D
60-63	D-
<60	F

Attendance Policy

n/a

Accommodations for Students with Disabilities

It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodations to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 232 Canfield Administration, 472- 3787 voice or TTY.

UNL CSE Department Policies

It is CSE Department policy that all students in CSE courses are expected to regularly check their email so they do not miss important announcements.

The CSE Department's Student Resource Center is located in Avery 12:
<http://cse.unl.edu/src>. The office hours by the TAs are held there.

The CSE Department has an anonymous contact form that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.

Academic Misconduct

All homework assignments, programs, and exams must represent your own work unless otherwise stated. No collaboration with fellow students, past or current, is allowed unless otherwise permitted on specific assignments or problems. The Department of Computer Science & Engineering has an Academic Integrity Policy. All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis

Technology Policy

Technology is allowed in class only as it directly supports class learning and announced by professor. General internet surfing, texting, e-mailing, working on other coursework or personal items is not appropriate or acceptable. Out of respect for the class, students are expected to silence or turn off their phones and other notification devices during class and to refrain from texting, calling or using electronics, except as a part of classwork. If any of these behaviors occur during the class meetings the student will be asked to leave the class and will be counted as absent. This is also applicable to online zoom meetings.

Academic Honesty Policy

Student Code of Conduct, Section B. Conduct – rules and regulations, 1. Acts of Academic Dishonesty

Academic integrity is of the utmost importance at Nebraska. Be sure you understand expectations of you and your academic work. View the complete list of academic dishonesty violations in the Student Code of Conduct, specifically Article III: Proscribed Conduct, Section B. Conduct – Rules and Regulations, 1. Acts of Academic Dishonesty. For more information, please visit <https://studentconduct.unl.edu/>. If you are unsure what counts as academic dishonesty in this course, please visit me during office hours. The first instance of academic dishonesty will result in a score of zero for the assignment or exam. The second incidence of academic dishonesty will result in a failing grade for the course

<https://studentconduct.unl.edu/student-code-conduct>

Diversity and Inclusion Statement

It is my intent that students from all diverse backgrounds and perspectives be well-served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. I would like to create a learning environment for my students that supports thoughts, perspectives, and experiences, and honors your identities (including gender, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, culture, etc.). If you have any concerns, please feel free to contact me.

<https://diversity.unl.edu/student-resources>

Emergency Procedures

City Campus Emergency Information:

Please follow the link below for UNL's Emergency Preparedness Resources. Sign up for the UNL Alert service under the “**Stay Connected**” portion of the web-page.

Fire and Medical Emergencies: 911

University Police: 402-472-2222

Twitter: @NebPrepare

E-Mail: preparedness@unl.edu

<https://emergency.unl.edu/>

Scott Campus Emergency Information:

A number of resources are available for distributing and receiving critical information and instructions during an emergency.

- All-campus email
- [UNO Alert text message and email alerts](#)
- Posts on UNO's official [Facebook](#) and [Twitter](#) accounts
- Emergency banner on the [UNO homepage](#)
- Overhead pages and indicator lights on campus
- Emergency information line (402.554.2255)
- Media outlets

<https://www.unomaha.edu/emergency/index.php>

Student Resource Information

(please double check, some of them may only have online services at this moment)

Well-Being (Home Campus – Lincoln Campus):

UNL offers a variety of options to students to aid them in dealing with stress and adversity.

- Counseling and Psychological Services (CAPS) is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological and emotional well-being when issues arise.
 - CAPS can be reached by calling 402-472-7450.
- Big Red Resilience & Well-Being (BRRWB) provides one-on-one well-being coaching to any student who wants to enhance their well-being. Trained well-being coaches help students create and be grateful for positive experiences, practice resilience and self-compassion, and find support as they need it.
 - BRRWB can be reached by calling 402-472-8770.

Well-Being (Home Campus – Scott Campus):

UNO offers a variety of options to students to aid them in dealing with stress and adversity.

- Counseling and Psychological Services (CAPS) is dedicated to working with students to provide that can assist with challenges that have impacted their overall well-being. These could include adjusting to life events, relationship issues and mental health changes. Counseling staff work closely with Nebraska Medicine Health Services to provide on-campus referrals and collaboration with their services.
 - CAPS can be reached by calling 402-554-2409.

Math Resource Center:

The Mathematics Resource Center (MRC) is a free tutoring service and is the primary facility for undergraduate students who are enrolled in an have questions related to any precalculus or calculus course offered by the department. The Center also provides an excellent location for students to meet and work together on assignments or group projects.

- No appointments or reservations are needed.
- Staffed by Graduate Teaching Assistants and by undergraduate math majors hired for the center.
- Ask for assistance with math courses 100A, 101, 102, 103, 104, 106, 107, and 107H.

<https://www.math.unl.edu/resources/undergraduate/mrc>

The Writing Center:

At the Writing Center, our undergraduate and graduate Writing Consultants work with writers at all levels, from all disciplines, at all stages of the writing process. All members of the UNL community (students, faculty, and staff) are welcome.

Whether you are brainstorming or organizing ideas or polishing a final draft, we look forward to discussing your writing with you.

All forms of communication are welcome, from essays, lab reports, research papers, and journal articles to presentations, cover letters, personal statements, and theses/dissertations.

<https://www.unl.edu/writing>

Engineering Study Shop:

The College of Engineering provides FREE walk-in tutoring services for all engineering students! The courses and areas-of-study that the Engineering Study Stop tutors are able to assist with include: Math, Physics, Chemistry, Mechanical Engineering, and a variety of other engineering-related coursework. Click on the button below to meet our tutors and learn more about which courses they are able to assist you with. Study Stop allows you to study with other engineering students and trained tutors to gain more confidence and a better understanding of your course material... No reservations needed!

<https://engineering.unl.edu/current-students/study-stop-city-campus/>

ADA and Accommodation:

Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements.

To receive accommodation services, students must be registered with the Services for Students

with Disabilities (SSD) office, 232 Canfield Administration, 472-3787 voice or TTY.

Office of Services for Students with Disabilities:

The University of Nebraska-Lincoln is committed to ensuring equal access to curricular and co-curricular opportunities for students with disabilities. Providing a range of services, SSD implements reasonable accommodations for students with disabilities and offers students the opportunity to contribute and participate in the diverse campus experience at the University of Nebraska–Lincoln. This site is intended to provide you with the information you need to enjoy your life as a University of Nebraska student. <http://www.unl.edu/ssd/home>.

Syllabus - CSCE 451/851 Operating Systems Principles

Syllabus - CSCE 451/851

Operating Systems Principles

Department of Computer Science and Engineering
University of Nebraska – Lincoln
Spring 2021

Instructor:

Justin Bradley 290 Schorr
Office hours: By appointment via Zoom
Email: justin.bradley@unl.edu
Phone: (402) 472-5072

TAs:

Minh Vu
Office hours: Mondays 9:30-11:30 via Zoom
Email: minh.vu@huskers.unl.edu

Brennan Rhoadarmer
Office hours: Wednesdays and Thursdays 17:30-19:30 via Zoom
Email: brhoadarmer@huskers.unl.edu

1 Course Overview

This course will introduce students to key concepts of modern operating systems. These concepts include system organization, processes, threads, process management, process and thread synchronization, memory management, and storage management.

Topics covered: Organization and structure of operating systems. Processes and concurrent programming. Inter-process communication. Process synchronization using reads/writes, semaphores, and monitors. CPU scheduling. Deadlocks including detection, avoidance, prevention and recovery. Intro to real-time systems. Memory organization and management including paging, segmentation, virtual memory, and page replacement algorithms. File system concepts and structure. Protection and security. Topics are conveyed through lectures and a series of hands on programming assignments.

2 Course Requirements

Prerequisites: A grade of “P” or “C” or better in CSCE 230, CSCE 230H or CSCE 231 and CSCE 310, CSCE 310H, CSCE 311, SOFT 260, SOFT 260H or RAIK 283H.

Text: Bic, Lubomir. [*Operating Systems Principles*](#). zyBooks, 2020, zyBook ISBN: 978-1-394-06717-6

3 Class Details

The two sections of the course will be identical with the (minor) differences indicated below.

3.1 Section 001 (CSCE 451/851-001)

This is a “Web Conferencing” version of the course. Asynchronous video lectures and readings in the book will be used to introduce concepts to students. Weekly synchronous, live Zoom lectures will be used to supplement those lectures with a focus on successfully completing the programming assignments. This allows students in this section to directly interact with me, ask questions, and receive feedback.

Class: Tuesdays 09:30 - 10:45 (all other lectures asynchronous)

Final exam: 00:00-23:59 Tuesday, May 4, 2021

3.2 Section 700 (CSCE 451/851-700)

This is an “Online” version of the course. Like Section 001 above, asynchronous video lectures and readings in the book will be used to introduce concepts to students. The weekly live Zoom lectures from Section 001 will be recorded and disseminated in this section.

Class: Asynchronous video lectures

Final exam: 00:00-23:59 Thursday, May 6, 2021

4 Website, Communication, and Other Logistics

4.1 Canvas

We will use Canvas to disseminate materials, host the class schedule, maintain all links, etc. To simplify everyone’s life the front page of the course is a module containing two pages:

1. **Announcements:** this will be a regularly updated, running list of announcements. You need to check this regularly. Canvas, depending on your settings, may send you a notification that the page has been updated. You’ll then need to go check it. I do it this way to reduce the number of Canvas announcement emails you receive. But it’s on you to check the announcements so you’re informed!
2. **Main Schedule:** this page holds a weeky-by-week schedule of the topics, resources, links, assignments, etc. Look here to find out what you should do, and where to get materials and resources. This will be updated over the course of the semester as topics, resources, and assignments are updated.

4.2 Other Resources

Other external materials, links, etc. will be used as follows (all links to these will be in the “Main Schedule” page in Canvas:

- **Piazza** will be used for all online discussion of homework, programming assignments, and logistics.
- **YouTube** will be used to host lecture videos.
- **Microsoft OneDrive** will be used to host programming assignments and other handouts.
- **Zoom** will be used for synchronous lectures (section 001 only).
- **zyBooks** will be used as our book. Readings and homework assignments will be given from this book, and homework assignments, based on the readings, will be given. You will have to subscribe to the book (it is much cheaper than the previous book).
- **CSE Handin** will be used to turn in programming assignments.

5 Policies

5.1 Grading Policies

This course will consist of 6 (graded) programming assignments, weekly reading check assignments, and a final exam. The grade breakdown will be:

- Programming Assignments: 60%
 - PA0 – 5%
 - PA1 – 11%
 - PA2 – 11%
 - PA3 – 11%
 - PA4 – 11%
 - PA5 – 11%
- Exams: 20%
 - Final Exam: 20%
- Homework - 20%
 - 7 reading/homework assignments – $20\%/7 = \sim 2.86\%$
- Extra Credit:
 - Evaluation: 2% bump to final percentage (only if >80% of class fills out evaluation)

5.1.1 Late Work Policy

Late work will **NOT** be accepted. I *will* make exceptions for [UNL-approved reasons](#). I *may* make exceptions for other reasons **if** you discuss it with me before the deadline.

Explanation: I have tried several variations on accepting late work. Accepting late work means more work for me and the TAs, and this class already has a high grading burden. My anecdotal evidence suggests the rate of students turning things in is about the same. Not accepting late work is simplest and encourages students to get started on assignments earlier.

5.1.2 Differences between 451 and 851

CSCE 851 students will take a different, more comprehensive final exam. Otherwise, assignments between 451 and 851 are identical.

5.1.3 A very important note on grading of programming assignments:

- Programming assignments will be graded on a 100 point scale.
- **Your program *must* compile and execute on the CSE servers (cse.unl.edu) - else you get 0/100. Make sure you check this prior to handing in.**
- All programming assignments must be handed into [CSE web handin](#) - else you get 0/100.
- Each programming assignment will indicate precisely how it will be scored. **Most** of the time points are gained by passing the test suites. Test suites are often broken into different parts so that you can get at least partial credit for partial implementations. You won't get any points for tests that you don't pass. This means there's no such thing as "being close" or "almost works."
- Despite the previous point, I also reserve the right to take off points for poor programming practices: poor commenting, function names, variable names, magic numbers, etc.

5.1.4 Explanations

Students have a variety of reactions to these policies. Here are some common ones, and associated explanations:

- “*That’s stupid!*”: When you develop code in industry you don’t get “credit” if you break the nightly build (i.e., your code doesn’t compile), or your program simply doesn’t do what it is supposed to do (i.e., test suite fails). Even so, most PAs have a suite of tests that allow you to get partial credit for implementing as much as you can.
- “*That doesn’t accurately reflect what I learned!*”: I’m in the business of preparing our students to be strong employees. As 4th year undergraduates, or graduate students, good programming practice is simply an expectation of a computer scientist. Writing programs that meet specifications is the *bare minimum* any computer scientist can do. In most PAs you will have access to the test suites or an approximate set of test suites, so it should be straightforward to test all your code, and make sure it compiles and runs on the CSE servers.
- “*Why not partial credit?*” or “*I was so close...*” You will get the indicated points for passing various tests within the test suite. If your code doesn’t pass a test you get 0 points for that particular test. There are several reasons for this:
 - I don’t know how to give partial credit. If the output of your code doesn’t match the required output, I (or the TA) would need to look at your code and make an assessment of how “close” you were to being correct. I’ve no idea how to do that fairly or efficiently.
 - I don’t have the time to do this sort of partial credit grading and I won’t ask the TAs to spend their time looking through and deciphering your code.
 - To my knowledge, there is no agreed upon metric for determining how close code is to being “correct.”

Bottom line: write *good* code that fulfills the specifications (i.e., passes all the tests) and you’ll get all the points.

5.1.5 Grading Appeals Policy

Sometimes you do your very best and something doesn't go right. You forgot a “\n” character in a printed statement, etc. In these minor cases, where you were *very* close, please write me an “appeals” email describing the scenario, submitting your new code, and I will, at my discretion, re-grade the assignment with a 10% penalty. All appeals must be filed within a week of grades being posted.

5.1.6 Grade Assignment

Grades will then be assigned using the following table:

Grade	Percent
A+	[98-10]
A	[93-98)
A-	[90-93)
B+	[88-90)
B	[83-88)
B-	[80-83)
C+	[78-80)
C	[73-78)
C-	[70-73)
D+	[68-70)
D	[63-68)
D-	[60-63)
F	[0-60)

5.2 Class Policies

1. Attendance is not mandatory, even at live sessions, but you are responsible for anything that transpires during class or a live

conferencing session. **Please** be on time to class!! If you're consistently late and/or disruptive in class I reserve the right to dock you points on an assignment.

2. Put your name, course, and section on everything you turn in. Sometimes we print things out for grading. Also, this is generally just good practice.
3. Exchange of ideas and techniques is **highly** encouraged but **your work must be your own**. If someone helps you, please give them credit in your code (even if it's the TA). Myself and the TAs will follow up with suspicions of academic dishonesty in accordance with department and university policy (see below).
4. Students have one week from time of grade posting to challenge a grade.
5. If you have *technical content* questions please follow this order of operations for getting help:
 - a. Look on Piazza and check for similar questions.
 - b. Post your question on Piazza – then everyone benefits from the answer!
 - c. Check the internet for similar questions or concepts.
 - d. Attend the TAs' office hours.
 - e. Email the TA asking your question or requesting a Zoom session.
 - f. Email me asking your question or requesting a Zoom session.
6. If you have *personal questions* about logistics or similar please skip straight to contacting me by email explaining the situation and your question.
7. Please use professional email communication. A salutation, description, request, closing, and concise subject line are appropriate for professional communication.
8. I'm particularly concerned with the mental and emotional distress our current circumstances could have on each of us. Don't hesitate to reach

out and let me know how things are going. Large doses of patience are in order during the global pandemic.

5.3 CSE Policies

1. The CSE Department has an [anonymous contact form](#) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.
2. CSE Department policy dictates that students in CSE courses are expected to regularly check their email so they do not miss important announcements.
3. Consider the [Student Resource Center](#) in Avery 12 if myself or the TAs are not available, or you otherwise need help.
4. All homework assignments, quizzes, exams, etc. must be your own work. No direct collaboration with fellow students, past or current, is allowed unless otherwise stated. The Computer Science & Engineering department has an [Academic Integrity Policy](#). All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself. The UNL College of Engineering also has an [academic integrity policy](#) you should read and understand.

5.4 UNL Policies

1. The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let me know immediately so that we can discuss options privately. To establish reasonable accommodations, I may request that you register with Services for Students with Disabilities (SSD). If you are eligible for services and register with their office,

make arrangements with me as soon as possible to discuss your accommodations so they can be implemented in a timely manner. SSD contact information: 232 Canfield Admin. Bldg.; 402-472-3787.

2. Academic honesty is essential to the existence and integrity of an academic institution. The responsibility for maintaining that integrity is shared by all members of the academic community. The University's Student Code of Conduct addresses academic dishonesty. Students who commit acts of academic dishonesty are subject to disciplinary action and are granted due process and the right to appeal any decision.
3. UNL offers a variety of options to students to aid them in dealing with stress and adversity. [Counseling and Psychological Services \(CAPS\)](#) is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological and emotional well-being when issues arise. CAPS can be reached by calling 402-472-7450. [Big Red Resilience & Well-Being](#) provides one-on-one well-being coaching to any student who wants to enhance their well-being. Trained well-being coaches help students create and be grateful for positive experiences, practice resilience and self-compassion, and find support as they need it. BRRWB can be reached by calling 402-472-8770.

CSCE 474/874: Introduction to Data Mining

Spring 2021

General Information

Instructor: [Ashok Samal](#)

Avery Hall, Room 359

Phone: 472-2217

Email: samal@cse.unl.edu

Office Hours: TR (12:15 – 1:15)

Teaching Assistant: Atharva Tendle (atharva.tendle@huskers.unl.edu).

Office Hours: M/T 10:00 am – 11:00 am (<https://unl.zoom.us/j/96441158653>)

Lectures: TR 11:00 am – 12:15 pm ([Web Conferencing](#))

Text: Introduction to Data Mining (Second Edition) by Tan, Steinbach, Karpatne, and Kumar, Pearson.

Prerequisites: CSCE310, STAT 380 or permission or equivalent.

1. **Mastery** of a high-level programming language.
2. **Familiarity** with calculus and statistics.
3. **Exposure** to linear algebra.

Course Objectives: Data mining is the process of automatically finding useful *hidden* patterns in data. The objective of data mining is to use discovered patterns to help *explain* current behaviors or to *predict* the future outcomes. Upon completing this course, students will understand:

- **Mastery** of basic data mining algorithms, including association rule mining, clustering, prediction and classification.
- **Familiarity** with data warehousing and its applications.
- **Familiarity** with role of data mining in knowledge discovery.
- **Familiarity** with data preprocessing and reduction techniques.
- **Exposure** to applications of data mining to practical problems.

Communication

To keep our inboxes clean and to ensure emails don't get lost, all communication should be done using Piazza. Piazza is an online communication tool where instructors can post notes, and students can ask questions. The Piazza site for this course is located at piazza.com/unl/spring2021/csce474874. You should have received an invitation to join. If you have not, you may sign yourself up at: piazza.com/unl/spring2021/csce474874.

Again, any general or homework questions you might have, or if you have questions about a graded assignment, please use Piazza. When you post a question, the instructor and the TA will see your question and respond as needed. You are encouraged to check-in and respond to the questions posed by other students. Extra credit will be given to students who actively participate in this activity, particularly to students who answer the questions posed by a student.

Please remember to use professional language and letter salutations in your emails to the instructor and the TA (no “Hey” or any kind of text jargons).

Assignments

There will be four assignments for this course. Together they will count for 40% of the overall grade. The assignments are due at the class time on the indicated dates. We will use a flexible, slip date system for the assignments. Each student is given an automatic extension of four (4) calendar days. You can use these on any assignment(s) during the course. However, the total number of late days cannot exceed 4. Slack days cannot be used for the project. After you have used your “late” days, if you submit an assignment late, you will not get any credit for it. Then, it is better to submit a partially completed homework than a late one. If you have a special reason for being late, get permission well ahead of the due date.

The assignments will be done in pairs and your partner will be randomly assigned and will change for each assignment.

You are required to submit the assignment online on Canvas. Do not turn in any hard copies. You are required to submit all source code, executable, report, etc.

While it can be informative to discuss the assignments with others, you should write your final programs independently. Make sure that all your files are protected. You are also responsible if somebody copies your files and hands them in.

Attendance

Attendance in class is required. 5% of the overall grade is reserved for class attendance and participation.

Tests and Quizzes

There will be no exams in this class. However, there will be periodic quizzes (both announced and unannounced). 20% of the overall grade is devoted to quizzes.

ICDM/KDD Session Summary and Presentation

Each student will be required to study a paper (from a selected set) presented either at IEEE International Conference on Data Mining (ICDM) or ACM Conference on Knowledge

Discovery and Data Mining (KDD), two leading conferences in data mining and present a summary to the class. This will be worth 5% of the overall grade.

Group Project

Students are required to work on a group project on some aspect of data mining. You must select a project from a list assigned in the class. Each team should consist of three to four students. All project documents will be kept online in Blackboard. The group project that will be worth 30% of the overall grade.

Students with Disabilities

Students with disabilities, including mental health disabilities like depression and anxiety, are encouraged to contact the instructor for a confidential discussion of their individual academic accommodation needs. The University of Nebraska-Lincoln provides individualized accommodations to students with documented disabilities that may affect their ability to participate in course activities fully or meet course requirements. Students must register with the Services for Students with Disabilities (SSD) office, 232 Canfield Administration, 472-3787 voice or TTY to receive the accommodations.

UNL CSE Department Policies

The CSE Department expects all students in CSE courses to check their email regularly, so they do not miss important announcements.

The CSE Department's Student Resource Center is located in Avery 12: <http://cse.unl.edu/src>. The office hours by the TAs are also listed there.

The CSE Department has [an anonymous contact form](#) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.

Academic Misconduct

All homework assignments, quizzes, exams, etc. must be your own work. No direct collaboration with fellow students, past or current, is allowed unless otherwise stated. The Computer Science & Engineering department has an [Academic Integrity Policy](#). All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case by case basis and may result in a failing assignment or a failing grade for the course itself.

Those who share their code, and those who copy other's code will be penalized in the same way; both parties will be considered to have plagiarized.

The penalty for the violations are as follows:

- 1st violation — award a zero in homework or exam
- 2nd violation — award an F in the course
- 3rd violation — recommend the student be dismissed from the program

Final Grade

The distribution of points for the course is as follows.

Category	Total Points
Participation	50
ICDM/KDD Presentation	50
Quizzes	200
Homework	400
Project	300
Total	1000

The final grade will be based on the following scale.

≥ 95%: A	≥ 87%: B+	≥ 77%: C+	≥ 67%: D+	
≥ 90%: A-	≥ 84%: B	≥ 74%: C	≥ 64%: D	< 60%: F
	≥ 80%: B-	≥ 70%: C-	≥ 60%: D-	

CSCE 478/878 – Introduction to Machine Learning

Fall 2021

COVID-19: Policy on Face Coverings

Given current Center for Disease Control (CDC) and Lancaster county health guidance, and the current transmission level of COVID-19 in our community, I respectfully request that you join me in wearing a face covering during our classes.

Student conduct expectations: <https://studentconduct.unl.edu/covid-19-related-expectations-and-non-compliance>

Is this course suitable for you?

The Introduction to ML course (CSCE 478/878) is designed mainly for Computer Science and Engineering students, thus provides a rigorous theoretical treatment of the topics. It requires implementing various ML algorithms from scratch (using vanilla python and its scientific non-ML libraries). Students must have strong programming skills in Python as well as a background in probability & statistics, linear algebra, calculus, and algorithm complexity analysis. The assignments are programming-heavy and time-consuming. No support will be provided on coding or debugging for the programming assignments.

If you are a non-CS student and are only interested in how to use ML for practical applications, then this course is not suitable for you. You should instead do the “Practical Machine Learning” course, which is designed for Engineering students. It emphasizes less on the theory and mathematical foundation of the ML models as well as implementation of the ML algorithms. Also, the assignments are required to be done using ML APIs (e.g., Scikit-Learn, TensorFlow). So, a moderate level of programming skill in Python is enough. This course is offered in Spring.

Course Info

Instructor	Dr. M. R. Hasan hasan@unl.edu Avery Hall 123E https://cse.unl.edu/~hasan/
Office Hours	See Canvas
Textbook	See Canvas
Teaching Assistants	See Canvas

Prerequisite

Probability & Statistics, Linear Algebra, Calculus, Data Structure & Algorithm, Algorithm Complexity Analysis. **Strong programming skill in Python is a must.** If you don't know Python you will have to learn it independently. No help will be provided on Python programming (e.g., learning, coding, debugging).

Course Topics

This is a rough sequence of topics that we will cover in this course.

- Probabilistic Reasoning
- Linear Algebra for Machine Learning
- Classification & Regression: (Nonparametric Methods) Nearest Neighbors & K-Nearest Neighbors (K-NN)
- Classification: Performance Measures
- Kernel Density Estimator & K-NN
- Statistical Estimation of Distributions: Binomial, Bernoulli, Multinomial, Gaussian
 - Frequentist Learning (MLE)
 - Bayesian Learning (MAP)
- Regression: Linear Regression (Frequentist & Bayesian) – Closed form & Iterative Optimization Algorithm (gradient descent)
- Classification: Naïve Bayes Classifier
- Classification: Logistic Regression (Frequentist & Bayesian), Newton's Method
- Classification: Support Vector Machine; Kernel Trick
- Classification & Regression: Perceptron Algorithm & Artificial Neural Network
- Classification & Regression: Decision Tree & Random Forest
- Bagging & Boosting
- Unsupervised Learning:
 - K-means
 - Expectation Maximization (EM)
 - Gaussian Mixture Model (GMM)
- Dimensionality Reduction:
 - Principle Component Analysis (PCA)
 - Singular Value Decomposition (SVD)
 - Linear Discriminant Analysis (LDA)
- Graphical Models; Hidden Markov Model (HMM) (time permitting)
- Learning Theory (time permitting)
 - PAC
- Advanced Topics
 - Sampling Methods, MCMC, Reinforcement Learning, etc. (time permitting)

Text Resources

- Lecture slides (thorough and extensive) should provide a detailed account of the topics.

Though there is no required text for this course. However, my lectures will draw references substantially from the following three books.

- Machine Learning: A Probabilistic Perspective by Kevin P. Murphy
- Pattern Recognition and Machine Learning by Christopher M. Bishop
- Introduction to Machine learning by Ethem Alpaydin

For understanding implementation issues and hands-on insights, following text will be used:

- Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (2nd Edition, 2019) by Aurélien Géron

Following books are useful as Python refresher/introduction:

- Data Science from Scratch by Joel Grus (O'Reilly)
- Python for Data Analysis (2nd Edition) by Wes McKinney (O'Reilly)
- Python Machine Learning by Sebastian Raschka (Packt Publishing)

Statistics, Linear Algebra & Calculus:

- Advanced Engineering Mathematics by Erwin Kreyszig
- All of Statistics: A Concise Course in Statistical Inference by Larry Wasserman
- Convex Optimization by Boyd and Vandenberghe

Discussion Through Piazza

We will be using Piazza for class discussion. The system is highly catered to provide you help fast and efficiently from classmates, TAs, and myself. Rather than emailing questions to the teaching staff, I encourage you to post your questions on Piazza. Please note that posted questions or discussion will be moderated if appeared to be inappropriate or irrelevant. Also, if you have a **question for a specific instructor (me or the GTA)**, you must send an email to that person, otherwise don't expect to receive a quick response. See the signup/login link for the Piazza on the Canvas Home page.

878 Level

There will be higher expectations with regard to the quality and quantity of work for the 878 level students compared to the 478 level students. As a student in this course, you will be expected to meet or exceed those expectations, which may require more of your time and a more proactive approach to studying and greater diligence in completing assignments. In particular, each assignment contains additional requirements or activities.

Recitations

There will be weekly recitations. A set of problems will be posted prior to the recitation. During the recitation hour the GTA will be present to help you. However, **recitations are not graded and no submission is required**. You may want to use the recitation sessions and tasks as an opportunity to deepen your understanding of the Machine Learning models that we won't have time to go over in detail during the lecture.

Homework Assignments

There will be 4/5 assignments. Assignments are due at the beginning of class on the due date. Code and other relevant files must be submitted using CSE's webhandin (<http://cse.unl.edu/handin>). If there are

written portions of the homework, they should be typed. You should typeset code snippets using a monotype font (Courier for example) for readability.

In this course, the homework assignments are built on prior assignments. Therefore, it is extremely important for you to stay on task and not fall behind with assignments. You should start on assignments early and resolve any and all issues (bugs, design flaws) as soon as possible so they do not carry over to the next assignment.

Coding

There will be **no support for the coding, compilation, or debugging**. You are expected to use vanilla Python and some of its scientific libraries such as Pandas, NumPy, and Matplotlib for programming assignments. However, no Machine Learning libraries such as Scikit-Learn or TensorFlow are allowed.

Exams

There will be a midterm and a final exam. See the exam schedule on Canvas Home page (Schedule section). The questions will be multiple-choice type. The exams are open book/note/computer.

Grading

Grading will be based on homework assignments, project, and quality of presentation. Note that your course total score will be computed based on the stipulated weights (see below) associated with the following graded components. **The Canvas default total score does not reflect your weighted total score.**

Assignments	70%
Midterm Exam	15%
Final Exam	15%

Scale

Letter grades will be awarded based on the following scale. I will use an **absolute grading policy**. For example, scores within the range between 89.00 and 89.49 will be rounded to 89.00 while scores within the range between 89.50 and 89.99 will be rounded to 90.00. Therefore, a score of 89.49 will earn you a B+ grade but 89.50 will earn an A-(minus) grade.

A+	>= 97	B+	>= 87	C+	>= 77	D+	>= 67	F	<60
A	>= 93	B	>=83	C	>= 73	D	>= 63		
A-	>= 90	B-	>= 80	C-	>= 70	D-	>= 60		

Grading Policy

We make every effort to grade and return materials within one academic week of the due date. You are expected to pick up your graded material in a timely manner. If you have questions about grading or believe that points were deducted unfairly, you must first address the issue with the grader it to see if it can be resolved. Such questions should be made within a reasonable amount of time after the graded

assignment has been returned. **No further consideration will be given to any assignment a week after it has been graded and returned (regardless of whether you fail to pick it up when handed back).**

It is your responsibility to **check Canvas regularly** and report missing/incorrect entry within one week of submission of your assignment or exam.

You need to report any missing/incorrect Canvas grade entry or questions related to grading **before I post your final grade** in Canvas. **I will not take any question/complaint after the final grade is posted.**

Late Work

In general, late submission will not be allowed. Exceptions may be made in certain circumstances such as health or emergency, but you must make every effort to get prior permission. Documentation may also be required.

Homework assignments have a strict *in-class* (at the beginning) due date. Any written portions should be handed in hardcopy in class while softcopies should be handed in using webhandin as specified in individual homework assignments. The webhandin program that you will use enforces a *strict* handin time based on the CSE server's clock. Programs that are even a few seconds past the due date/time will be considered late.

It is understandable that unforeseen events may interfere with your ability to submit all homework assignments on time. As such, this course allows the following late work policy: you may hand in **any one assignment up to 7 days late**. Any submissions after 7 days will not be considered and will be given an automatic zero. These 7 "free passes" can be used on multiple assignments.

Exceptions: You *CANNOT* use a late pass for the first assignment as well as on the last assignment.

Academic Integrity

All homework assignments, programs, and exams must be your own work unless otherwise stated. No collaboration with fellow students, past or current, is allowed unless otherwise permitted on specific assignments or problems. The Computer Science & Engineering department has an Academic Integrity Policy. All students enrolled in any computer science course are bound by this policy. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case-by-case basis and may result in a failing assignment or a failing grade for the course itself. **If any student performs academic dishonesty (complete or partial), as defined by the UNL academic integrity policy, in any graded component of the course, her/his grade will be set to zero for that assignment/exam and she/he will be reported to the academic integrity committee of the CSE department.** The most recent version of the Academic Integrity Policy can be found at

<http://cse.unl.edu/academic-integrity-policy>

Communication

The best way to communicate with your Professor is through email. The Professor and teaching assistants will communicate with you either directly or through the Canvas email system. You are responsible for ensuring that the email associated with your Canvas account is up-to-date and that you are regularly

checking it. Generally speaking, you should talk to Dr. Hasan for questions about course material and lecture. You should talk to the TA about grading and homework assignments.

In addition, the Department of Computer Science & Engineering also maintains an anonymous suggestion box that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified. It is available at the following URL:

<http://cse.unl.edu/department/suggestion.php>

Finally, I will hold regular office hours and will make myself available by appointment; please email me to set one up.

Disabilities

Students with disabilities are encouraged to contact Christy Horn for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 472-3787 voice or TTY.

[CS136](#)[Home](#)[Syllabus](#)[Course Materials](#)[Resources](#)[Staff & Office Hours](#)

Syllabus

This is a class about the digital economy, specifically the interplay between economic thinking and computational thinking as it relates to electronic commerce, incentives engineering, and networked systems. Topics covered vary each year, but include a subset of:

- game theory (including algorithmic game theory),
- auctions (including internet advertising and combinatorial auctions),
- incentive compatible mechanism design (including theoretical and algorithmic approaches),
- human computation, crowd sourcing, and peer prediction,
- matching algorithms,
- trust and reputation,
- cryptoeconomics,
- networks (network formation, cascades, games on networks),
- privacy (including differential privacy, privacy-protected advertising), and
- ethical considerations.

Emphasis is given to fundamental topics, and the class involves the discussion of theoretical and empirical content, as well as algorithm design. We hope to convince you that economic considerations matter in many computational settings, and that computation matters in many economic ones.

Prerequisites

We will assume familiarity with probability theory (as taught by, e.g., Stat 110), single-variable calculus (as taught by, e.g. Math 1b), linear algebra (as taught by, e.g., Math 21b/22a/25a/AM 21b), and the basics of theoretical computer science (e.g., complexity theory, asymptotic runtime analysis, as taught by CS 120, 121, or 124). You should also be prepared to write multi-file programs in Python (and prep from a course like CS 51 can help). No background in economic theory is assumed, although students without some economics background would benefit from a course that provides exposure to the basic ideas of utility maximization and rational behavior (e.g., from an AI course). Generally, we suggest students have completed *one of* Computer Science 181, Computer Science 182, Economics 1011a, Economics 1052, Economics 1071, or Economics 2056.

Book

The instructor is developing a textbook with Sven Seuken (U. Zurich), hopefully to be published by Cambridge University Press in early 2024, and we will follow a draft of the book in class. Chapters will be distributed during the course. [Networks, Crowds, and Markets: Reasoning About a Highly Connected World](#), by David Easley and Jon Kleinberg, also provides a good introduction to many topics (this is not required reading).

beginning of class. It is very important to keep on top of the reading, and familiarity with the reading will be assumed during lecture. **You should set aside two hours before each class for the reading.**

I do not expect you to fully understand everything before coming to class. Rather, the goal you should set for yourself is to familiarize yourself with new terminology and definitions, and be ready to ask about things that you find especially confusing. **It is strongly recommended to attend class** (lecture recordings will only be available 2-3 days after class).

Support Resources

Online Q+A. We will be using Ed. **All announcements will be made on Ed**, and we encourage you to answer each other's questions (but without giving away answers to homework questions).

Office hours. Please make use of staff office hours! You are welcome to come with specific questions about the material, to discuss final project ideas, or just chat about things you find interesting.

Philosophy

You can expect the course staff to work hard to make the course useful for you, be available throughout the semester and look forward to meeting you in person, promptly answer your questions, and return assignments and midterms to you in a timely manner. We expect that you will, in turn, be prepared to play an active role in your learning, and come to class prepared (having done the assigned reading), and participate in classroom discussions. In this way, we can use class time to work together to develop a deeper understanding of the material.

CS 136 is a challenging class because it covers interdisciplinary material. But we'll work together to try to get as much understanding as possible of the underlying principles and methodologies. If you feel confused about something and you have given yourself some time to try to understand on your own, you should seek help by discussing it with other students or with the staff, or asking a question in the classroom. Other students might well have the same confusion!

Learning outcomes

After successful completion of this course, you will be able to...

Apply principles of economics and computation to

- Understand, model and analyze the interplay between incentives and computation in the design of online platforms
- Solve mathematical and conceptual problems involving such systems, including problems you have not seen before

Organize your learned knowledge in Econ/CS, by

- Articulating the big ideas from each lecture
- Making connections between different concepts
- Synthesizing concepts with ideas from other branches of computer science, including machine learning, complexity theory, and artificial intelligence
- Having an appreciation for the way in which computational systems influence, and are influenced by, the aggregate behavior of groups of people

Collaborate and communicate clearly, in particular

- Justify your thinking by clearly articulating the principles that apply in a given setting
- Explain your solutions, either written or orally to peers and instructors

Logistics

Sections

There will be a recommended, but optional, 60 minute section each week. During section, you will work in groups to solve exercises that are designed to prepare you for the homework and midterms. The TF is there to help guide you and clarify confusions.

Assignments

There are two types of assignments: theory and programming.

Theory assignments include qualitative discussion questions as well as formal, theoretical analysis. The qualitative questions will help you gain an understanding of the assumptions behind the models that we discuss, and the connections to real-world problems. The theoretical questions will teach you to formally analyze simplified economic and computational systems, and to reason precisely. Some problems will require you to write proofs. **You should work by yourself on the theory assignments.**

Programming assignments are mainly in Python, and will let you experiment with simplified models of real problems. These flow into contests (and very good performance can lift up a grade at a grade boundary). where we pit submissions against each other. The assignments are designed to give you the chance to be creative and experiment with different ideas. **You are strongly encouraged to work in pairs on the programming assignments.**

Midterms

T. ⓘ will be **two in-class midterms**--- one is half-way through the semester, and the second in the last week of the semester (and non-cumulative).

There is a class project. The goals of the project are to allow you to explore independent interests, learn more about a particular area, and practice teamwork and presentation skills. The project will result in a final paper that is due during reading week. **Projects should be done in groups of 2 or 3, and may be primarily theoretical or primarily programming-based.**

A project can also involve an **exposition of a topic** related to topics covered in class, but in this case should **be done by a single student.**

Grading

Problem sets are **due at 5pm on Fridays (with [two exceptions](#))**. Students have a maximum of six late days during the semester. There is a two day maximum on any pset, with no credit after two late days. For pair assignments, only one student needs the late days, but they will be counted against both students (if you have remaining credit!).

Submissions are to Gradescope. We will only give extensions for emergencies, and will need either a note from a doctor or your Resident Dean. In the case of regrade requests, the work will be regraded fully, and your grade may go up or down. **Regrade requests are due 1 week after the grades are released.**

We will evaluate the programming assignments according to the following criteria:

- Correctness: does it work?
- Write-ups: what did you do? Why? How well did it work? Why did you get the results you did? Is the writing clear?
- In class contests: How does your approach compare to those of your classmates? Doing well in contests will be worth a large amount of glory (!) but will only factor into your grade at grade boundaries.

We will evaluate your solutions to theory assignments according to the following criteria:

- Correctness: are your answers correct? This is precisely defined for mathematical questions, but includes having valid arguments in qualitative responses.
- Clarity: is your explanation clear? Is the main argument muddled by a lot of irrelevant discussion?

Your final grade will be determined as follows:

- Assignments: 45%
- In-class midterms: 30%
- Final project: 15%
- Participation/completing reading comments: 10%

E  theory assignment will be equally weighted. The bottom theory assignment will not be counted in the final tally. Every student must attempt every assignment!

The course philosophy is that we want you to work together and feel comfortable talking about the problem sets. Discussion and the exchange of ideas are essential to doing academic work. However, at the same time, we want to be able to grade your own work, and not the work of your friends. Please reach out to David Parkes if anything is unclear or you have concerns.

Comprehension Questions: You can discuss the reading with others but your responses should be your own.

Single-person Assignments (theory): For the theory assignments, you are encouraged to discuss with your classmates as you work on problem sets. **However, you should not share answers**, and after discussions with peers, you should make sure that you can work through the problem yourself and ensure that any answers you submit for evaluation are the result of your own efforts. In addition, you must cite any books, articles, websites, lectures, and such that have helped you with your work. Similarly, you must list the names of students with whom you have collaborated on problem sets.

Group assignments (programming): For the programming assignments, we expect you and your partners **to design and implement the solutions together**. You may also discuss with your classmates in other groups, **but not share code or answers**. Make sure that your group can work through the problem yourself, and ensure that any answers you submit are the results of your own efforts. In addition, you must cite any books, articles, websites, lectures, and such that have helped you with your work, and list the names of students from other groups with whom your group has collaborated.

Class project: You are encouraged to consult with your classmates on the choice of topic and to share resources. You may also find it useful to discuss your class project with other students. Otherwise, the guidelines are as above, depending on whether your project is a single-person or multi-person effort.

NEW Generative AI/LLM policy (1 of 2): If you find it helpful, you may use LLMs in answering problem sets, both theoretical and programming based, just as you may consult a text book or web source. As there, you should cite any tools that you use, and as there, **you should be able to provide a full explanation of your answers, so that these tools are used as a learning aid but not a replacement for your own learning**. You will not be able to access LLMs during the in-class midterms.

NEW Generative AI/LLM policy (2 of 2): For your end of class projects, you may choose to use LLMs as a tool to help with your writing, but as with all other work, you are responsible for the content of your paper, including its correctness and that it does not plagiarize other sources. In particular, you must have a full understanding of the paper that you turn in, and be able to explain its content (away from your AI helper!). Also, it is essential that you cite any tools that you use, **including a few sentences to explain how you used them**.

Math 106, Calculus I, University of Nebraska-Lincoln, Fall 2015

Text *Calculus: Single and Multivariable, Sixth Edition* Hughes-Hallett, *et al.*

ACE Outcome 3: This course satisfies ACE Outcome 3: “Use mathematical, computational, statistical, or formal reasoning (including reasoning based on principles of logic) to solve problems, draw inferences, and determine reasonableness.” Your instructor will provide examples, you will discuss them in class, and you will practice with numerous homework problems. The exams will test how well you’ve mastered the material. The final exam will be the primary means of assessing your achievement of ACE Outcome 3.

Placement: You may take Math 106 if you satisfy one of the following conditions:

1. You have passed UNL’s Math 102 or 103 with a grade of C, P or better.
2. You have a prior grade of D, D+, or C- in Math 106 from UNL.
3. You have passed the prerequisite courses in high school or at another college and have a qualifying score on the Math Placement Exam dated after March 2015. The Math Placement Exam is given at the College Testing Center (Burnett 127); see <http://www.math.unl.edu/resources/undergraduate/mpe/>.

Calculators: A graphing calculator can be useful for this course, and the TI-83, TI-84 and TI-86 are recommended. However, calculators that have a built-in computer algebra system (CAS), such as the TI-89, TI-92, TI-Nspire, HP-40, HP-41, Casio ALGEBRA FX 2.0, Casio ClassPad 300, and Casio ClassPad 330, **will not be permitted** during the final exam. Your instructor will determine the calculator policy for hour exams.

Cell Phones: As a courtesy to others, please turn off your cell phones and similar devices when you come to class.

Scheduling: A tentative schedule is included in this syllabus, as a guide. Your instructor will notify you of any changes.

Reading, Exercises, and Assignments: Your instructor will be planning class activities assuming you have done previously assigned reading and exercises. Keeping up is an essential prerequisite for getting the most out of class time. There may be graded projects, computer assignments, paper assignments, and quizzes, at the discretion of your instructor.

Mathematics Resource Center: You are encouraged to visit with your instructors when you have questions about the material or course and to use the Mathematics Resource Center (MRC) in Avery 13 as a meeting place for the course and a resource for assistance. The hours for the MRC are MTWR 12:30–8:30 pm, Fri 12:30–2:30 pm, and Sun 1:00–5:00 pm.

Calculus Readiness Activity: There is a mastery exam given on paper on **Thursday, August 27** in recitation. This exam covers prerequisite material essential to your success in the course. There are 15 questions and if you have a score of 13 or better, you get full points for the CRA. If you do not pass on paper, then you can take it again online at the Arts and Sciences Testing Center (Burnett 127; student ID required) up to once per day until **Wednesday, September 9**.

Gateway Exam: This exam consists of 8 questions in which you are asked to find the derivative without using calculators, notes, or tables. You must get at least 7 questions completely right to pass, with no partial credit and no points awarded for less than a passing mark. You may repeat the exam up to once a day during the exam period. The Gateway exam will be given once in class on **October 1**. Retakes will be given in the College Testing Center (Burnett 127, student ID required) from **October 2** through **October 23**.

Exams: Mid-term exams will be given on **September 24, October 29**, and **between November 30 and December 3**, all from 6pm to 7:30pm, and a Final Exam will be given on **Wednesday, December 16** from 6pm to 8pm, at locations TBA. (The precise date of Exam 3 will be announced as soon as it is scheduled, which will be before Labor Day.) You are expected to take these exams at the scheduled time, with exceptions made for students who have a conflict with another scheduled exam or who have three or more finals on one day. Makeups should be arranged with Lori Mueller in Avery 203 as soon as possible. Under no circumstances will exams be given early. **You are not allowed to have on your person during exams any device that can access the internet or communicate in any way. Cellphones, smart watches, etc. should be put away in backpacks/purses.**

Disability Accommodation: The University of Nebraska-Lincoln is committed to providing flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 472-3787 voice or TTY.

Department Grading Appeal Policy: Students who believe their academic evaluation has been prejudiced or capricious have recourse for appeals to (in order) the instructor, the department vice chair, the department chair, the departmental appeals committee, and the college appeals committee.

Course Evaluation: The Department of Mathematics Course Evaluation Form will be available through your Blackboard account during the last two weeks of class. You'll get an email when the form becomes available. Evaluations are anonymous and instructors do not see any of the responses until after final grades have been submitted. Evaluations are important—the department uses evaluations to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

Date	Section	Topic(s)	Exercises
Aug 24	M 1.2,4	Exponentials and Logs	p17: 7, 9, 13, 16, 32, 41; p33: 7, 10, 21, 25, 30, 31, 39, 42, 47, 54
Aug 26	W 1.5,3	Trig & PreCalc	p42: 1, 15, 20, 29, 31, 42, 43; p26: 1, 5, 11, 13, 18, 23, 28, 33, 57
The CRA will be given in recitation on August 27.			
Aug 28	F 1.7	Intro to Continuity	p56: 1, 3, 5, 8, 9, 11, 12, 20, 33, 35, 37
Aug 31	M 1.8	Limits	p64: 1, 3, 29, 31, 35, 36, 54, 55, 57, 65
Sept 2	W 2.1	Measuring Speed	p80: 2, 3, 5, 7, 15, 17, 18, 20, 24, 25
Sept 4	F 2.2	Derivative at a Point	p87: 1, 2, 8, 11, 13, 15, 17, 18, 20, 21, 24, 29, 35, 37, 41, 45
September 4 is the last day to drop the course with no transcript record.			
September 7 : Labor Day			
Sept 9	W 2.3, 4	Derivative Function	p95: 1, 3, 4, 5, 19, 28, 36, 44, 47; p101: 1, 3, 10, 14, 15, 27
September 9 is the last day to take the CRA.			
Sept 11	F 2.5, 6	Second Derivative	p108: 2, 4, 9, 10, 11, 19, 20, 26, 28; p114: 1–4, 8, 9, 16
Sept 14	M 3.1,2,5	Derivative Formulas	p129: 4–5, 32, 42, 46, 58, 61, 73; p135: 1, 5, 21, 43, 44; p154: 63
Sept 16	W 3.3	Product Rule	p139: 1, 3, 6, 8, 31; p153: 3, 12, 25–27, 36
Sept 18	F 3.3	Quotient Rule	p139: 10, 15, 19, 27, 32, 35, 39, 45, 52; p153: 42, 44

Sept 21	M	3.4	The Chain Rule	p146: 3, 5, 10, 25, 34, 37, 51, 57, 59, 61; p153: 11, 16, 17, 24, 33
Sept 23	W		Review for Exam	
Sept 24	R		EXAM 1	6 PM - 7:30 PM, Location TBA
Sept 25	F	3.6	Inverse Functions	p159: 3–10, 17–20, 24, 29, 33, 38–40, 44, 47, 50, 52, 64; p153: 45
Sept 28	M		Review for Gateway Exam	
Sept 30	W	3.7	Implicit Differentiation	p164: 1, 5, 10, 13, 14, 17, 23, 24, 26, 28, 29, 31, 35, 41
The Gateway Exam will be given in recitation on October 1.				
Oct 2	F	3.9	Linear Approximation	p172: 1-4, 7, 12, 16, 17, 25, 26, 40, 42
Oct 5	M	3.8	Hyperbolic Functions	p167: 1–8, 23, 24, 29, 33;
Oct 7	W	3.10	Mean Value Theorem	p178: 1–9, 12–15, 19, 20
Oct 9	F	4.1	Local Extrema	p192: 1, 3, 10–14, 16–18, 21, 26–28, 33, 35, 37, 45
Oct 12	M	4.2	Global Extrema	p202: 1, 4, 6–8, 11, 13, 15, 18, 26, 31, 37
Oct 14	W	4.3	Optimization	p210: 1–4, 7, 9, 11, 12, 18, 20,
Oct 16	F	4.3	Optimization	p210: 29, 34, 39, 41, 45, 46, 48
October 16 is the last day to change your grade option to or from Pass/No Pass.				
October 19-20 : Fall Break				
Oct 21	W	4.4	Families of Functions	p220: 1, 3, 5, 9–11, 13, 14, 30, 32, 43, 49
Oct 23	F	4.6	Related Rates	p237: 1, 3, 22, 24, 26–28, 33, 37, 44, 45, 50
October 23 is the last day to take the Gateway Exam.				
Oct 26	M	4.7	L'Hôpital's Rule	p247: 1–8, 16, 21, 25, 27, 28, 30–32, 34, 38, 45, 46, 49, 56
Oct 28	W		Review for Exam	
Oct 29	R		EXAM 2	6 PM - 7:30 PM, Location TBA
Oct 30	F	4.8	Parametric Equations	p256: 1, 2, 12, 16, 18, 19, 27, 29, 38, 44
Nov 2	M	4.8	Parametric Equations	p256: 5–8, 37, 45–47, 49, 50
Nov 4	W	5.1	Distance Traveled	p277: 1–4, 9, 12, 13, 15, 17–19, 26, 27
Nov 6	F	5.2	The Definite Integral	p286: 1, 3, 4, 11 14, 18, 19, 29–31, 35, 39, 41, 43
Nov 9	M	5.3	Fundamental Theorem I	p294: 1, 2, 4, 5, 9, 12, 15, 19, 21, 30, 31, 38
Nov 11	W	5.4	Properties of Definite Integrals	p305: 1–7, 10–13, 18, 21, 27, 30, 31, 38–43
Nov 13	F	6.1	Antiderivatives from Graphs	p323: 1, 3–5, 14, 16, 17, 20, 22, 24, 25, 30
November 13 is the last day you can withdraw from the course.				
Nov 16	M	6.2	Antiderivatives from Formulas	p330: 1-9 (odd), 20, 22, 31, 38, 45, 54, 58, 62, 66, 69, 74
Nov 18	W	6.3	Differential Equations	p337: 1–3, 14, 16, 19, 23, 24, 33
Nov 20	F	6.4	Fundamental Theorem II	p342: 1, 4, 7–12, 14, 17, 20, 23, 25, 26, 35, 36
Nov 23	M	7.1	Substitution	p360: 1, 2, 4, 6, 7, 9, 11, 15, 25, 27–31, 36, 42, 46, 59–64
November 25-27 : Thanksgiving Break				
EXAM 3 will be given from 6 to 7:30 PM one evening between Nov. 30 and Dec. 3				
Nov 30	M	7.1	Substitution	p360: 66, 69, 70, 77, 80, 81, 85, 92, 114, 125, 128, 133, 142
Dec 2	W		Review for Exam	
Dec 4	F		Review for Final Exam	
Dec 7–11			Review for Final Exam	
Dec 16	W		FINAL	6 PM - 8 PM, Location TBA

COURSE SYLLABUS

MATH 107H: Honors Calculus II

UNL, Fall 2016, Section: 002, CRN: 4388

Dr. Adam Larios

Lecture: MTWF, 9:30 am–10:20 am, Oldfather Hall 208

Email: alarios@unl.edu

Office: Avery Hall 305

Math Dept. Phone: (402) 472-7250

Office Hours: M,W,F, 1:30 pm–2:20 pm, or by appointment

Text: *Calculus: Single and Multivariable*, 6th ed. by Hughes-Hallet, et al., ISBN: 978-0470-88861-2.

ACE Outcome 3: “Use mathematical, computational, statistical, or formal reasoning (including reasoning based on principles of logic) to solve problems, draw inferences, and determine reasonableness.” Your instructor will provide examples, you will discuss them in class, and you will practice with numerous homework problems. The exams will test how well you’ve mastered the material. The final exam will be the primary means of assessing your achievement of ACE Outcome 3.

Prerequisites: Students who take Math 107H must have passed Math 106 or an equivalent course with a grade of P or C or better. Students who are new to UNL must also pass a readiness test for Math 107H. Any students who do not meet this requirement will be dropped from the course.

Advanced Placement Program: If this is the first college mathematics course that you have attempted, then you may be eligible for 5 hours of free credit for Math 106, provided you get a grade of C, P or better in Math 107H this semester. To be considered for this credit, you should visit the Arts & Sciences Advising Center in 107 Oldfather to make a request by the end of the third week of classes, i.e., by Friday, September 9.

Calculators, Phones, etc.: A graphing calculator may be useful for this course, and the TI-83, TI-84 and TI-86 are standard devices, although there are many phone or computer applications that are equivalent or better. No calculators are allowed during quizzes or exams. **No phones or any devices capable of wireless communication** including smart-watches are permitted at any time. As a courtesy to others please silence your phones when you come to class.

Scheduling: A tentative schedule of assignments and exams is included in this syllabus. Your instructor may change the order of the topics, modify the list of exercises and introduce new assignments. It is your responsibility to keep track of the course details and the schedule for your section.

Attendance: Daily attendance for class lectures is expected and is extremely important. While attendance is not recorded, missing even one class will put you behind. Note that there is a strong correlation between class absences and poor grades. You are responsible for all material and announcements in class regardless of whether or not you attended. **You are also responsible for making arrangements with another classmate to find out what you missed. You should not ask me to go over material you missed (due to tardiness or absences) during office hours or over email.**

Reading and Homework: You are expected to read the appropriate sections of the text BEFORE coming to the class in which the topic is scheduled. You are expected to work the assigned exercises after the corresponding material has been presented in class, and BEFORE the next class meeting (lecture or recitation). The suggested exercises at the end of this syllabus are not graded, but are there to give extra practice. Some students may have to work additional exercises from the text to attain sufficient mastery of the material. It is HIGHLY recommended (thorough not technically required) that students work on these exercises in addition to the WebWork exercises (see below).

WebWork: Graded homework will be assigned through WebWork. It is your responsibility to let your instructor know as soon as possible if you are having any issues logging into WebWork, or submitted your assignments. Please work on assignments well before the deadline so that any problems are detected early enough to fix them. Your lowest two WebWork assignment scores (including missed assignments) will be dropped. Homework must be properly submitted to WebWork before the deadline.

Quizzes: Weekly quizzes will be given in the lab section. These quizzes will have mathematical information, but will also have brief questions about upcoming topics in the course. (For example: Which topics do we plan to cover next week?) Be sure to read ahead so that you are prepared for these.

Midterm Exams: There will be 3 midterm exams in class, given on the indicated days (see the schedule in this syllabus). Please let your instructor know immediately if you will be unable to make any of the exams, and be sure to get the necessary documentation to prove your absence was an excused one.

Grading: Your minimal course grade will be computed as follows.

Homework:	10%	A-/A/A+	90%	–	100%
Quizzes	5%	B-/B/B+	80%	–	89.99%
Midterms:	$3 \times 20\% = 60\%$	C-/C/C+	70%	–	79.99%
Final Exam:	25%	D-/D/D+	60%	–	69.99%
Total:	100%	F	0%	–	59.99%

A “curve” (grade scale adjustment) may be applied in favor of the students if deemed necessary by your instructor.

Tentative Schedule

Date	Day	Section and Topic	Exercises
Aug 22	M	5.2 The definite integral	p.??? : 1, 2, 5, 7, 15, 16, 20, 32
Aug 24	W	Review of foundations	Worksheet
Aug 26	F	5.3 The fundamental Theorem	p.??? : 2, 3, 4, 5, 6, 19, 21, 23
Aug 29	M	6.2 Review of basic integration	p.330: 1-15 (odd), 21, 25, 26, 28, 33, 36, 41, 43, 46, 48-51, 53, 58
Aug 31	W	7.1 Integration by Substitution	p.360: 1, 2, 3, 6, 7, 9, 11, 15, 25, 27-31, 33, 40, 42, 43, 57-62, 71, 72, 74 76, 77, 80, 81
Sept 2	F	7.2 Integration by Parts	p.368: 1-3, 6, 7, 8, 10, 15, 20, 21, 25, 27, 28, 30, 31, 33-39 (odd), 41, 47, 48, 49, 53
Last day to drop without a W			
Sept 5	M	Labor Day, No classes	
Sept 7	W	7.3 Table of Integrals	p.??? : 1, 7, 10, 11, 12, 15, 16, 20, 33, 34, 45, 46, 49
Sept 9	F	7.4 Partial Fractions & Trig. Substitutions	p.384: 1, 3, 5, 10, 12, 16, 17, 18, 19, 33, 35, 41, 42, 45, 47, 49
Sept 12	M	Catch up/Review	
Sept 14	W	Gateway practice exam (in class)	
Sept 16	F	7.4 Partial Fractions & Trig. Substitutions	p.384: 52, 20,21, 55, 57, 55-63(odd), 62, 67, 69
Sept 19	M	7.5 Numerical Integration	p.392: 1, 2, 7, 16, 29, 30
Sept 21	W	7.6 Improper Integrals	p.401: 1, 2, 5, 7, 8, 9, 11, 12, 15, 16, 20, 22, 23, 26, 28, 29, 35, 48
Sept 23	F	7.7 Comparison of Improper Integrals	p.406: 1-7(odd), 8, 11, 15-25 (odd)
Sept 26	M	Catch up/Review	
Sept 28	W	EXAM 1 (in class)	
Sept 30	F	8.1 Areas & Volumes	p.419: 2, 3, 7, 9-17 (odd), 18, 21, 24, 28, 29, 31, 32, 34, 36
Oct 3	M	8.2 Volumes by Slicing & Arc Length	p.427: 5-13(odd), 13,14, 15-17 (numer. integr.), 18, 20, 21-23, 25, 26, 29, 31, 32 43, 44, 45, 49, 53
Oct 5	W	8.3 Area & Arc Length in Polar Coordinates	p.438: 1, 2, 5-8, 13, 15, 17, 18, 19, 25, 26, 27, 32,33, 34, 39, 40, 43, 44
Oct 7	F	Catch up/Review	
Oct 10	M	8.4 Density (only)	p.446: 1-3, 5(c) (use trapez. rule), 13, 15, 16(b)
Oct 12	W	8.5 Applications to Physics (work only)	p.456: 1, 4-6, 11-15 (odd), 18, 19, 21, 23, 26(b)
Oct 14	F	Catch up/Review	
Last day to change to or from Pass/No Pass			
Oct 17,18	M,T	Fall break, No class	
Oct 19	W	9.1 Sequences	p.495: 1, 2, 7, 9, 11, 13, 16, 20-25, 28, 29, 41, 43
Oct 21	F	9.2 Geometric Series	p.502: 8-14, 19, 20, 23, 25-30, 40, 41
Oct 24	M	9.3 Convergence of Series & the Integral Test	p.510: 4-7, 10, 11, 15, 16, 17, 21, 25, 26, 27, 28, 29, 46
Oct 26	W	9.4 Tests for Convergence	p.518: 1, 3, 4-7, 8, 9, 10, 11, 12, 14, 15, 17, 18, 20, 21-24
Oct 28	F	9.4 Tests for Convergence (Include the Root Test)	p.518: 25-28, 29-35 (odd), 36, 38-40, 41-47(odd), 53, 55, 61, 62, 65 66, 67, 69-77(odd), 79, 80, 81, 84-86, 94, 95
Oct 31	M	9.5 Power Series	p.527: 1-3, 4, 5, 7, 10, 12,13, 15, 17-20, 21, 28-32, 40, 43, 45
Nov 2	W	Catch up/Review	
Nov 4	F	EXAM 2 (in class)	
Nov 7	M	10.1 Taylor Polynomials	p.544: 1-3, 6, 7, 12, 18, 19, 29, 31
Nov 9	W	10.2 Taylor Series	p.550: 1, 4, 7, 9, 15, 17, 18, 19, 21, 22, 32, 33, 35, 36, 37, 44, 45
Nov 11	F	Defining Logs and Exp the right way	To be announced
Last day to drop with a W			
Nov 14	M	10.3 New Taylor Series from Old	p.557: 1, 3, 4, 6, 8, 9, 10, 11, 12, 14, 27, 32, 33, 37
Nov 16	W	10.4 Error in Taylor Polynomials	p.563: Use Theorem 9.9 to work: 2, 3, 5, 17, 19
Nov 18	F	10.4 Error in Taylor Polynomials	1, 5, 6, 9, 15
Nov 21	M	Catch up/Review	16, 20, 21, 22
Nov 23	W	Thanksgiving, No class	
Nov 25	F	Thanksgiving, No class	
Nov 28	M	10.5 Fourier Series	5, 6, 8, 12, 17
Nov 30	W	Catch up/Review	
Dec 2	F	EXAM 3 (in class)	
Dec 5	M	Catch up/Review	
Dec 7	W	Catch up/Review	
Dec 9	F	Catch up/Review	
Dec 15	R	FINAL EXAM, 10:00 am–12:00 pm	Location: OLDH 208 (our usual classroom)

Course Schedule: The tentative schedule in this syllabus (given above) is a rough guide to the material covered in the course, but is subject to change. **Updates and changes to the content will be announced in class, over email, or on the course website. You are responsible for all material and announcements in class regardless of whether or not you attended class.**

Getting help: When you need help with something in the course, you have several options:

- Try discussing the material with your classmates; learning from and explaining to your peers often contributes most to your own understanding.
- Talk to your TA or the primary instructor during their office hours.
- Last but not least, there is the **Math Resource Center (MRC)** in Avery 13. The hours for the MRC are 12:30–8:30 p.m. Monday through Thursday, 12:30–2:30 p.m. on Friday, and 1:00–5:00 p.m. on Sunday.

Final Exam Policy: Students are expected to arrange their personal and work schedule to allow them to take the final exam at the scheduled time. Students who have conflicting exam schedules may be allowed to take an alternate final, which is always given **after the regularly scheduled final**. No student will be permitted to take the final exam early. The final exam for this course will be on:

Thursday, Dec 15, 10:00 am–12:00 pm (Location: OLDH 208, the usual classroom).

Incompletes: A grade of “incomplete” may be considered if all but a small portion of the class has been successfully completed, but the student in question is prevented from completing the course by a severe, unexpected, and documented event. Students who are simply behind in their work should consider dropping the course.

Disclaimer: While this syllabus was prepared carefully and according to information available at the beginning of the semester, changes may be necessary in the interest of good teaching. Changes to any of the information above will be announced in class and posted on the class web site. This includes in particular possible updates or corrections to the syllabus, and changes of exam dates.

Course Evaluation: The Department of Mathematics Course Evaluation Form will be available through your Blackboard account during the last two weeks of class. You’ll get an email when the form becomes available. Evaluations are anonymous and instructors do not see any of the responses until after final grades have been submitted. Evaluations are important—the department uses evaluations to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

ADA Language: Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 472-3787 voice or TTY.

Departmental Grading Appeals Policy: Students who believe their academic evaluation has been prejudiced or capricious have recourse for appeals to (in order) the instructor, the departmental chair, the departmental appeals committee, and the college appeals committee.

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COURSE SYLLABUS

Text: *Calculus: Single and Multivariable, 6th ed.* by Hughes-Hallett, et al., ISBN: 978-0470-88861-2. Note, this syllabus works also for the 7th edition.

ACE Outcome 3: This course satisfies ACE Outcome 3: “Use mathematical, computational, statistical, or formal reasoning (including reasoning based on principles of logic) to solve problems, draw inferences, and determine reasonableness.” Your instructor will provide examples, you will discuss them in class, and you will practice with numerous homework problems. The exams will test how well you’ve mastered the material. The final exam will be the primary means of assessing your achievement of ACE Outcome 3.

In-Person/Remote Teaching: Please consult your instructor’s policy sheet on class attendance and remote teaching. The current UNL policy on health and safety states: “Those who are fully vaccinated are not required to wear face coverings, but are strongly encouraged to do so indoors. Anyone who is not fully vaccinated is expected to wear a face covering.” Also, you can consult the UNL policy on COVID19 (which will be updated from time to time) at: <https://covid19.unl.edu/>

Course Evaluation: The Department of Mathematics Course Evaluation Form will be available through your Canvas account during the last two weeks of class. You’ll receive an e-mail when the form becomes available. Evaluations are anonymous and instructors do not see any of the responses until after final grades have been submitted. Evaluations are important—the department uses evaluations to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

Scheduling: A tentative schedule of assignments and exams is included in this syllabus. The pace of this course is necessarily fast since it is a four credit which meets 3 times plus one recitation per week. It is your responsibility to keep track of the course details and the schedule of your section. You can also see the Faculty Senate Class Attendance Policy for Fall semester at: <https://www.unl.edu/facultysenate/policies/Class-Attendance-Policy-081121.pdf>

Daily Work: Do an initial reading of the section(s) expected to be covered before coming to class each day—even if you don’t understand the details, that reading will help you to better understand the lecture. Rereading more carefully after the class can also be helpful. The exercises listed below represent a minimum assignment and should be done as the material is covered.

Exams/Quizzes/Homework: Please consult your instructor’s course information sheet or policy.

Calculators: Calculators will not be allowed on any of the exams or quizzes in this course.

Final Exam: The time for the final exam is **6:00-8:00p.m., Wednesday, December 15**. The final exam will be a comprehensive exam. More details/instructions will be provided by your instructor later during the semester. You are expected to arrange your personal and work schedule to allow you to take the exam at the scheduled time. Students with conflicting exam schedules may be allowed to take an alternate final, which is always given after the regularly scheduled final. No student will be allowed to take the final exam early.

Advanced Placement: If this is the first college mathematics course that you have attempted, then you may be eligible for 10 hours of free credit for Math 106 and Math 107, provided you earn a grade of P, C or better in Math 208 this semester. To be considered for this credit, you should register with the Department of Mathematics, 203 Avery Hall, by Friday, September 10, 2021.

ADA Language: Students with disabilities are encouraged to contact the instructor for confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 472-3787 voice or TTY.

Department Grading Appeals Policy: The Department of Mathematics does not tolerate discrimination or harassment on the basis of race, gender, religion or sexual orientation. If you believe you have been subject to such discrimination or harassment in this or any math course, please contact the department. If, for this or any other reason, you believe that your grade was assigned incorrectly or capriciously, appeals should be made to (in order) the instructor, the department chair, the department grading appeals committee, and the college grading appeals committee.

Dates	Sections/Topic	Exercises
Aug 23–27	12.1 Functions of Two Variables	1–4, 6, 10, 11, 13, 14, 17, 21, 23, 25, 29, 30
	12.2 Graphs of Functions of Two Variables	1, 2, 3, 7, 8, 10, 15, 16, 18–20, 23–25
	12.3 Contour Diagrams	1, 2, 4, 5, 7–9, 13, 16, 17, 24, 27(a,b)
	12.4 Linear Functions	1–5, 7–11, 13, 21, 23, 26, 27
Aug 30–Sep 3	12.5 Functions of Three Variables	1–4, 8–11, 13, 15, 16–18, 23, 31
	12.6 Limits and Continuity	1, 2, 3, 6, 7, 9, 11–14, 19, 23
	13.1 Vectors	1, 2, 5, 7, 12, 15, 24, 25, 29, 31, 32, 39(b), 40, 41
	13.2 Vectors	1–5, 7, 10, 11, 16, 20, 30, 36
Friday, Sep 3, is the last day to drop without a W.		
Sep 6–10	Labor Day Holiday, Sep 6	(no class)
	13.3 The Dot Product	1, 5–19(odd), 23, 25–27, 29, 31, 33, 38, 40, 41, 43, 45
	13.4 The Cross Product	2, 3, 7, 9, 11, 14, 15, 17, 19, 20, 23, 27, 29, 31, 33, 39
Sep 13–17	14.1 The Partial Derivative	1, 3, 5, 10–12, 17–19, 21, 22, 24, 25, 30
	14.2 Computing Partial Derivatives	1, 3–5, 9, 11, 18, 21–25, 31, 39–41, 44, 45, 49
	14.3 Local Linearity	1, 2, 3, 5, 6, 7, 9, 11, 13, 18, 20, 22, 23
	14.4 Gradients and Directional Derivatives	1–8, 15, 17, 20, 21, 23–26, 29–33, 37
Sep 20–24	14.4 Gradients and Directional Derivatives	39, 45, 46, 49, 51, 53, 56–59, 61, 67, 69–71, 75
	14.5 Gradients and Directional Derivatives	2, 3, 7, 9, 14, 17–19, 21, 25–27, 35, 37–41, 46, 47, 53, 56, 57
	14.6 The Chain Rule	1, 2, 3, 5, 7, 9, 11–15, 17, 20, 23, 24, 28, 33, 38
	14.7 Second-Order Partial Derivatives	1, 3, 6, 11, 13, 14, 16, 19–21, 23–31(odd), 34, 35, 37, 41, 44
Sep 27–Oct 1	15.1 Local Extrema	1–4, 6, 7–35(odd)
	Catch up/Review Exam 1	
	15.2 Optimization	2, 5, 7, 8, 9, 10–13, 15, 17, 19, 20
Oct 4–8	15.3 Constrained Optimization	1, 3, 5, 6, 7, 9–13, 17–19, 22, 24, 30, 31, 32, 45
	Catch up/Review	
	16.1 The Definite Integral in the Plane	1, 3, 6–11, 13, 15, 22, 23
Oct 11–15	16.2 Iterated Integrals	1–4, 9–23(odd), 29, 33–38
	16.2 Iterated Integrals	42, 43–51(odd), 52, 54, 55
	16.3 Triple Integrals	1–9(odd), 11–27(odd), 28–30, 33, 34–37, 39, 41, 44, 45, 50, 57
	21.2 Change of variables (optional)	1, 3, 4, 5–7, 9, 13, 15
	16.4 Double Integrals in Polar Coord.	1–8, 10–12, 14–17, 19, 21, 24, 25, 27, 28, 30, 31, 33
Friday, October 15, is the last day to change your grade option to or from Pass/No Pass.		
Oct 18–19	Fall Semester Break	
Oct 20–22	16.5 Triple Integrals in Cylindrical Coord.	1, 3–7, 9, 3–15, 21, 23, 27–37(odd), 48, 49, 57–59
	16.5 Triple Integrals in Spherical Coord.	8, 10, 11, 24, 30, 32, 39, 43, 47, 65
	17.1 Parameterized Curves	1–7, 10, 11, 13, 15, 19, 21, 26, 29, 33, 45–49(odd), 57, 62, 81
Oct 25–29	17.2 Motion, Velocity and Acceleration	1, 3, 5, 9, 10, 13, 15, 17, 21–23, 25, 27, 30, 36, 41, 45
	Catch up/Review Exam 2	
	17.3 Vector Fields	1–5, 7, 9, 11, 13, 15, 16, 20, 21–27(odd), 28, 30, 31, 33
Nov 1–5	18.1 The Idea of a Line Integral	1–8, 11, 13, 15–21(odd), 25–28, 31, 32, 36, 46
	18.2 Computing Line Integrals	1–23(odd), 26, 29–35(odd)
	18.3 Gradient Fields and Path-Indep.	1, 3, 4, 5, 7, 8, 9, 10, 13, 15, 17–21, 27, 29, 31, 32, 38, 39, 48
	18.4 Green's Theorem	1–17(odd), 19, 20, 22, 26, 27, 33, 34, 39
Nov 8–12	Catch up/Review	
	19.1 The Idea of a Flux Integral	1–9, 13, 14, 16–19, 21, 25, 26, 29–35(odd), 39, 48
	21.1 Parameterized Surfaces	1–8, 10–13, 17, 15, 17, 19, 20
	21.3 Flux Through Parameterized Surfaces	1–3, 5–9, 11, 12
Friday, Nov 12, is the last day to withdraw from the course and receive a grade of W.		
Nov 15–19	19.2 Flux Integrals through graphs	1–6, 7, 9, 11, 13, 15, 19, 29, 31, 41–43, 47
	19.2 Flux Integrals through graphs	16, 17, 21–23, 25, 27, 35, 37, 49, 51, 52
	Catch up/Review	
	19.3 The Divergence of a Vector Field	1, 3, 4, 5–11(odd), 12, 17–20, 24, 29, 36
	19.4 The Divergence Theorem	1, 2, 3–9(odd), 8, 14–17, 19, 23, 25, 28, 32
	Catch up/Review	
Nov 24–28	Thanksgiving Holiday	
Nov 29–Dec 3	20.1 The Curl of a Vector Field	1–5, 7, 11–16, 25
	Catch up/Review Exam 3	
	20.2 Stokes' Theorem	1–5, 7, 9, 11, 14, 17–19, 21, 23, 25–27
Dec 7–11	Catch up/Review	
Final Exam, 6:00-8:00p.m., Wednesday, December 15.		

Math 310 Introduction to Modern Algebra

Fall 2021 Syllabus

Instructor: Eloísa Grifo

email: grifo@unl.edu

Office: 339 Avery Hall

Class time and location: Tuesdays and Thursdays 12:30 – 1:45 pm in 112 **Avery Hall**

Office Hours: **Tuesdays 3:30 – 4:30 pm**, **Thursdays 2 – 3 pm**, and by appointment **OH zoom:** **97297545735**

What are office hours? During the scheduled office hours, I'll be in my office and available to you, so you can either come in person to my office or attend on zoom, and you do not need an appointment — just show up! If the scheduled times are not compatible with your schedule, please email me to schedule an alternative time. Office hours are an excellent opportunity to discuss any questions you may have, including homework problems.

Face coverings: Per CDC guidelines, everyone is strongly encouraged to wear masks indoors, though for now masks are not required in our class. If you have vulnerable individuals in your household and would like to request that everyone in our class wears a mask, please contact me immediately.

Prerequisite: A grade of P, C, or better in Math 107 or Math 107H

Course Description: One central goal of this course is to teach you to read, write, and understand mathematical proofs. The only way to learn to write good proofs is by doing so, and thus there will be a heavy emphasis on problem sets in this course. Many of the assigned problems will require you to develop original, rigorous proofs of mathematical statements. This is a difficult skill to master and success in this course will require a sustained effort on your part.

Another goal is to learn about modern algebra, specifically focusing on the mathematical concept of a *ring*. The collection of integers (both positive and negative) along with the usual rules for addition and multiplication are an example of a ring; the collection of polynomials also forms a ring. In this course, we will explore the features that these two examples have in common, and we will meet other examples of rings.

Attendance: This is an in person class. I expect everyone to attend class every day; however, you should **definitely not** come to class if you are feeling ill or have the need to quarantine. I will forgive all absences for which *you notify me by e-mail in advance of your need to miss class*. Depending on the circumstances, I might require you to do something to make up for your absence.

Textbook: *Abstract Algebra — an introduction*, by Thomas W. Hungerford, 3rd Edition.

The textbook is recommended but **not required**; it provides many examples of well written proofs.

Class format: Rather than a more traditional lecture format, this class will employ the methodology of Inquiry Based Learning (IBL). This means that the vast majority of class time will be spent with you working together in small groups, on activities I have carefully designed. You will formulate conjectures, propose definitions, develop examples, and construct proofs of mathematical statements together with your peers. Sometimes I will have you share your work with the class as a whole, and throughout the process you will receive feedback from your peers and from me. On occasion, I will give short mini-lectures to clarify matters, as appropriate. Let me quote from the Conference Board of the Mathematical Sciences (CBMS) position statement:

Classroom environments in which students are provided opportunities to engage in mathematical investigation, communication, and group problem-solving, while also receiving feedback on their work from both experts and peers, have a positive effect on learning. (...) A wealth of research has provided clear evidence that active learning results in better student performance and retention than more traditional, passive forms of instruction alone.

Grade Breakdown: We will have homeworks, quizzes, 2 midterms, a final exam, and iClicker questions.

Component	Value
Quizzes	10%
Problem Sets	40%
Midterms (two)	30% (15% each)
Final Exam	20%

We will have problem sets, two midterm exams, quizzes, and a final exam.

- We will have a short **quiz** (almost) every Tuesday. In every quiz, you will be asked to write down a definition or cite a theorem from the previous week. Some quizzes might also contain short proofs.
- The **problem sets** will be assigned and collected approximately once per week. You are strongly encouraged to work together on the problem sets, but each of you will hand in your own solutions, written in your own words, and your work must demonstrate a true understanding of the material. Never hand in something that you do not completely understand. I ask that at the top of each assignment you list the students with whom you collaborated. Please hand in all assignments on time.
- There will be two **midterm exams** and a **final exam**.
 - * Midterm 1: September 28.
 - * Midterm 2: November 2.
 - * The final exam will be on Tuesday, December 14, 4–6 pm, in Avery 117.

Academic Dishonesty: Academic dishonesty includes cheating on any test, plagiarism, fabricating an otherwise justifiable excuse to avoid or delay timely submission of academic work, and helping or attempting to help another student commit academic dishonesty. For a comprehensive list, see Section 4.2 of the Student Code of Conduct. A student observed cheating in any manner during an exam or quiz will earn a score of 0 for that assignment and will be referred to the Vice Chair for academic sanction. A second offense will earn the student an F in the course.

Departmental Grading Appeals Policy: Students who believe their academic evaluation has been prejudiced or capricious have recourse for appeals to (in order) the instructor, the departmental chair, the departmental appeals committee, and the college appeals committee.

Accommodations: The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let me know immediately so that we can discuss options privately. To establish reasonable accommodations, please register with the [Services for Students with Disabilities](#) (SSD) office. If you are eligible for services and register with their office, make arrangements with me as soon as possible to discuss your accommodations so they can be implemented in a timely manner.

Resources: Here are some resources available to you:

Counseling and Psychological Services (CAPS) is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological, and emotional well-being.

Big Red Resilience and Well-Being provides one-on-one well-being coaching to any student who wants to enhance their well-being. Trained well-being coaches help students create positive experiences, practice resilience and self-compassion, and find support as they need it.

Husker Pantry: Provides food and hygiene items for free to students who might need them.

Equipment checkout: Free 7 day laptop and iPad loans.

First Gen Nebraska. For first generation college students.

You'll find many other useful resources on the [Student Affairs website](#).

MATH 314 – Section 005
Spring Semester 2020
MWF 11:30 am–12:20 pm, Avery Hall 019

1 Instructor

Dr. Xavier Pérez-Giménez
Assistant Professor
Department of Mathematics
University of Nebraska – Lincoln
Lincoln, NE 68588-0130, USA

Email: xperez@unl.edu

Website: <http://www.math.unl.edu/~xperezgimenez2/>

Office: Avery Hall 333

Office Hours: TR 8:30 am – 9:30 am (or by email appointment)

Grader: TBA

2 Textbook

Linear Algebra and Its Applications, by David C. Lay, Steven R. Lay, and Judi J. McDonald, 5th Edition.

3 ACE Outcome 3

This course satisfies ACE Outcome 3: “Use mathematical, computational, statistical, or formal reasoning (including reasoning based on principles of logic) to solve problems, draw inferences, and determine reasonableness.” Your instructor will provide examples, you will discuss them in class, and you will practice with numerous homework problems. The exams will test how well you’ve mastered the material. The final exam will be the primary means of assessing your achievement of ACE Outcome 3.

4 Contacting me

The best way to contact with me is by email, xperez@unl.edu. Please put “[MATH 314]” at the beginning of the title and make sure to include your whole name in your email. Using your official UNL email to contact me is strongly recommended. My office is Avery Hall 333. My office hours are TR 8:30 am – 9:30 am. If you want to meet at another time, please email me in advance, and we will try to schedule a time to meet.

5 Course Description

One can say that Linear Algebra is a mathematical branch studying system of linear equations, for example

$$\begin{cases} 4x + 5y + 6z = 7 \\ 2x + 6y + z = 2 \\ x + y + z = 4 \end{cases}$$

Linear Algebra has many applications to diverse scientific areas, and even has powerful uses in computer graphics, linguistics, audio engineering, artificial intelligence, humanitarian aide, and legal studies etc.

This course is a transition course from computational courses, like calculus, to more theoretical ones. You will need to understand definitions and theorems, be able to apply them, and sometimes, prove theorems. *The material in the course will tend to be more mathematically subtle than that encountered in your previous math courses, and will consequently require a significant effort on your part to master.*

The course covers sections from Chapters 1 to 7 of the text: systems of linear equations, matrix algebra, determinants, vector spaces, eigenvalues and eigenvectors, orthogonality, and inner product spaces, and quadratic forms.

6 Homework and quizzes

Homework is designed to help students understand the materials and to prepare them for exams. We would have homework almost every week. Pop quizzes will be given if needed. There are **no make-up quizzes**.

Collaboration is encouraged in this course. However, copying someone else's work and submitting it as your own is unacceptable. This act of academic dishonesty will be prosecuted in accordance with university policy.

Besides homework and quizzes, you are expected to read the appropriate sections of the text before coming to the class. You are also expected to work through the indicated exercises after the corresponding material is presented in class, and **before** the next class meeting.

7 Calculators and Electronics

You are **not** allowed to have on your person during exams or quizzes any devices that can access the internet or communicate in any way. Cell phones, Apple watches, etc. should be put away in backpacks/purses. Calculators, laptops, tablets, cell phones, and other non-medical electronic devices are not permitted during exams unless otherwise stated. During class, cell phones should be set on vibrate or off. If you need to take a call, send a text message, etc., please quietly leave the classroom to do so, so that you do not distract other students. You are welcome to return to class quietly when you are finished. If you wish to take notes using an electronic device, you must first demonstrate to me that you can type or write fast enough to do so properly, and that you can do it without distracting others, before the privilege to use such devices may be granted. If you are found to be abusing this privilege, you risk forfeiting it.

8 Grading

Your minimal course grade will be computed as follows.

Homework+Quizzes:	30%
Midterms:	$2 \times 17.5\% = 35\%$
Final Exam:	35%
—	—
Total:	100%

A+	95.00%	—	100%	B-	76.66%	—	79.99%	D	60.00%	—	63.32%
A	90.00%	—	94.99%	C+	73.33%	—	76.65%	D-	56.66%	—	59.99%
A-	86.66%	—	89.99%	C	70.00%	—	73.32%	F	0%	—	56.65%
B+	83.33%	—	86.65%	C-	66.66%	—	69.99%				
B	80.00%	—	83.32%	D+	63.33%	—	66.65%				

9 Attendance

Daily attendance for class lectures is expected and is **extremely important**. While attendance is not recorded, missing even one class will put you behind. Note that there is a strong correlation between class absences and poor grades. You are responsible for all material and announcements in class regardless of whether or not you attended. **You are also responsible for making arrangements with another classmate to find out what you missed. You should not ask me to go over material you missed (due to tardiness or absences) during office hours or over email.**

10 Exams

There are three exams: two midterms and a final. Students are expected to arrange their personal and work schedule to allow them to take the final exam at the scheduled time. No student will be permitted to take the final exam early.

Make-up exams will only be given with written evidence of an official university excused absence.

11 Incompletes

A grade of “Incomplete” may be considered if all but a small portion of the class has been successfully completed, but the student in question is prevented from completing the course by a severe, unexpected, and documented event. Students who are simply behind in their work should consider dropping the course.

12 ADA Statement

Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 472-3787 voice or TTY.

13 Grade Questions

Any questions regarding grading/scoring of homework, exams, or projects must be made **within two class days** from when they were handed back, or no change in grade will be made.

Because of privacy rights, **I cannot discuss grades over email or telephone. Please do not email or call me asking about your grade. I will not be able to give you any information.** Of course, I am happy to discuss grades in my office.

14 Special Dates

- | | |
|----------------------------|--|
| January 24, 2020 (Friday): | Last day to withdraw from this course and not have it appear on your transcript. |
| March 6, 2020 (Friday): | Last day to change your grade option to or from Pass/No Pass. |
| April 3, 2020 (Friday): | Last day to drop this course and receive a grade of W. (No permission required.) After this date, you cannot drop. |

15 Departmental Grading Appeals Policy

Students who believe their academic evaluation has been prejudiced or capricious have recourse for appeals to (in order) the instructor, the departmental chair, the departmental appeals committee, and the college appeals committee.

16 Course Evaluation

The Department of Mathematics Course Evaluation Form will be available during the last two weeks of class. You will get an email when the form becomes available. Evaluations are anonymous and instructors do not see any of the responses until after final grades have been submitted. Evaluations are important—the department uses evaluations to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

17 Tentative schedule:

The following table shows the material expected to be covered for each week of the semester. The exercises listed here are only recommended problems, they are *not* your official assignments. Note that what is shown here is approximate; your instructor may change the dates for each assignment and/or exam. It is your responsibility to keep track of the course details and schedule for your section.

Week of	Section	Recommended (but not necessarily mandatory) Exercises
January 13	1.1 Systems of Linear Equations	1, 3, 5, 9, 10, 11, 15, 18, 19, 20, 23, 24, 25, 31
	1.2 Row Reduction and Echelon Forms	1, 3, 7, 11, 13, 15, 17, 19, 21, 22, 23, 24, 25, 26
	1.3 Vector Equations	1, 3, 5, 7, 9, 11, 13, 14, 15, 17, 18, 19, 23, 24, 25, 28
January 20	Martin Luther King Day	
	1.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$	1, 3, 7, 9, 11, 13, 14, 15, 17–24
	1.5 Solution Sets of Linear Systems	2, 5, 6, 7, 8, 9, 11, 12, 13, 15, 16, 20, 23, 24, 25, 40
January 27	1.6 Applications	3(a,b), 7, 14
	1.7 Linear Independence	1, 3, 5, 7, 8, 9, 13, 14, 15, 17, 19, 21, 22, 23, 24, 28, 30
	1.8 Introduction to Linear Transformations	1, 2, 3, 5, 7, 9, 11, 13–16, 19, 21, 22, 32, 33, 34
February 3	1.9 The Matrix of a Linear Transformation	1, 5, 7, 8, 13, 15, 17, 22–25, 38
	2.1 Matrix Operations	1, 3, 5, 7–11, 15, 16, 19, 22, 24
	2.2 The Inverse of a Matrix	1, 3, 5, 7, 8, 9, 10, 13, 20, 21, 23, 24, 29, 31, 32, 33
February 10	2.3 Characterization of Invertible Matrices	1–7(odd), 11, 12, 13, 16, 17, 19, 22, 33, 37
	2.5 Matrix Factorizations	3, 5, 9, 11, 19
	Catch Up and Review	
February 17	Midterm Exam I	
	3.1 Introduction to Determinants	1–13 (odd), 39, 40
	3.2 Properties of Determinants	1–8, 11, 15, 18, 19, 25, 27, 28, 31
February 24	4.1 Vector Spaces and Subspaces	1–15, 17, 19, 20, 21, 23, 24, 25, 27
	4.2 Null Spaces, Column spaces	1, 2, 3, 5, 7, 11, 12, 15, 17, 19, 20, 21, 25–28, 30, 35, 37
	4.3 Linearly Independent Sets; Bases	1–19 (odd), 21–25, 31, 32
March 2	4.4 Coordinate Systems	1, 3, 5, 7, 8, 11, 13, 15, 16, 27, 28, 29
	4.5 The Dimension of a Vector Space	1–5, 7–17 (odd), 19, 20, 21, 29, 30, 31
	4.6 Rank	1, 3, 4, 5–15 (odd), 17, 18, 19, 21, 25, 27–29
March 9	4.7 Change of Basis	1–9 (odd), 11, 12, 13, 15
	4.9 Applications/Catch Up	1, 3, 5, 9, 11
	5.1 Eigenvectors and Eigenvalues	1–15 (odd), 19, 21, 22, 23, 24, 25, 27, 31, 33
March 16	Classes cancelled	
March 23	Spring vacation	
March 30	5.2 The Characteristic Equation	1, 3, 7, 9, 11, 13, 17, 21, 22, 23, 24
	5.3 Diagonalization	1, 3, 5, 7, 11, 15, 16, 19, 21, 22, 23, 24, 25, 27, 29
	5.4 Eigenvectors and Linear Transformations	1, 3, 5, 8, 9, 11, 13, 19, 23, 27
April 6	5.5 Complex Eigenvalues	1, 5, 9, 13, 16
	6.1 Inner Product, Length and Orthogonality	1–19 (odd), 20, 25–31
	6.2 Orthogonal Sets	1, 5, 9, 11, 13, 15, 17, 23, 24, 27–29
April 13	Midterm Exam II	
	6.3 Orthogonal Projections	1, 5, 7, 9, 11, 13, 15, 17, 21, 22, 23, 24
	6.4 The Gram-Schmidt Process	1, 5, 9, 11, 15, 17, 18, 19, 22
April 20	6.5 Least-Squares Problems	1, 3, 5, 7, 11, 15, 17, 18, 19, 21
	6.6 Applications	1, 3, 7a, 9
	7.1 Diagonalization of Symmetric Matrices	1–19 (odd), 23, 25, 26, 28, 29, 36
April 27	7.2 Quadratic Forms	1–13 (odd), 21, 22, 23, 24
	7.4 Singular Value Decomposition	1, 3, 5, 7, 9, 11, 12, 13, 17, 18, 23
	Catch Up and Review	
May 4	Final Exam Week	
The Final Exam is on Monday, May 4, 10:00 am – 12:00 pm		

Course Documentation for Spring 2022 Math 325-002

Elementary Analysis

TR 11:00-12:15pm, Burnett Hall 121

Instructor:	Yvonne Lai
Office:	Avery Hall 306
Office hours:	Tuesdays 12:30-1:30pm after class (and by appointment)
Contact information:	ylai3@unl.edu

Overview

Analysis. Analysis is the wellspring of much of modern mathematics. It studies (very carefully) the properties of the real numbers, and of functions taking real values. There are some surprises lurking (enough to trip up several generations of the world's best mathematicians) and the careful resolution of these difficulties was simultaneously one of the crowning mathematical achievements of the 19th century, and the foundation for the 20th century's mathematical flowering.

We will start by developing a clear understanding of the real numbers (which will require us to solidify our understanding of rational numbers also) and then progress to studying the behavior of real-valued functions. Throughout the course we will be working on developing the skills crucial to a young mathematician—those of reading and writing proofs. These activities lie at the heart of mathematics.

Themes. Throughout this course, we will practice:

- Understanding the rigorous foundations of functions, derivatives, and integration
- Connecting visual heuristics with formal argument
- Writing clear, logical, rigorous proofs

Course requirements

Weekly Homework (270 points). This course has regular homework and it will be carefully graded. I expect to collect homework once a week, mostly on Thursdays. There will be two kinds of homework questions. Most will be graded in the usual way, with partial credit for partially correct reasoning.

Homework is typically scored out of 20 points each. If the homework does not total to 270 points, it will be scaled out of that many points (i.e., your total will be multiplied by $\frac{270}{\text{maximum possible weekly homework points}}$).

Mastery Homework (40 points). Other questions on the homework will be graded for mastery. You will not get credit until you have written a correct proof. Responses to these questions can be submitted until you get credit for them, but not exceeding 1 month after it is assigned. The reason for the 1 month deadline is that proof communication is essential for this course (and mathematics), and I would like to provide incentive for you to improve these skills, and for us to work together on them, within an amount of time that you can use these skills for as much of the course as possible.

Mastery homework is typically scored out of 5 points each. If the mastery homework does not total to 40 points, it will be scaled out of that many points (i.e., your total will be multiplied by $\frac{40}{\text{maximum possible mastery homework points}}$).

Midterm 1, 2, 3 and Final Exams (100 + 100 + 100 + 150 points). There will be three midterms and one final exam. They are scheduled as shown below in Course Timeline. The midterms take place in class, and they will typically have a take-home component due the next course meeting. Each take-home component will be worth 20% of the total midterm to which it is associated.

Required Materials

Main Reference. Abbott, S. (2016). *Understanding Analysis*. New York, NY: Springer.

Google Drive. We will have a course Google Drive on which notes and supplemental materials will be posted. Please make sure you can access this drive.

<https://tinyurl.com/UNL325Spring2022>

(The tinyurl points here: <https://drive.google.com/drive/folders/1zZ1x-w7gzjwXi17fcgMoB5AAJWVt-0-M>)

Course Timeline (approximate)

Week	Ch.	
W1, Jan 17-21	§1.2	Mathematical statements, quantifiers, and negation; set notation; equivalent statements. Proof techniques; good proof communication. (See: http://www.math.toronto.edu/preparing-for-calculus/3_logic/logic.html)
W2, Jan 24-28	§1.1-1.4	Induction. Archimedean Principle. Existence of square roots. Completeness. Infimums, supremums.
W3, Jan 31-Feb 4	2	Limits and limit theorems of sequences, Monotone and Cauchy sequences.
W4, Feb 7-11	2	Subsequences, lim sup and lim inf. Series. Open and closed sets in $\mathbb{R}$ (§3.2).
Thursday, Feb 11 : Exam 1 : Material from Weeks 1-3		
W5, Feb 14-18	4	Continuous functions and their properties
W6, Feb 21-25	4	Limits of functions, Uniform continuity
W7, Feb 28-Mar 4	5	Derivatives
W8, Mar 7-11	6	Power series, Uniform convergence, Weierstrauss's Approximation Theorem
Thursday, Mar 10 : Exam 2 : Material from Weeks 4-7		
March 13-20, 2022 : Spring Break		
W9, Mar 22-25	6, 7	Mean Value Theorem, Taylor's Theorem
W10, Mar 28-Apr 1	7	The Riemann integral and its properties
W11, Apr 4-8	7	Fundamental Theorem of Calculus
W12, Apr 11-15	8	Improper integrals
(Week 14) April 14 : Exam 3 : Material from Weeks 8-11		
W13, Apr 18-22	7	Exponents and Logarithms
W14, Apr 26-29	7	The constant e
W15, May 2-6	7	Continuous nowhere but differentiable functions
Tuesday May 10: Cumulative Final Exam: 3:30-5:30 p.m.		
Location: Our Regular Classroom		

Departmental and University Policies

Course Evaluations. The Department of Mathematics course evaluation form will be available through Canvas during the last two weeks of class. Evaluations are anonymous and instructors do not see the responses until after final grades have been submitted. Evaluations are important – the department uses them to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

Academic Honesty Policy. Please see Student Code of Conduct, Section B. Conduct – Rules and Regulations, 1. Acts of Academic Dishonesty.

Accommodations for Students with Disabilities Policy. The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let me know immediately so that we can discuss options privately. To establish reasonable accommodations, I may request that you register with Services for Students with Disabilities (SSD). If you are eligible for services and register with their office, make arrangements with me as soon as possible to discuss your accommodations so they can be implemented in a timely manner. SSD contact information: 232 Canfield Admin. Bldg.; 402-472-3787.

Participation in pandemic times. This class builds your knowledge through listening, writing, and speaking mathematics. Given current CDC guidance and the current transmission level of COVID-19 in our community, I respectfully request that you join me in wearing a face covering during our classes, particularly when we are working in close proximity.

Grading Appeals Policy. The Department of Mathematics does not tolerate discrimination or harassment on the basis of race, gender, religion, or sexual orientation. If you believe you have been subject to such discrimination or harassment, in this or any other math course, please contact the department. If, for this or any other reason, you believe your grade was assigned incorrectly or capriciously, then appeals may be made to (in order) the instructor, the vice chair, the Department grading appeals committee, the College of Arts and Sciences grading appeals committee, and the University grading appeals committee.

Assessment Criteria

You have the potential to earn 760 points through the activities of this term.

A course grade of *A* in this course is likely to result from 93% or higher of these 760 points, an *A⁻* from 90% to 93%, a *B⁺* from 86% to 89%, *B* from 83% to 85% etc. An *A⁺* is reserved for exceptional, sustained performance. Point potentials are summarized below.

Point potentials for Spring 2022 Math 325-002

Homework		
Weekly homework	270	} 310 points
Mastery homework	40	
Exams		
Midterm exams	100 + 100 + 100	} 450 points
Final exam	150	
Total		760 points

Statistics and Applications

Instructor: Ruizhi Zhang

Email: rzhang35@unl.edu

Lecture Video: The lecture video will be posted on Canvas at 12:30 PM on Tuesdays and Thursdays

Office Hours: 2:00 PM - 3:00 PM Tuesdays online via Zoom, or By appointment

TA: Rachel Rogers

Email: rachel.rogers@huskers.unl.edu

Office Hours: 3:00 PM - 4:00 PM Monday, and 4:00 PM-5:00 PM Thursday online via Zoom, or By appointment

TA: Liangrui Sun

Email: liangrui2016@huskers.unl.edu

Office Hours: 3:00 PM - 4:00 PM Wednesday, and 3:00 PM - 4:00 PM Friday online via Zoom, or By appointment

Textbook: *PROBABILITY AND STATISTICS FOR ENGINEERS AND SCIENTISTS*, 9th edition; Walpole, Myers, Myers and Ye.

Prerequisites: MATH 107 (MATH 107H): Analytic Geometry and Calculus II

Course Description: The STAT/MATH 380: Introduction to Probability and Statistics Concepts is the second-level course in “undergraduate statistics and mathematics” sequence. The course provides an introduction to probability concepts, i.e., random variables, probability distributions, expectation, variance, covariance, correlation; and statistical concepts, i.e., fundamental sampling distributions and data descriptions, one- and two-sample estimation and testing problems, and simple linear regression. This course lays the foundation for many 400-level courses in Engineering, Mathematics and Statistics, and is also offered for Honors credit.

ACE Outcome 3: The STAT/MATH 380 course is accredited as Achievement Centered Education (ACE) course and satisfies ACE outcome 3: to use mathematical, computational, statistical, or formal reasoning (including reasoning based on principles of logic) to solve problems, draw inferences, and determine reasonableness. Therefore, the reinforced skill for STAT/MATH 380 is Critical Thinking.

Course Goals:	This course will help you learn to think and reason statistically, and to construct arguments based on numerical evidence. My role as the instructor is to facilitate this type of learning by providing you with a variety of meaningful activities and opportunities to learn, as well as creating an environment conducive to learning. This will manifest in a variety of ways: direct instruction, individual practice, exploration and discovery activities, writing, discussions panel. Ultimately, you are responsible for your own learning, so please put into the class what you hope to get out of it. • Students will understand essential theoretical concepts (and logic behind them) including common discrete and continuous probability distributions, sampling distributions and data description techniques, hypothesis testing problems, and modeling approaches. • Students will be able to recognize a classical distribution model when are presented with details of an experiment, translate a research question into a logical probability statement in terms of a mass/density or cumulative probability function, perform calculations, and comment on the result. • Students will be able to state a hypothesis testing (or estimating) problem (out of the ones considered in the course) for a given research question, perform testing (construct the corresponding confidence interval) and draw statistical inferences. • For a given data set, students will be able to fit a simple linear regression model (obtain regression coefficients), provide an equation of a fitted model, interpret the observed relationship in words, and use the model to make predictions.
Course Content:	Introduction to Statistics and Data Analysis (sample, population, observational and experimental studies, measures of location and variability, basics of data cleaning, graphical methods and data description) Probability Random Variables, Probability Distributions and Expectation Discrete and Continuous Probability Distributions Fundamental Sampling Distributions and Data Descriptions (random sampling, key statistics, sampling distributions of a sample mean and sample variance) One- and Two-Sample Estimation Problems (statistical inference, confidence intervals for the mean, difference between two means, proportion and the difference between two proportions: their derivations and applications) One- and Two-Sample Testing Problems (statistical hypotheses and their testing, tests for a single mean and the difference between two means: cases with known and unknown variances; large sample tests for a single proportion and the difference between two proportions, applications) Simple Linear Regression

Course Expectations: In this course, you are expected to have professional behavior. You are expected to listen to all lecture videos, be curious, ask questions, seek opportunities to learn, and be open and responsive to constructive feedback. You are also expected to exhibit a professional demeanor (language, attitude) toward others. Disagreement during discussions is welcome and often productive in developing a deeper understanding of the concepts being discussed. However, disagreement does not warrant yelling or disrespectful language or behavior. Unprofessional behavior will not be tolerated, and appropriate actions will be taken to prevent future occurrences.

Online Lectures: Lecture video and slides will be posted on Canvas at (or before) 12:30 pm on Tuesday and Thursday. Students are expected to make notes to supplement the slides. If a student plans to miss a lecture then he/she should notify the instructor in advance.

Grading:

Homework	15%	Exam 1	20%
Quizzes	15%	Exam 2	20%
		Final Exam	30%

Grading Scale:	Grade	Final Percentage Range
	A	94.0-100
	A-	90.0-93.99
	B+	88.0-89.99
	B	84.0-87.99
	B-	80.0-83.99
	C+	78.0-79.99
	C	74.0-77.99
	C-	70.0-73.99
	D+	68.0-69.99
	D	64.0-67.99
	D-	60.0-63.99
	F	<60.0

Homework: Each student should download/print out and complete each homework assignment, and also take an on-line homework test on Canvas. Answer keys are posted on Canvas after the due date. Students are expected to compare their solutions with the answer keys and discuss their questions and concerns with the instructor or TAs. Homework tests are equally weighted. All assignments should be submitted before the deadline. There is **no makeup possible** for any assignments unless prior arrangements with the instructor have been made.

Quizzes: All quizzes need to be done online on Canvas before the due date. Any missed quiz is given 0 points. All quizzes are equally weighted. There is **no makeup possible** for any quizzes unless prior arrangements with the instructor have been made.

Exams: Three exams will be given during the course: **EXAM1: 12:30pm -1:45pm, Sep. 17 (online); EXAM2: 12:30pm -1:45pm, Oct. 20 (online); FINAL: 7:30am -9:30am, Nov. 25(online).**
Exams are open-book and open-notes. But **NO discussions are allowed.** You are not allowed to share the questions or your answers with anyone. You are expected to take exams at the scheduled times on Canvas. If this is impossible due to extreme circumstances (illness, death in the family, previously scheduled activities vital to academic program), please notify me. **No make-up exams will be given if I am not notified prior to the examination.** You will be required to obtain a note from your physician or adviser explaining the nature of the conflict.

Discussion and Bonus : We will open the discussion panel on Canvas. Each week, we will open a new discussion session. Students are welcomed to ask any questions related to the lecture taught in that week (but should not related to the homework, quiz, and exams). Students who know the answers are encouraged to answer the questions. In the end of this semester, active students in answering questions in the discussion panel can receive **additional one point** to their final grades.

Department Grade Appeal Policy: Students who believe their academic evaluation has been prejudiced or is capricious have recourse for appeals to, in order: their instructor; the Chair of the Statistics Department; the undergraduate academic grading appeals committee; and lastly, the college grading appeals committee.

Disabilities: Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual needs for academic accommodation. It is the policy of the University of Nebraska-Lincoln to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 132 Canfield Administration, 404-472-3787.

Academic Integrity: You are encouraged to work together on problems and exercises, but the work you turn in must be your own (unless the assignment specifically states otherwise). Work on exams must be your own (no discussion allowed). University policy will be followed in cases of academic dishonesty:

In cases where an instructor finds that a student has committed any act of academic dishonesty, the instructor may in the exercise of his or her professional judgment impose an academic sanction as severe as giving the student a failing grade in the course. Before imposing an academic sanction the instructor shall first attempt to discuss the matter with the student. If deemed necessary by either the instructor or the student, the matter may be brought to the attention of the student's major adviser, the instructor's department chairperson or head, or the dean of the college in which the student is enrolled.

For additional details see <http://stuafs.unl.edu/ja/code/three.shtml>.

Important Dates:

Class Begins	Aug. 18
Last day for late registrations and adds	Aug. 24
Last day to drop with 100% refund	Aug. 28
Last day to file a drop to remove course from student's record	Aug. 28
Exam 1 12:30 PM-1:45 PM	Sep. 17
Exam 2 12:30 PM-1:45 PM	Oct. 20
Last Class	Nov. 19
Final Exam 7:30 AM-9:30 AM	Nov. 25

Disclaimer: Information contained in this syllabus was, to the best knowledge of the instructor, considered correct and complete when distributed at the beginning of the term. However, the instructor reserves the right, acting within the policies and procedures of UNL, to make changes in course content or instructional technique without notice or obligation. However, any changes will be explained to the class as a whole including reasons for the change.

TENTATIVE COURSE OUTLINE

Due to the natural variation in the progress of lectures, the schedule is **tentative**.

Week	Day	Topic	Book Section(s)
August. 17 Week 1	Tues Thurs	Syllabus, Introduction to Statistics Introduction to Data Analysis	Chapter 1 Chapter 1
August. 24 Week 2	Tues Thurs	Probability, counting techniques; Probability rules, Conditional probability	Chapter 2.1-2.4 Chapter 2.5-2.6
August. 31 Week 3	Tues Thurs	Random variables, Discrete distribution Continuous distribution, Joint distribution	Chapter 3.1-3.2 Chapter 3.3-3.4
September. 7 Week 4	Tues Thurs	Expectation, Variance Linear Combinations of Random Variables, Chebyshev's Theorem	Chapter 4.1, 4.2 Chapter 4.3, 4.4
September. 14 Week 5	Tues Thurs	Review EXAM 1 (Sep. 17)	
September. 21 Week 6	Tues Thurs	Discrete distributions, Bernoulli distribution Binomial distribution, Poisson distribution	Chapter 5.1, 5.2 Chapter 5.2, 5.5
September. 28 Week 7	Tues Thurs	Continuous distributions, Uniform, Normal distributions Gamma and Exponential Distribution	Chapter 6.1-6.4 Chapter 6.6
October. 5 Week 8	Tues Thurs	Sampling Distributions, Sample mean, Sample variance Sampling Distributions of Means, Central Limit Theorem	Chapter 8.1-8.2 Chapter 8.3-8.4
October. 12 Week 9	Tues Thurs	Sampling Distributions of Sample Variances Review	Chapter 8.5
October. 19 Week 10	Tues Thurs	EXAM 2 (Oct. 20) Statistical inference, Point estimator, Interval estimator	Chapter 9.1-9.3
October. 26 Week 11	Tues Thurs	Confidence interval Two-sample estimation	Chapter 9.4-9.5, 9.10 Chapter 9.8, 9.11
November. 2 Week 12	Tues Thurs	Test of Hypotheses Test on single mean, single proportion	Chapter 10.1-10.3 Chapter 10.4, 10.8
November. 9 Week 13	Tues Thurs	Tests on two means Simple linear regression, least squares estimators	Chapter 10.5 Chapter 11.1-11.3
November. 16 Week 14	Tues Thurs	Properties and inference of least squares estimators Review	Chapter 11.4, 11.5
November. 23 Week 15	Wed	FINAL EXAM (Nov. 25)	

Instructor: Dr. Tefjol Pllaha

Office: 238 Avery Hall, office hours Tuesday 4–5 pm (or by appointment)

Email: tefjol.pllaha@unl.edu

Class Times and Location: TR 11:00–12:15 pm, Avery Hall – Room 118.

Optional Text Book: *Contemporary Abstract Algebra*, 10th edition by Joseph A. Gallian.

Our primary source will be lecture notes provided by the instructor and worksheets, but the text will be helpful as a secondary resource to enhance your understanding.

ACE 10: This course satisfies ACE (Achievement-Centered Education) outcome 10, with which the learning outcomes include *the ability to generate a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.*

Course Description: The primary goal of this course is to understand groups, whatever they are. This course will be inquiry-based, meaning that there will be very little conventional lecturing. Rather, we will explore known objects – such as an equilateral triangle or Rubik’s cube – and try to discover associated features. The course will also be somewhat flipped, meaning that, often, you will have preparatory readings ahead of class time, which will then be used as a base for worksheets.

Grades: Grades for the course will be computed as follows:

Homework	50%
Midterms	30% (2×15%)
Final	20%

Guaranteed grade cutoffs are below; these may be lowered based on the course difficulty.

A 93 % A- 89 % B+ 85 % B 81 % B- 78 % C+ 74 % C 70 %

To be successful in this course, plan to budget at least 6 hours of work outside of class weekly.

Homework: There will be weekly homework assignments. The purpose of the homework is to simultaneously deepen and complement concepts discussed in class. The homework will be graded for logic, ability to reason abstractly and the rigorous presentation of your arguments, but also for clarity and appropriate use of language, including full sentences. Collaboration among students in this class on homework is not only allowed, but encouraged as the best way to come up with a solution and to understand it is to explain it to your peers and get their feedback. However, each student is required to turn in their own assignment, written in a way that indicates their individual understanding of the solutions. The (one) lowest homework score will be dropped.

Any student who posts homework questions online or turns in solutions to homework that are taken from any source outside of class material will be charged with academic dishonesty. Identical (or nearly identical) homework will receive 0.

Worksheets: Most class periods we will be working in groups using worksheets. It is your responsibility to take individual notes based on class work.

Midterm Exams: There will be two midterm exams. Makeup midterms are only allowed for reasons limited to serious illness or travel to university approved events. Proof of these circumstances will be required. The *tentative* dates for the midterm exams are

Thursday, October 6 and Thursday, November 10.

Final Exam: The final exam is cumulative. Students are expected to arrange their personal and work schedules to allow them to take the final exam at the scheduled time. Our final exam is on

Wednesday, December 14, 3:30 – 5:30 pm

Course schedule The following table shows the material expected to be covered each week and the important academic schedule events. The numbers in parentheses refer to corresponding sections of the optional textbook. Note that what is shown below is *tentative and subject to change*. Please be alert for changes. Any changes will be announced in class.

Date	Chapter covered
Week of August 22	Introduction to Groups (Chap. 1)
Week of August 29	Groups (Chap. 2)
<i>Friday, September 2 is the last day to remove a course from a student's record.</i>	
Week of September 5	Finite Groups and Subgroups (Chap. 3)
Week of September 12	Cyclic Groups (Chap. 4)
Week of September 19	Group homomorphisms, Isomorphisms (Chaps. 10 & 6)
Week of September 26	Permutation Groups and Cayley's Theorem (Chaps. 5 & 6)
Week of October 3	Exam 1: Thursday, October 6
Week of October 10	Cosets and Lagranges Theorem (Chap. 7)
<i>Friday, October 14 is the last day to change the grading option to Pass/NoPass.</i>	
<i>October 17–18 Enjoy Fall break!</i>	
Week of October 17	Normal Subgroups and Factor Groups (Chap. 9)
Week of October 24	Isomorphism Theorems (Multiple Chapters)
Week of October 31	External Direct Products (Chap. 8)
Week of November 7	Exam 2: Thursday, November 10
<i>Friday, November 11 is the last day to withdraw from the course.</i>	
Week of November 14	Fundamental Theorem of Finite Abelian Groups (Chap. 11)
<i>November 23–25 Enjoy Thanksgiving break!</i>	
Week of November 28	Group Actions (Notes)
Week of December 5	Applications (Notes)
The Final Exam is 3:30 – 5:30 pm on Wednesday, December 14.	

Course Evaluations: The Department of Mathematics course evaluation form will be available through Canvas during the last two weeks of class. Evaluations are anonymous and instructors do not see the responses until after final grades have been submitted. Evaluations are important – the department uses them to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

Grading Appeals Policy: The Department of Mathematics does not tolerate discrimination or harassment on the basis of race, gender, religion, or sexual orientation. If you believe you have been subject to such discrimination or harassment, in this or any other math course, please contact the department. If, for this or any other reason, you believe your grade was assigned incorrectly or capriciously, then appeals may be made to (in order) the instructor, the vice chair, the Department grading appeals committee, the College of Arts and Sciences grading appeals committee, and the University grading appeals committee.

Students with Disabilities: Students with disabilities are encouraged to contact me for a confidential discussion of their individual needs for academic accommodation. It is the policy of UNL to provide flexible and individualized accommodation to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office in 117 Louse Pound Hall.

UNL Course Policies and Resources: Students are responsible for knowing the university policies and resources found on this page <https://go.unl.edu/coursepolicies>:

- University-wide Attendance Policy
- Academic Honesty Policy
- Services for Students with Disabilities
- Mental Health and Well-Being Resources
- Final Exam Schedule
- Fifteenth Week Policy
- Emergency Procedures
- Diversity & Inclusiveness
- Title IX Policy
- Other Relevant University-Wide Policies

Math 428/828 Course Outline

Instructor: Steve Cohn (Avery 226, scohn@unl.edu)

Text: *Introduction to Operations Research*, 9th Edition, by F. Hillier and G. Lieberman. (**Note:** This book and the solutions manual is available at various sites as a free pdf file. You can also find inexpensive used hardcover copies. You do **not** need the additional online material that comes with new copies.)

Office Hours: Monday 2:30-3:20, Thursday 10:00-10:50 and by appointment.

Time and Place: 12:30-1:20 pm, Monday, Wednesday and Friday, in Avery 119.

Prerequisites: The nominal prerequisites are Math 314 and Math/Stat 380. To follow the lectures, you'll need calculus, linear algebra and basic probability.

Course Log: This is a daily course log in the Canvas/Files folder. If you miss class, you can check the update to find which topics and sections were covered, announcements, changes to the syllabus, etc.

ACE Outcome: Math 428 satisfies ACE outcome 10: Generate a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection. The scholarly product will consist of oral and written group project reports. Your instructor will provide you with background material and guidance when needed.

UNL Covid Policies and Updates can be found here.

UNL Policies and Resources for

- Attendance,
- Academic Honesty,
- Services for Students with Disabilities,
- Mental Health and Well-Being,
- Final Exams (15th Week),
- On Campus Emergency Procedures,
- Diversity and Inclusiveness,
- Title IX,

can be found here.

Special Dates: Some dates to bear in mind are:

- The last day to file a drop to remove a full semester course from student's record is **January 28**.
- The last day to change your grade option to or from **Pass/No Pass** is **March 11**. If you choose the Pass/No Pass grading option, you need to earn a C or better in order to pass.
- The last day to **withdraw** from the course is **April 15**.

You can find more detail here.

Course Evaluation: The Department of Mathematics Course Evaluation Form will be available through your Canvas account during the last two weeks of class. You'll be notified when the form is made available. Evaluations are anonymous and instructors do not see any of the responses until after final grades have been submitted. Evaluations are important—the department uses evaluations to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

Course Outline: We'll try to cover all or part the following topics and sections:

- Formulation of Linear Programs: Sections **3.1, 3.2, 3.4, 3.5**
- Solution of Linear Programs: Sections **4.1-4.7**
- Theory of the Simplex Method: Sections **5.1-5.3**
- Duality, Sensitivity Analysis and the Simplex Method: Sections **6.1-6.5**
- Dynamic Programming: Sections **10.1-10.4**
- Queueing Theory: Sections **17.1-17.6**

Additional Material: In the unlikely event that we have time left over, we will choose additional material from these sections:

- Integer Programming: Sections **12.1-12.5**
- Inventory Theory: Sections **18.1-18.6**
- Decision Analysis: Sections **16.1-16.4**
- Simulation: Sections **20.1-20.5**

Math 828 Students in 828 will be graded on the same as those in 428, but should expect longer and more demanding exams and homework and project assignments.

Project: There will be a project, possibly broken into up into two or more parts. You may work on the project alone or in a group of at most three students.

Midterms: We'll have three 100-point midterms. The tentative dates are **February 23, March 23** and **April 27**. The tests will be given in class. Exam problems will be based *mostly* on the homework, lectures and assigned reading.

Final Exam: A 200-point, comprehensive final exam will be given **3:30 - 5:30 PM on Monday, May 9**. Students are expected to arrange their personal and work schedule to allow them to take the final exam at the scheduled time. No student will be permitted to take the final exam early.

Missed Exams: If you must miss a midterm because of UNL-mandated activities, injury or illness, you may be allowed to make it up. The rules governing makeups are:

- You must provide documentation.
- A request to make up a midterm can be made no later than 10:00 am on the day of the midterm.
- A missed exam that is not made up receives a score of zero.
- The foregoing rules may be suspended in the event of an emergency or extended crisis or illness.

Calculators: You may use a simple scientific calculator during an exam, but nothing else. In particular you **may not** use a graphing or programmable calculator, or any internet-enabled device. This includes smart phones and smart watches. For homework and the project, you may use any software and any device.

Homework: The facts and policies are:

- Homework will be assigned regularly. You are responsible for all assigned problems, though only a few in each week will be collected for grading. Exam problems will be based mostly on the homework.
- There are free **solutions manuals** in pdf form available online, as well as inexpensive paper copies. Even if you take a solution from a such a written source, I urge you to work the problem yourself. That is the best way to learn the material and prepare for exams.
- To receive full credit for a homework problem, you must have a correct, **clearly written** solution that is supported by appropriate work. A poorly written solution earns you one point at most. The point will be awarded if the answer is correct and supported by sufficient work.
- Late homework will not be accepted.
- You may be excused from a homework assignment because of illness, injury, some personal or family crisis, etc. I may ask for documentation.

Semester Grades: Your semester grade depends on your three midterms, the final, the project and the homework.

Activity	Value
Midterms	300
Final Exam	200
Project	100
Homework	100
Total	700

Letter grades correspond to percentages as follows:

Percentage	96	90	88	86	80	78	76	70	68	66	60	58
Letter Grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-

Grading Appeals: Students who believe their academic evaluation has been prejudiced or capricious have recourse for appeals in order, to: their instructor; the Chair of the Mathematics Department; the Mathematics Department grading appeals committee; and lastly, the College grading appeals committee.

Class Times and Location: MWF 2:30 - 3:20 p.m. in Brace Laboratory.

Professor: Richard Rebarber, Avery 330, rebarber@unl.edu. If you have any questions, e-mail me to make a Zoom appointment (you can also contact me through Canvas email). Also, I'll be available after most class periods.

Teaching Assistants: Lawrence Seminario-Romero (lseminario-romero2@huskers.unl.edu), Stephen Becklin (sbecklin2@huskers.unl.edu)

Prerequisites: Two of MATH 221, MATH 221H, MATH 314, MATH 314H, RAIK 270H, STAT 380. This semester Math 221 (or any introductory differential equations class) will be especially useful.

Course Documents: All Course Documents will be kept on Canvas.

Textbook: We will occasionally use *An Introduction to Mathematical Epidemiology* by Maia Martcheva. This book is available on Canvas, and also for download at the UNL Libraries website.

Course description: This semester in Math In The City we will focus on modeling infectious diseases, especially COVID-19. Approximately the first six weeks of the course will consist of lecture, homework and in-class activities designed to get you comfortable with epidemic modeling. During the rest of the course you will be doing a group project.

Journals: There are electronic files that you will continuously update throughout the semester. A template will be provided and you will submit your journals electronically every Monday starting January 24. During the first part of the course you will use your Journal entries to let us know what material you have questions about - if you don't have any questions, say so. Once we start the projects, your Journal entries will give details about what work you personally did on the project; we will use this part of the journals to determine your project participation.

Homework: During the first half of the course, there will be occasional homework assignments. You are encouraged to discuss the problems with your colleagues and instructors, but you have to write up anything you hand in by yourself. Sometimes homework will be done in class so you can talk to us and your fellow students. You are expected to upload a clear file (PDF, JPEG, or PNG) with organized and clean work. Usage of an app to scan your work is recommended, for example, Microsoft Office Lens or Evernote Scannable are free. A sloppy, convoluted work that is not readable for grading may (shall) be returned without grades and you need to arrange for a resubmission.

Project: Around 6 weeks into the semester, the class will work together to formulate some projects, and you will choose which project you wish to work on. You and your group will be to find references and learn material that is specific to your project, and to find relevant data. We will be available to answer questions and give suggestions. About two weeks before the project is due, you will hand in a rough draft so that we can give you pointers about how to improve the report.

Final Report: Part of understanding advanced concepts and doing research is describing these concepts and that research in writing. Therefore, the quality of your written exposition will be graded. Guidelines for good technical writing will be discussed in class. The Final Report

will be due Wednesday of Finals Week. The grade on the Final Report will be the same for all members of a group. The project participation grade may vary between members of a group.

Data: Most of the projects will use data. There is a huge amount of data that is publicly available, and part of the project will be to find and use relevant data.

Technology : You are allowed to use any software/programming language. You will write your reports in LaTeX, using the online environment Overleaf so that everybody in your group can work on the same file easily. We will provide LaTeX training.

During the second half of the semester, it might be helpful to bring a laptop to class. During the first half of the semester I might suggest bringing your laptop to some classes.

Do not use cellphones or other technology in class unless you are looking up information relevant to projects.

Final presentation: During Dead Week we will have a mini-conference where all groups can present their work. This will be advertised in the Math and Biology Departments, and will be open to anybody who is interested. Each group will be given 30 minutes to present their work. Taking part in the group presentation is compulsory and you are expected to adjust your schedule accordingly. Please attend the talks from the other groups if you can. Later in the semester details about what should be in the presentation will be given.

Illness Policy: If I cannot come in for class because of illness, I will give the class via Zoom (I'll send a link via email) or ask Lawrence and/or Stephen to take over.

If you cannot attend class because of illness *before* we start working on projects, please use the detailed written course material (notes, description of class activities) that will be on Canvas. Also, make sure you let me know if you miss a class, so I can give you a rundown of what you missed. If you have trouble with any of the material, find a time to Zoom with me, Lawrence or Stephen.

If you cannot attend class because of illness *after* we start working on projects, if you're feeling well enough you should Zoom in to your group meetings.

Grading: Grades for the course will be computed as follows:

- Homework = 25%
- Final project report = 25%
- Project participation = 20%
- Final presentation = 22%
- Attendance and Class Participation = 8%

University Policies: Policies on Academic Honesty, Services for Students with Disabilities, Mental Health and Well-Being Resources, Inclusiveness, and Title IX can all be found at <https://executivevc.unl.edu/academic-excellence/teaching-resources/course-policies>. You can ignore the parts about the Final Exam, since we don't have a Final Exam.

Course Evaluation: The Department of Mathematics Course Evaluation Form will be available through your Canvas account during the last two weeks of class. You'll get an email when the form becomes available. Evaluations are anonymous and instructors do not see any of

the responses until after final grades have been submitted. Evaluations are important - the department uses evaluations to improve instruction. Please complete the evaluation and take the time to do so thoughtfully.

ACE Outcome: This course satisfies ACE Outcome 10.

- Generate a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.

The outcome objectives will be demonstrated by a written report, an oral presentation, group meetings, and assignments or quizzes. The lectures and group meetings will provide the skills necessary for these assignments. Anonymous copies of your group written project and a sampling of journal entries will be kept as examples of the scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation, and reflection.

ECONOMICS 215-001
Statistics
Spring 2023
12:30PM – 1:45PM, M/W
HLH 018

Instructor: Dr. Yuxi Yao
Office: HLH 525B
Email: yyao10@unl.edu
Office Hours: M/W: 3:30PM-5:00PM or by appointment

Prerequisites: Sophomore standing; MATH 104 or equivalent or MATH 106/108H; BSAD 50; 2.5 GPA

Course Description

The course is an introduction to basic probability and statistical methods that are used in conjunction with economic and business information. Topics include descriptive statistics, probability foundations, probability distributions, sampling distributions, methods of statistical inference, and bivariate relationships. This course lays the foundation for subsequent courses.

Class Materials

1. Essentials of Statistics for Business and Economics; Anderson, et al., 9th edition, 2020 (textbook).
2. PowerPoint Slides: As the semester progresses, you will be able to view/download the slides and other materials used in the lecture in [Canvas](#).
3. Assignments: Assignments and grades will be posted in Canvas. Submissions of assignments should be done through Canvas as well.
4. Microsoft Excel: We will use Microsoft Excel for group assignments. UNL students have subscriptions to Office 365, which includes Microsoft Excel. Check [this link](#) for installation instructions.
5. Calculator: You will need a calculator for assignments and exams.

Textbook Purchase Options

Option 1. Through the UNL Bookstore's Inclusive ACCESS program.

Note: The bookstore's Inclusive ACCESS program is provisioning eBook access for every student enrolled in this course; every student receives an email from "donotreply@redshelf.com" to their @huskers.unl.edu email address with information for how they access the digital material, and an email from "noreply@follett.com" explaining ACCESS and referencing the opt out process. For Spring 2023 students have through **February 1st** to opt out (or to opt back in after initially opting out) if choosing to obtain the required material through another means. If a student opts out of the eBook access, the digital access will be revoked and UNL will remove the eBook fee from the student's account.

Option 2. eBook from Cengage: Anderson/Sweeney/... 9th ©2020, Essentials of Statistics for Business & Economics, 9780357045435. Click [here](#) to access eTextbook.

Grading

Total course points for this course will be a combination of the following:

i) 4 Individual Assignments (5 Points each, 15 Points total)

Your lowest score will be dropped and only the highest 3 will count towards your grade. *Late submissions will not be accepted.*

ii) 3 Group Excel Assignments (5 Points each, 10 Points total)

Your lowest score will be dropped and only the highest 2 will count towards your grade. You need to form your own groups with a maximum group size of 5 students and inform me by **February 10th**, 2022. Students who do not form groups by **February 10th** will be randomly assigned to a group. For each group, one student should send me an email with the names and UNL emails of all group members. Please include all your group members in the mailing list. Upon receiving the email, I will reply to all students in the group to confirm. For each assignment, each group needs to make one and only one submission. *Late submissions will not be accepted.*

All students in the group receive the same grades for the assignments. *Exception: In the case where a student does not respond to other group members and/or refuses to work on a group assignment (as confirmed by other group members), the student's grade for that group assignment will be zero and this student will be put in a separate group for all remaining group assignments.* Please keep in mind that, while I will let you decide exactly what and how much work each group member needs to do, the expectation is that all students should contribute to the completion of the group assignments.

iii) 4 Module Exams (15 Points each, 45 Points total)

Your lowest score will be dropped and only the highest 3 will count towards your grade. Module Exams will be held in the Digital Learning Center. The (tentative) exam dates are provided below.

iv) A Final Cumulative Exam (30 points)

Bonus Points: To help you learn the material and prepare for the exams, I will provide one practice quiz for each chapter covered in this class (11 quizzes in total). You are encouraged to work on those quizzes as the semester progresses. To provide some extra incentive, for each practice quiz completed, you will obtain 0.5 extra course point.

Tentative Module and Final Exam Dates

Module Exam 1: February 17 (Friday) – February 19 (Sunday)

Module Exam 2: March 9 (Thursday) – March 11 (Saturday)

Module Exam 3: April 14 (Friday) – April 16 (Sunday)

Module Exam 4: May 5 (Friday) – May 7 (Sunday)

Final Cumulative Exam: May 16 (Tuesday) – May 18 (Thursday)

The final cumulative exam will also be held in the Digital Learning Center. A review session for the final exam will be scheduled during the week before the final exam. Details will be confirmed and communicated later in the semester.

Make up exams:

There will be no makeup exams after the exam period in this class. If you need to write the exam

before the exam period for valid reasons, please contact me at least one week before the exam period. If you miss one module exam, it will be considered as the module exam with the lowest score and will be dropped. Should you miss the final exam or miss more than one module exam, I will require proof of the reason and will decide the validity of the excuse on a case-by-case basis. Should I find your excuse valid, 1) if you miss two module exams, the remaining two module exams will be scaled up to count for 45% of your grade; 2) if you miss the final exam and no more than one module exam, the module exams that you take will be scaled up to count for 75% of your grade; 3) if you miss a) three or more module exams or b) the final exam and two or more module exams, you may be given the grade “Incomplete.” The final decision rests with me.

Standard Grading Scale:

Your Score	Grade	Your Score	Grade
90% to 100%	A/A+	68% to 75%	C
88% to 90%	A-	65% to 68%	D+
85% to 88%	B+	61% to 65%	D
81% to 85%	B	58% to 61%	D-
78% to 81%	B-	Below 58%	F
75% to 78%	C+		

Achievement-Centered Education (ACE) Designation

Institutional Objective and Learning Outcome #3: Use mathematical, computational, statistical, or formal reasoning (including reasoning based on principles of logic) to solve problems, draw inferences, and determine reasonableness.

Course Objectives: (1) Acquaint students with the roles that probability and statistics play in examining and evaluating economic and business information; (2) Provide students with opportunities to become proficient with selected statistical methods that are commonly encountered in the business world; (3) Expose students to problem-solving environments where inferences and decisions must be made on the basis of statistical evidence, calculations, and probabilistic reasoning.

Learning Outcome Assessments: Exams, computer-based assignments and regular assignments.

NOTES

1. Students at the University of Nebraska are members of an academic community in which academic integrity and responsible conduct are essential for the community to function. To ensure that students know what is expected of them, the University has adopted the [Standards of Academic Integrity and Responsible Conduct](#) (“Standards”).
2. If you need special accommodations in order to meet any of the requirements of this course, please contact me as soon as possible so we can make the necessary arrangements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office. Please visit the [SSD webpage](#).
3. Stress, depression, anxiety, alcohol abuse, eating disorders, and other mental health challenges affect at least one in five college students. Resources for supporting student mental health can be found [here](#).
4. If in-person classes are canceled and the campus follows instructional continuity plans, the

- information will be communicated via Canvas announcements.
5. For other university-wide information and policies, please use this [link](#).

TENTATIVE COURSE OUTLINE

Module 1:

- **Topic 1**

Data and Statistics (Chapter 1)

- Applications in Business and Economics
- Data
- Data Sources
- Descriptive Statistics
- Statistical Inference
- Ethical Guidelines for Statistical Practice

- **Topic 2**

Descriptive Statistics: Tabular and Graphical Displays (Chapter 2)

- Summarizing Data for a Categorical Variable
- Summarizing Data for a Quantitative Variable
- Summarizing Data for Two Variables Using Tables
- Summarizing Data for Two Variables Using Graphical Displays

- **Topic 3**

Descriptive Statistics: Numerical Measures (Chapter 3)

- Measures of Locations
- Measures of Variability
- Measures of Distribution Shape
- Measures of Association Between Two Variables

Module 2:

- **Topic 4**

Introduction to Probability (Chapter 4)

- Random Experiments, Counting Rules, and Assigning Probabilities
- Events and Their Probabilities
- Some Basic Relationship of Probability
- Conditional Probability

- **Topic 5**

Discrete Probability Distributions (Chapter 5)

- Random Variables
- Developing Discrete Probability Distributions
- Expected Value and Variance
- Bivariate Distributions and Covariance
- Binomial Probability Distributions

- **Topic 6**

Continuous Probability Distributions (Chapter 6)

- Uniform Probability Distribution
- Normal Probability Distribution

Module 3:

- **Topic 7**
Sampling and Sampling Distribution (Chapter 7)
 - Selecting a Sample
 - Point Estimation
 - Introduction to Sampling Distributions
 - Sampling Distribution of the Mean
 - Sampling Distributions of the Proportion
- **Topic 8**
Interval Estimation (Chapter 8)
 - Population Mean: Known Population Variance
 - Population Mean: Unknown Population Variance
 - Population Proportion
 - Determining the Sample Size
- **Topic 9**
Hypothesis Tests (Chapter 9)
 - Developing Null and Alternative Hypothesis
 - Type I and Type II Errors
 - Population Mean: Known Population Variance
 - Population Mean: Unknown Population Variance
 - Population Proportion

Module 4:

- **Topic 10**
Inference About Means and Proportions with Two Populations (Chapter 10)
 - Difference Between Two Population Means: Known Population Variances
 - Difference Between Two Population Means: Unknown Population Variances
 - Difference Between Two Population Means: Matched Samples
 - Difference Between Two Population Proportions
- **Topic 11**
Simple Linear Regression (Chapter 14)
 - Simple Linear Regression Model
 - Least Squares Method
 - Coefficient of Determination
 - Model Assumptions
 - Inferences and Predictions

TENTATIVE COURSE SCHEDULE

Module	Areas of Focus	To do list and due dates
Module 1	Chapters 1, 2 and 3	Individual Assignment 1: Due on Feb 14 at 11:59PM Module Exam 1: Feb 17 – Feb 19
Module 2	Chapters 4, 5, and 6	Individual Assignment 2: Due on Mar 7 at 11:59PM Group Excel Assignment 1: Due on Mar 8 at 11:59PM Module Exam 2: Mar 9 – Mar 11
Module 3	Chapters 7, 8, and 9	Individual Assignment 3: Due on Apr 11 at 11:59PM Group Excel Assignment 2: Due on Apr 19 at 11:59PM Module Exam 3: Apr 14 – Apr 16

Module 4	Chapters 10 and 14	Individual Assignment 4: Due on May 2 at 11:59PM Group Excel Assignment 3: Due on May 10 at 11:59PM Module Exam 4: May 5 – May 7
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Disclaimer:

I reserve the right to modify this syllabus as the semester progresses to meet what I believe to be the needs of the class.

ECON 417-002: INTRO TO ECONOMETRICS

Spring 2021

INSTRUCTOR

Daniel Tannenbaum

Email: dtannenbaum@unl.edu

Class Lecture:

Location: HLH 032 and Zoom (both live and recorded)

Meeting Time: 2:00PM-3:15PM

Zoom Lectures:

Meeting ID: 951 0287 4295

Passcode: 4321

Course Webpage: <https://canvas.unl.edu>

Office Hours:

Meeting Time: Wednesdays, 4:00-5:30PM

Location: Zoom

Meeting ID: 920 1719 6473

Passcode: 4321

These office hours are open and do not require an appointment. Students should feel free to join and leave the Zoom meeting as desired. I am also available by appointment, and students should feel free to email me to set up a time to meet outside the scheduled office hours.

TEACHING ASSISTANT

Anupam Ghosh

Email: aghosh4@huskers.unl.edu

Office Hours: Tuesdays, 4-5:30PM

Location: Zoom

Meeting ID: 965 9608 4078

Passcode: 199

COURSE DESCRIPTION AND GOALS

This course presents an introduction to univariate and multivariate regression and its uses in economics. These methods lay the foundation for empirical work in economics and public policy. Students will be introduced to Stata, an econometric software package, which we will use to apply the tools we develop in class and to answer empirical questions using data from the real world.

By the end of the course students will be able to read empirical economics papers and answer two questions: (1) What are the needed statistical assumptions to interpret the results in the way the author is advocating? (2) What would pose a threat to these assumptions, in the context of the

problem the author is studying, and how would the potential threats change the interpretation of the results? These two questions will be a recurring theme throughout the course.

An additional theme of the course is for students to learn how to take a public policy question and to develop a precise econometric framework for answering the question, which can then be implemented with data.

I will be assuming the mathematical background provided in the prerequisites for the class. The class will make use of calculus and basic probability and statistics. Some of this material will be reviewed during the course, but this review will be done very quickly. If you have any questions about your own level of preparedness, come to office hours.

TEXTBOOK

The main textbook will be *Introduction to Econometrics, 3rd Edition*, by James H. Stock and Mark W. Watson. Other good undergraduate econometrics texts are *Introductory Econometrics*, by Jeffrey M. Wooldridge and *Basic Econometrics*, by Domodar N. Gujarati and Dawn C. Porter.

You are strongly advised to supplement class lectures with your own personal study of an econometrics text. My lectures will not follow Stock and Watson exactly but I will adopt their notation. I will often provide more mathematical details than found in the texts, so you are also strongly advised to attend each of the lectures and try to follow them closely. If you need to miss a class, contact one of your classmates and arrange to get a copy of their notes.

A class schedule will be uploaded to Canvas, containing a rough outline of the lectures and exact dates for the exams. Each lecture topic is referenced with chapter numbers from each of the three books above. **Anything covered in lecture is testable material.**

PROBLEM SETS

Problem sets will be assigned semi-weekly and are to be turned in at the beginning of the lecture during which it is due.

Computational Problems: Some of the problem sets will involve applied empirical problems with data using the econometric software package Stata. This software is widely used by applied researchers, both academic and professional. You will not be tested on Stata commands or syntax. However, there may be Stata-related exam questions – for example, asking you to interpret Stata regression output. You should be familiar with this aspect of the software and be able to interpret it. We have a license for Stata available for students. I will provide a class roster with names and NUID numbers to the CoB IT department to obtain permissions for each student to access Stata.

Problem set policies:

- Late problem sets will be penalized.

- You may work in groups of up to 4 students, but each student must submit their own set of solutions. On each problem set, you must write the names of the other students with whom you worked. **No two students' answers should be identical on any question.**
- Code for computational problems should be turned in via Canvas. We will create a folder for each assignment to which you can upload your code. Non-computational problems can be written by hand or typed but please print out a copy and bring to class. (The few students who have exemptions from attending in-person classes may upload a pdf of their problem sets to Canvas.)

Please keep in mind that it is in your interest to work through the problem sets on your own, as the majority of your grade will be determined by tests, for which collaboration is not possible.

GRADING

There will be 8-10 problem sets, two in-class midterm exams, and one in-class final exam. The midterms are tentatively scheduled for Week 5 (Thursday, February 25) and Week 10 (Thursday, April 1st). The final exam is scheduled by the Registrar. Final grades will be calculated by applying the following weights:

Problem Sets: 15%

Midterm Exams: 40%

Final Exam: 35%

Attendance and Participation: 10%

Grading policies:

- The lowest two problem set grades will be dropped.
- All grades are final except for obvious clerical errors. This rule applies equally to problem sets and exams. We award partial credit systematically, so we cannot revise up one student's grade without implicitly penalizing their classmates. If you would like guidance on missed points, feel free to discuss questions first with the TA and then with me during office hours.
- We will be happy to go through your exam or homework and help explain how you can improve your performance going forward.

The final course grades will be curved. Below is the grading scale that will serve as the starting point for final grades. I reserve the right to be more generous with the assignment of letter grades to percentages.

93-100: A
 90-92: A-
 87-89: B+
 83-86: B
 80-82: B-
 77-79: C+
 73-76: C

70-72: C-
65-69: D+
50-64: D
<50: F

ACADEMIC HONESTY

Students are expected to create and submit original research and written work. Plagiarism is taken very seriously and students will be held to UNL's Student Code of Conduct, section B.1.1c, which defines plagiarism as: "Presenting the work of another as one's own (i.e., without proper acknowledgment of the source) and submitting examinations, theses, reports, speeches, drawings, laboratory notes or other academic work in whole or in part as one's own when such work has been prepared by another person or copied from another person." Both undergraduate and graduate students will be held to the University's "zero tolerance" policy with respect to all aspects of the Student Code of Conduct, including plagiarism. Anyone caught violating the code of academic honesty on any written assignments will fail the course and be referred to the Dean's office for further disciplinary action.

UNL RESOURCES FOR STUDENTS WITH DISABILITIES

The University of Nebraska-Lincoln is committed to ensuring equal access to curricular and co-curricular opportunities for students with disabilities. Providing a range of services, SSD implements reasonable accommodations for students with disabilities and offers students the opportunity to contribute and participate in the diverse campus experience at the University of Nebraska-Lincoln.

Services for Students with Disabilities (SSD)

Office Location: 232 Canfield Administration

Building Phone: (402) 472-3787

UNL NON-DISCRIMINATION POLICY

The University of Nebraska does not discriminate based on race, color, ethnicity, national origin, sex, pregnancy, sexual orientation, gender identity, religion, disability, age, genetic information, veteran status, marital status, and/or political affiliation in its programs, activities, or employment. The following office has been designated to handle inquiries regarding non-discrimination policies:

Title IX or Discrimination Inquiries

Institutional Equity and Compliance

128 Canfield Administration Building

(402) 472-3417

equity2@unl.edu

COVID-19 POLICY

Face Covering Policy

All University of Nebraska–Lincoln (UNL) faculty, staff, students, and visitors (including contractors, service providers, and others) are required to use a facial covering at all times when indoors on the UNL campus, with the following exceptions:

- When eating; however, social distancing practices must be observed.
- When alone in a room.
- When alone in a motor vehicle.
- When an alternative is necessary as indicated by a medical professional during patient care.*
- When exercising either in a campus recreation facility or during a university-sponsored activity when use of a campus recreation facility is not practicable.
- When in one's assigned apartment or residence hall room.
- When the task requires the use of a NIOSH-approved respirator.
- When pertaining to children under the age of 2 years.

Facial coverings are also required in outdoor settings on the UNL campus if safe social distancing and gathering practices are not possible unless subject to an exception. This policy shall remain in effect until further guidance is provided from the Office of the Chancellor.

Rationale

An overarching principle to which the University of Nebraska–Lincoln ascribes is that Every Person and Every Interaction Matters. This facial covering policy has been established because we respect and care for ourselves and every person that we encounter on campus.

The Centers for Disease Control and Prevention (CDC) has acknowledged that use of facial coverings can help to slow the spread of COVID-19 in our communities. A facial covering can reduce the trajectory of expelled respiratory droplets, which pose risk to those around you. When you wear a face covering, you demonstrate care about the health and safety of those in our campus community.

What constitutes a face covering?

There are no universal design standards for facial coverings. However, the covering must extend from the bridge of the nose to below the chin.

- Reusable cloth facial coverings are acceptable, as are disposable paper masks.
- A face shield is an acceptable alternative.
- Regardless of the type used, a mask should not be designed with an exhalation valve.

*Requesting Alternatives

Individuals whose unique and individual circumstances require an alternative face covering, as indicated by a medical professional during patient care, may request one. To request an alternative,

students should contact the Office of Services for Students with Disabilities. Faculty and staff should contact Faculty/Staff Disability Services.

Wearing and Caring for Your Facial Covering

Each individual is responsible to care for their own facial covering. Disposable facial coverings are to be discarded as ordinary refuse after a single day's use. Reusable cloth facial coverings should be washed regularly. Reusable face shields should be cleaned thoroughly with soap and hot water after each day's use. Facial coverings that are visibly soiled or in poor condition should not be used.

Instruction for properly caring for and donning (putting on) and doffing (taking off) a face mask is provided at the Environmental Health & Safety website.

Additional Protective Measures

This policy on facial coverings complements other important community and self-protection measures which are to be consistently practiced by all faculty, staff, students, and visitors. These measures include:

- Self-monitoring for symptoms and staying home when ill or after having been exposed to someone known or suspected of having COVID-19.
- Washing hands often and thoroughly.
- Refraining from touching the face, eyes, nose, and mouth.
- Practicing social distancing, by maintaining 6 feet of distance from others.
- Regularly cleaning and disinfecting surfaces.

Compliance

Any employee, student or visitor who fails to abide by these guidelines may be asked or directed to leave the campus space. Employees and students who are directed to leave a campus space for failure to comply with these guidelines may be taken off duty and/or subject to disciplinary action.

Procedures for ensuring compliance in classrooms and similar places are described in the required syllabus statement, "Required Use of Face Coverings for On-Campus Shared Learning Environments."

Counteracting Existing Marginalization

In implementing this policy, we must ensure protocols to optimize safety and provide a sense of well-being for all members of our community. We recognize that a policy to wear facial coverings may exacerbate implicit biases and existing racial stereotypes. Thus, we will provide training and professional development for members of campus units regarding face coverings in order to prevent racial profiling and targeting.

WEEK	TOPIC	READINGS		
		Stock & Watson	Gujarati & Porter	Wooldridge
1	Introduction to Empirical Methods in Economics			
	STATA PRIMER			
2	Probability & Statistics: Bivariate and multivariate distributions	Ch 1-3	Appendix A	Appendix B
3	Estimators, small sample properties	Ch 1-5	Appendix A	Appendix C
3	Estimators, large sample properties	Ch 1-6	Appendix A	Appendix C
4	Central limit theorem (asymptotic distribution of the sample mean), convergence. MLE and analogy principle			
5	Simple linear regression: Intro and interpretations	Ch 4	Ch 1-3	Ch 2
	MIDTERM I			
6	Simple linear regression, ordinary least squares			
6	Simple linear regression: R^2 , properties of OLS	Ch 4	Ch 3, 4	Ch 2
7	Simple linear regression: Inference	Ch 5	Ch 5	Ch 2
7	Simple linear regression: Inference,	Ch 5	Ch 6	Ch 2, 8
8	Intro to Multivariate Linear Regression		Appendix B,	Appendix D,
8	Multivariate Linear Regression: Estimation	Ch 6	Ch 7.4, 7A.1	Ch 3
9	Multivariate Linear Regression: R^2 , adjusted R^2 , properties of OLS	Ch 6	Ch 7.8	Ch 3, 5, 6
	MIDTERM II			
11	Multivariate Regression: Inference	Ch 7	Ch 8	Ch 4
12	Dummy Variables and Non-Linear (In Variables) Models	Ch 8, 9, 11	Ch 9, 6.4-6.8, 7.10	Ch 6, 7
13	Differences-in-Differences			
14	Instrumental Variables (IV) I: Using IV to solve omitted variables	Ch 12		Ch 15, 16
14	Instrumental Variables II	Ch 12		Ch 15, 17
15	Instrumental Variables IV (time permitting) Heterogeneity and LATE			
	IN-CLASS REVIEW: Examples and Applications			
	FINAL EXAM (Time and Location Scheduled by the University)			